Knowledge Base

Synthetic Financial Time Series Generation

From Arithmetic to GANs — Every Step Explained

Feynman Technique Throughout

Situation → Problem → Worked Arithmetic
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Chapter 1

Numbers, Algebra and Functions

Before and After

Before this chapter you need: arithmetic with whole numbers — adding, subtracting, multiplying and dividing. Nothing else.
After it you will understand: what a variable is, how to rearrange an equation without breaking it, and what people mean when they call something a *function*. Every later chapter is written in the language of functions, so this is the vocabulary the rest of the book assumes.

This chapter contains no finance and almost no notation. Its job is to make sure that when Chapter 3 writes $f(x)$ and asks how fast it changes, the symbols are already familiar rather than a second obstacle stacked on top of the idea.

1.1 Fractions, Ratios and Percentages

Situation — the everyday picture

A share costs \$40 on Monday. On Tuesday it costs \$46. You want to describe the change to someone who does not know the starting price. Saying "it went up \$6" is useless to them — \$6 on a \$40 share is a large move, \$6 on a \$4,000 share is a rounding error. What they want is the change *relative to where it started.*

Problem — where it breaks or why it matters

Absolute differences cannot be compared across things of different size. A \$6 move in Brazil and a \$6 move in China tell you nothing about which market had the bigger day. To compare, every quantity must first be put on a common scale, and the natural common scale is "fraction of the starting value".

Worked Arithmetic — concrete numbers

Start 40, end 46.

Change: $46 - 40 = 6$.

As a fraction of the start: $\dfrac{6}{40} = \dfrac{3}{20} = 0.15$.

As a percentage: $0.15 \times 100 = 15\%$.

Check it the other way round. If 40 grows by 15%, the new value is $40 \times (1 + 0.15) = 40 \times 1.15 = 46$. ✓

Now the same move on a bigger share. Start 4,000, end 4,006: $\frac{6}{4000} = 0.0015 = 0.15\%$. Same \$6, one hundredth of the move.

Formal Name — the mathematics

For a quantity that starts at P_0 and ends at P_1, the **simple return** is

$$R = \frac{P_1 - P_0}{P_0} = \frac{P_1}{P_0} - 1.$$

A **percentage** is a fraction multiplied by 100; the symbol % means "divide by 100". The quantity P_1/P_0 is called the **gross return** or **price relative**: a gross return of 1.15 and a simple return of 0.15 say the same thing.

Where the Analogy Breaks

"15% up" and "15% down" are not opposite moves. Going up 15% from 40 gives 46; coming back down 15% from 46 gives $46 \times 0.85 = 39.1$, not 40. The percentage is always taken relative to *wherever you currently are*, and after the first move that is a different place. This asymmetry is not a curiosity — it is the reason Chapter 2 replaces percentages with logarithms for everything in this thesis.

Why It Matters Here — link to the thesis

Every number in this thesis is a return, not a price. The five BRICS markets trade at wildly different index levels, and only by converting to returns can BOVESPA and SHANGHAI be placed on the same axis, fed to the same model, and scored by the same metric.

1.2 Variables and Rearranging Equations

Situation — the everyday picture

A recipe says: to find the cost of a taxi, take \$3 for the pickup and add \$2 for every kilometre. You could write this out in words every time, or you could write $C = 3 + 2k$ and let k stand for however many kilometres you happen to travel. The letter is shorthand for "a number I have not decided yet".

Problem — where it breaks or why it matters

Once the relationship is written down, you often need to run it backwards. The formula gives cost from distance, but the question you actually have is "I have \$15 — how far can I go?". That means solving for k instead of C, and doing that safely requires a rule for what you are allowed to do to an equation.

Worked Arithmetic — concrete numbers

The rule is: *whatever you do to one side, do to the other.* An equation is a claim that two things are equal, and equals stay equal if treated identically.

Start with $C = 3 + 2k$ and set $C = 15$:

$$15 = 3 + 2k$$

Subtract 3 from both sides:

$$15 - 3 = 3 + 2k - 3 \quad \Longrightarrow \quad 12 = 2k$$

Divide both sides by 2:

$$\frac{12}{2} = \frac{2k}{2} \quad \Longrightarrow \quad 6 = k$$

Check by substitution: $C = 3 + 2 \times 6 = 3 + 12 = 15$. ✓

Substituting the answer back into the original equation is the only check you ever need, and it costs ten seconds.

Formal Name — the mathematics

A **variable** is a symbol standing for a number that may take different values. A **constant** is a fixed number. An **equation** asserts that two expressions are equal; to **solve** it for a variable is to produce an equivalent equation with that variable alone on one side.
The operations that preserve equality are: adding or subtracting the same quantity from both sides, and multiplying or dividing both sides by the same *non-zero* quantity.

Where the Analogy Breaks

The "do the same to both sides" rule has one trap: multiplying both sides by zero. From $5 = 5$ we may multiply both sides by 0 and get $0 = 0$, which is true but has lost all information — and the step cannot be undone. This is why division by zero is forbidden: it is the reverse of a step that destroys information. The same caution reappears in Chapter 5, where a matrix with determinant zero cannot be inverted for exactly this reason.

Why It Matters Here — link to the thesis

Rearranging is how the definition of a log return, $r = \ln(P_1/P_0)$, is turned into the reconstruction $P_1 = P_0 e^r$ used whenever a synthetic return series must be turned back into a synthetic price path.

1.3 What a Function Is

Situation — the everyday picture

A vending machine has a keypad. You press A4 and a specific bar of chocolate comes out. You press A4 again tomorrow and the *same* kind of bar comes out. Press B2 and something else comes out. The machine is a rule that turns each input into exactly one output, reliably, every time.

Problem — where it breaks or why it matters

The reliability is the whole point. A machine that sometimes gave chocolate and sometimes gave crisps for the same button would be useless — you could not reason about it or plan around it. Mathematics needs the same guarantee before it can say anything about how outputs change when inputs change.

Worked Arithmetic — concrete numbers

Take the rule "double it, then add one", written $f(x) = 2x + 1$.

Input x	Working	Output $f(x)$
0	$2 \times 0 + 1$	1
1	$2 \times 1 + 1$	3
2	$2 \times 2 + 1$	5
3	$2 \times 3 + 1$	7
-1	$2 \times (-1) + 1$	-1

Each input appears once and produces one output. Note that outputs *may* repeat for different inputs without breaking anything: for $g(x) = x^2$ we get $g(2) = 4$ and $g(-2) = 4$. Two doors, same room. That is allowed. What is not allowed is one door opening onto two rooms.

Formal Name — the mathematics

A **function** f from a set A to a set B assigns to each element $x \in A$ exactly one element $f(x) \in B$. We write $f : A \to B$, read "f maps A to B". The symbol $\in$ means "is an element of".

The value $f(x)$ is the **image** of x. The notation is deliberately mechanical: f is the machine, x is what goes in, $f(x)$ is what comes out.

Where the Analogy Breaks

The vending-machine picture suggests the machine is simple and you could open it up and see the mechanism. Many functions in this thesis are not like that. A trained neural network is a function — one input gives one output, reliably — but it has hundreds of thousands of internal numbers and no human-readable mechanism. "Function" guarantees only the input–output discipline, never simplicity or interpretability.

Why It Matters Here — link to the thesis

A generator in a GAN is a function: it takes a vector of random noise and returns a synthetic return series. A discriminator is a function: it takes a series and returns a score. Training changes *which* function these are, but they are functions at every moment, which is what makes it meaningful to ask how their outputs respond to their inputs.

1.4 Domain and Range

Situation — the everyday picture

The vending machine has buttons A1 to D5 and nothing else. Pressing a letter that is not on the keypad does nothing — the question "what does the machine give for Z9?" has no answer, because Z9 was never a legal input.

Problem — where it breaks or why it matters

Ignoring which inputs are legal produces confident nonsense. A formula that divides by x is silent about what happens at $x = 0$ until someone asks; a formula involving $\sqrt{x}$ has nothing to say about $x = -1$ if we are working with ordinary numbers. Stating the legal inputs up front prevents the error.

Worked Arithmetic — concrete numbers

Example 1: $f(x) = 1/x$.
Legal inputs: every number except 0. At $x = 2$, $f = 0.5$; at $x = 0.1$, $f = 10$; at $x = 0.001$, $f = 1000$. As x shrinks toward zero the output grows without bound, and at $x = 0$ there is no output at all.

Example 2: $f(x) = \sqrt{x}$ over ordinary (real) numbers.
Legal inputs: $x \geq 0$. Outputs: $\sqrt{0} = 0$, $\sqrt{9} = 3$, $\sqrt{2} \approx 1.414$. Every output is ≥ 0, so although the legal inputs run from 0 upwards, so do the outputs.

Example 3: $f(x) = x^2$.
Legal inputs: all numbers. Outputs: never negative, because a number times itself is never negative. So the outputs occupy only half the number line even though the inputs occupy all of it.

Formal Name — the mathematics

The **domain** of f is the set of inputs on which f is defined. The **codomain** is the set the outputs are declared to live in. The **range** (or image) is the set of values f actually attains. For $f(x) = x^2$ with domain $\mathbb{R}$ (all real numbers), the codomain may be declared as $\mathbb{R}$, but the range is only $[0, \infty)$ — the non-negative numbers. Range is a subset of codomain, sometimes a strict one.

Where the Analogy Breaks

Domain is often presented as a technicality to be recited and forgotten. It is not. In Chapter 2 the domain of ln is exactly the strictly positive numbers, and that single restriction is what forces prices — never returns — to be the thing we take logarithms of. The restriction carries real content about what the mathematics is willing to model.

Why It Matters Here — link to the thesis

The generators in this thesis end with a tanh activation, whose range is $(-1, 1)$. That is a deliberate choice of range: it guarantees the network cannot emit an arbitrarily large value, and it is why the training data must first be squashed into the same interval before the two can be compared.

1.5 Linear Functions and Slope

Situation — the everyday picture

You are walking up a straight ramp. For every 4 metres you move forward, you rise 1 metre. The ramp does not get steeper or shallower — the trade-off between forward and upward is the same everywhere on it. One number, 1/4, describes the entire ramp.

Problem — where it breaks or why it matters

Most relationships are not ramps. But a great deal of mathematics consists of finding the ramp that best matches a curve near a point of interest, because ramps are the only shape we can reason about effortlessly. Before that programme can start, "steepness" has to be pinned to a number.

Worked Arithmetic — concrete numbers

Take $f(x) = 3x + 2$.

x	$f(x)$	Change in x	Change in f
1	5	—	—
2	8	1	3
4	14	2	6
7	23	3	9

From $x = 1$ to $x = 4$: heights 5 and 14, change 9, horizontal distance 3, ratio $9/3 = 3$.
From $x = 4$ to $x = 7$: heights 14 and 23, change 9, distance 3, ratio 3.

The ratio is 3 no matter which two points we pick. That constancy is precisely what makes the function linear, and the 3 is exactly the number multiplying x in the formula. The $+2$ shifts the whole line up without tilting it: at $x = 0$ the value is 2.

Formal Name — the mathematics

A **linear function** (strictly, an *affine* one) has the form

$$f(x) = mx + c,$$

where m is the **slope** or **gradient** and c is the **intercept**, the value at $x = 0$. For any two distinct points (x_1, y_1) and (x_2, y_2) on the line,

$$m = \frac{y_2 - y_1}{x_2 - x_1},$$

often read "rise over run". Positive m tilts upward, negative m downward, and $m = 0$ is flat.

Where the Analogy Breaks

Slope as "rise over run" silently assumes the line is not vertical. A vertical line has zero run, and the ratio would require dividing by zero — so vertical lines simply have no slope, rather than an infinite one. Refusing to write $m = \infty$ here is the same discipline that will matter in Chapter 9, where a distribution with an undefined fourth moment is genuinely undefined rather than merely large.

Why It Matters Here — link to the thesis

The derivative in Chapter 3 is nothing more than this slope, computed over a vanishingly small run. Gradient descent — the algorithm that trains every model in this thesis — reads that slope and steps downhill.

1.6 Quadratics and Curvature

Situation — the everyday picture

Throw a ball straight up. It rises, slows, stops, and falls. Its height against time is not a ramp: the steepness changes continuously, from strongly positive at the throw to zero at the top to strongly negative on the way down.

Problem — where it breaks or why it matters

A single slope cannot describe this path, because the path has a different slope at every instant. We need a shape with built-in bend, and the simplest such shape is the one produced by squaring.

Worked Arithmetic — concrete numbers

Take $f(x) = x^2$.

x	$f(x)$	Interval	Change in f	Average slope
0	0	—	—	—
1	1	$0 \to 1$	1	1
2	4	$1 \to 2$	3	3
3	9	$2 \to 3$	5	5
4	16	$3 \to 4$	7	7

The average slope over each unit step is $1, 3, 5, 7$ — rising by 2 every time. The function is not merely increasing; the *rate* at which it increases is itself increasing at a steady rate. That steady second-level change is what "curvature" means, and here it equals 2.

Note also the symmetry: $f(-3) = 9 = f(3)$. The curve is a mirror image about the vertical line $x = 0$, which sits at the bottom of the bowl.

Formal Name — the mathematics

A **quadratic function** has the form

$$f(x) = ax^2 + bx + c, \qquad a \neq 0.$$

Its graph is a **parabola**. If $a > 0$ it opens upward and has a single lowest point; if $a < 0$ it opens downward and has a single highest point. That turning point sits at $x = -b/(2a)$. For $f(x) = x^2$ we have $a = 1$, $b = 0$, $c = 0$, so the turning point is at $x = 0$, matching the table above.

Where the Analogy Breaks

A parabola has exactly one turning point, which makes "find the lowest value" easy: there is nowhere else it could be. Almost none of the functions minimised in this thesis are like that. A neural network's loss surface has enormous numbers of local dips, and an algorithm that walks downhill will stop at whichever one it happens to reach. The single-minimum picture is the exception, not the rule — Chapter 4 returns to this.

Why It Matters Here — link to the thesis

Squared error — the quantity minimised when the TimeGAN autoencoder learns to reconstruct its input — is a quadratic in the reconstruction error. Its bowl shape is why the reconstruction loss has a well-defined best answer, and why a reported R^2 of $+0.995$ means the bottom of that bowl was very nearly reached.

1.7 Composition of Functions

Situation — the everyday picture

A factory line has two stations. The first paints the part; the second boxes it. A raw part enters, a painted part passes between them, a boxed painted part comes out. Nobody inspects the middle stage — from outside, the line is one machine turning raw parts into finished goods.

Problem — where it breaks or why it matters

Deep networks are precisely this: dozens of simple stations in a row. To train them we must answer "if I adjust station one slightly, how much does the final output move?" — a question about the whole line, asked at a single station. Answering it requires first being clear about what stacking functions even means.

Worked Arithmetic — concrete numbers

Let $g(x) = x + 3$ (the painter) and $f(u) = 2u$ (the boxer). Apply g first, then f.

Input x	After g: $x + 3$	After f: $2(x + 3)$
1	4	8
2	5	10
5	8	16

The combined rule is $f(g(x)) = 2(x + 3) = 2x + 6$. Check at $x = 5$: $2 \times 5 + 6 = 16$. ✓

Order matters. Reverse the stations — box first, then paint: $g(f(x)) = 2x + 3$. At $x = 5$ that gives 13, not 16. Painting a boxed part is not the same as boxing a painted one.

Formal Name — the mathematics

Given $g : A \to B$ and $f : B \to C$, the **composition** $f \circ g$ is the function $A \to C$ defined by

$$(f \circ g)(x) = f(g(x)).$$

It is read "f after g": the function written on the *left* is applied *last*. Composition requires the range of g to lie inside the domain of f — the parts leaving the first station must be things the second station can accept.

Where the Analogy Breaks

The factory picture suggests each station is independent, so you could swap or remove one freely. In a trained network that is false. Each layer's weights have been fitted assuming exactly what the previous layer sends it; extract a middle layer and it produces nothing meaningful on raw input. The stations are separable in notation only.

Why It Matters Here — link to the thesis

A neural network is one long composition, and the chain rule of Chapter 4 is the tool that differentiates it. Everything called "backpropagation" is the chain rule applied to a composition of this kind. The TimeGAN recovery function composed with its embedder is a composition whose round-trip fidelity we measure directly — and require to be accurate to within 10^{-3} before training is allowed to proceed.

Chapter 2

Exponentials and Logarithms

Before and After

Before this chapter you need: percentages and gross returns (§1.1), functions and their domains (§1.3–§1.4), composition (§1.7).
After it you will understand: why every quantity in this thesis is a *log* return rather than a percentage return, why that choice makes multi-day returns add up exactly, and why a log-scaled histogram is the honest way to look at the tail of a distribution.

This chapter earns its length. The decision to work in log returns propagates into every model, every metric and every figure in the project, and the reasons are worth seeing in full rather than accepting on authority.

2.1 Compound Growth

Situation — the everyday picture

You put \$100 in an account paying 8% a year, and you leave the interest in. After one year you have \$108. The second year's 8% is charged on \$108, not on the original \$100, so you gain \$8.64 rather than \$8. Growth feeds on itself.

Problem — where it breaks or why it matters

Because each year's gain depends on the running total, growth over n years is not n separate additions. It is n multiplications, and multiplication is far harder to reason about than addition — especially when you want to compare periods of different length, or split a long period into pieces.

Worked Arithmetic — concrete numbers

\$100 at 8% per year, interest retained.

Year	Start	Gain	End
1	100.00	8.00	108.00
2	108.00	8.64	116.64
3	116.64	9.33	125.97

As repeated multiplication: $100 \times 1.08 \times 1.08 \times 1.08 = 100 \times 1.08^3 = 125.97$. ✓

Compare to simple (non-compounding) interest: $100 + 3 \times 8 = 124$. The gap of \$1.97 after only three years is the compounding effect, and it widens with time.

Now compound more often. Take 100% per year, but split it into n instalments of $1/n$ each:

$$\begin{aligned} n = 1: &\quad (1 + 1/1)^1 = 2.00000 \\ n = 2: &\quad (1 + 1/2)^2 = 1.5^2 = 2.25000 \\ n = 4: &\quad (1 + 1/4)^4 = 1.25^4 = 2.44141 \\ n = 12: &\quad (1 + 1/12)^{12} = 2.61304 \\ n = 365: &\quad (1 + 1/365)^{365} = 2.71457 \end{aligned}$$

The numbers climb, but ever more slowly, and they are converging on something just above 2.718. They do not run away to infinity.

Formal Name — the mathematics

An **exponential function** has the form $f(x) = a^x$ with $a > 0$. Growth at rate R per period, compounded over n periods, multiplies the starting value by $(1 + R)^n$.
The limit reached by the table above defines **Euler's number**:

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \approx 2.718281828\ldots$$

It is irrational: its decimal expansion never terminates or repeats. The function $f(x) = e^x$ is called *the* exponential function, written $\exp(x)$.

Where the Analogy Breaks

The bank analogy suggests e arises from a peculiar accounting convention, as if it were an artefact of how often interest is credited. It is not. The quantity e^x is singled out by a property with nothing to do with money: it is the only function that is its own rate of change, a fact Chapter 3 makes precise. Continuous compounding is one place e shows up, not the reason it exists.

Why It Matters Here — link to the thesis

Continuous compounding is the natural setting for asset prices, which are quoted continuously through the trading day rather than at fixed interest dates. Writing $P_1 = P_0 e^r$ instead of $P_1 = P_0(1 + R)$ is what makes r additive across time, which is the property the rest of this thesis is built on.

2.2 The Logarithm as the Inverse

Situation — the everyday picture

A machine doubles whatever you feed it: put in 3, get 6; put in 5, get 10. Now someone hands you a 32 and asks what was fed in. You need the machine run backwards. Since $2^5 = 32$, the answer is 5 — you are asking "how many doublings?".

Problem — where it breaks or why it matters

Running exponentials backwards is exactly the question "what power was this?". It cannot be answered by any combination of the four arithmetic operations, so it needs a name and a function of its own. Without one, every compound-growth calculation is a one-way street.

Worked Arithmetic — concrete numbers

Base 2, so we are counting doublings.

x	2^x	$\log_2$ of it
0	1	$\log_2 1 = 0$
1	2	$\log_2 2 = 1$
2	4	$\log_2 4 = 2$
3	8	$\log_2 8 = 3$
5	32	$\log_2 32 = 5$

Read the table left to right for exponentials, right to left for logarithms. The two columns hold identical information.

The characteristic move. Note that $4 \times 8 = 32$, and in the log column $2 + 3 = 5$. Multiplying the numbers corresponds to adding their logarithms. This is not a coincidence of these particular values; it is the defining property, and it is the entire reason logarithms are useful.

With base e. The natural logarithm ln undoes e^x: $\ln(1) = 0$, $\ln(e) = 1$, $\ln(e^2) = 2$. Numerically $\ln(2) \approx 0.693$ and $\ln(10) \approx 2.303$.

Formal Name — the mathematics

The **logarithm to base a**, written $\log_a$, is the inverse of $x \mapsto a^x$:

$$\log_a(y) = x \iff a^x = y.$$

The **natural logarithm** $\ln = \log_e$ satisfies

$$e^{\ln x} = x \quad \text{and} \quad \ln(e^x) = x.$$

Its domain is the strictly positive reals: $\ln(0)$ and $\ln(-1)$ are undefined, because no real power of e produces zero or a negative number. As $x \to 0^+$, $\ln x \to -\infty$.

Where the Analogy Breaks

Calling ln "the inverse" invites the assumption that it undoes e^x in every sense, including numerically. It does not. Computing e^{-40} gives roughly 4×10^{-18}, which in ordinary floating-point arithmetic may round to something whose logarithm is no longer -40. The mathematical inverse is exact; the computational one is not, which is why numerical code so often carries quantities in log form from the start rather than converting back and forth.

Why It Matters Here — link to the thesis

The strictly-positive domain is the reason we take logarithms of *prices* and never of returns. A price is always positive; a return is negative on roughly half of all days, and ln has nothing to say about those.

2.3 The Log Rules

Situation — the everyday picture

Before electronic calculators, engineers multiplied large numbers by looking up two logarithms in a printed table, adding them by hand, and looking the result back up. A hard operation was traded for an easy one at the cost of two lookups.

Problem — where it breaks or why it matters

The same trade is worth making today, for a different reason. Products of many terms overflow or underflow the range a computer can represent, and they resist algebraic manipulation. Sums do neither. Anything that converts products to sums buys both numerical stability and analytical tractability.

Worked Arithmetic — concrete numbers

Verify each rule on numbers small enough to check mentally, using base 2.

Product rule. $\log_2(4 \times 8) = \log_2 32 = 5$, and $\log_2 4 + \log_2 8 = 2 + 3 = 5$. ✓

Quotient rule. $\log_2(32/4) = \log_2 8 = 3$, and $\log_2 32 - \log_2 4 = 5 - 2 = 3$. ✓

Power rule. $\log_2(4^3) = \log_2 64 = 6$, and $3 \times \log_2 4 = 3 \times 2 = 6$. ✓

Log of 1. $\log_2 1 = 0$, because $2^0 = 1$. This holds in every base, and it is the fact that makes "no change" correspond to a log return of exactly zero.

Formal Name — the mathematics

For $x, y > 0$ and any real k:

$$\ln(xy) = \ln x + \ln y, \qquad \ln\left(\frac{x}{y}\right) = \ln x - \ln y, \qquad \ln(x^k) = k \ln x, \qquad \ln 1 = 0.$$

To change base: $\log_a x = \ln x / \ln a$.

Where the Analogy Breaks

There is no rule for $\ln(x + y)$. The logarithm converts multiplication into addition, not addition into anything simpler — and beginners routinely invent $\ln(x + y) = \ln x + \ln y$, which is false. Check it: $\ln(2 + 2) = \ln 4 \approx 1.386$, while $\ln 2 + \ln 2 \approx 1.386$. Those agree, which is exactly the sort of accident that entrenches the error. Try $\ln(1 + 3) = \ln 4 \approx 1.386$ against $\ln 1 + \ln 3 = 0 + 1.099 = 1.099$. They differ.

Why It Matters Here — link to the thesis

The product rule is the engine of the next section. It is also why the log-likelihood, rather than the likelihood, is what gets maximised whenever a GARCH model is fitted to a return series in this project: a product of thousands of small probabilities underflows to zero, while the corresponding sum of logarithms is perfectly well behaved.

2.4 Why Log Returns Are Additive

Situation — the everyday picture

You invest \$100. It rises 10%, then falls 10%. You are back where you started — or so the symmetry of the two numbers suggests. You are not. You have \$99.

Problem — where it breaks or why it matters

Percentage returns do not add across time. Over two days the errors are small enough to ignore; over 5,000 trading days — the span of this project's data — they compound into nonsense. Any quantity that is to be summed, averaged, or modelled as a sequence of increments must be additive by construction.

Worked Arithmetic — concrete numbers

The asymmetry, in full. Start at 100.

$$100 \times 1.10 = 110, \qquad 110 \times 0.90 = 99.$$

Net change -1%, not 0%. In gross terms $1.10 \times 0.90 = 0.99$.

Now in logs:

$$\ln(1.10) = +0.09531, \qquad \ln(0.90) = -0.10536.$$

Sum: $+0.09531 - 0.10536 = -0.01005$. And directly, $\ln(99/100) = \ln(0.99) = -0.01005$. ✓
The log returns sum to exactly the right answer. The percentages do not.

The genuinely symmetric case. Take a rise and fall that really do cancel: up by a factor 1.10, then down by a factor $1/1.10 = 0.909091$.

$$\ln(1.10) = +0.09531, \qquad \ln(0.909091) = -0.09531.$$

They sum to exactly zero, and indeed $100 \times 1.10 \times 0.909091 = 100$. In log space, "up then back down" is $+a$ then $-a$ — perfectly symmetric. In percentage space it is $+10\%$ then -9.09%, which looks lopsided but is not.

Three days from the data-shaped example. Prices 4,000, 4,080, 4,050.

$$r_1 = \ln(4080/4000) = \ln(1.02) = +0.019803$$

$$r_2 = \ln(4050/4080) = \ln(0.992647) = -0.007380$$

Sum: $+0.012423$. Direct: $\ln(4050/4000) = \ln(1.0125) = +0.012423$. ✓
By percentages: $+2.000\% - 0.735\% = +1.265\%$, whereas the true change is $+1.250\%$. An error of 1.5 basis points over two days — and it accumulates.

Formal Name — the mathematics

The **log return** on day t is

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln P_t - \ln P_{t-1}.$$

Summing over T days telescopes:

$$\sum_{t=1}^{T} r_t = (\ln P_1 - \ln P_0) + (\ln P_2 - \ln P_1) + \cdots + (\ln P_T - \ln P_{T-1}) = \ln P_T - \ln P_0 = \ln\left(\frac{P_T}{P_0}\right),$$

every interior term cancelling with its neighbour. The relationship to the simple return R_t is $r_t = \ln(1 + R_t)$, and for small moves $r_t \approx R_t$: at $R = 0.02$ we get $r = 0.019803$, agreeing to three decimal places.

Where the Analogy Breaks

Log returns are additive across *time* but not across *assets*. The log return of a portfolio is not the weighted sum of its constituents' log returns, because a sum of prices does not decompose under a logarithm — there is no rule for $\ln(x+y)$, as §2.3 insisted. Portfolio work therefore often returns to simple returns, which *are* additive across assets and not across time. Each convention buys one form of additivity and forfeits the other.

Why It Matters Here — link to the thesis

Every model, metric and figure in this thesis consumes r_t. Prices are never used directly: they trend, they live on incomparable scales across the five BRICS markets, and they violate the stationarity that Chapter 8 requires. Converting to log returns fixes all three problems in one step.

2.5 Reading Tails on a Log Scale

Situation — the everyday picture

A histogram of daily returns is a spike at zero with two barely visible smears at the edges. The bar at zero is thousands of days tall; the bar at -8% holds three days. Drawn on ordinary axes, the three-day bar is invisible — less than one pixel — and the picture says only "returns are usually small", which everybody knew.

Problem — where it breaks or why it matters

The rare days are the ones that matter. A risk model is judged almost entirely on how it handles the far edges of the distribution, and those edges are exactly what a linear axis erases. We need a way of drawing the picture in which a count of 3 and a count of 3,000 are both legible.

Worked Arithmetic — concrete numbers

Suppose four bins hold counts 3,000, 300, 30, 3.

On a linear axis, with the tallest bar 10 cm high, the others are 1 cm, 1 mm and 0.1 mm. The last is thinner than the ink.

On a log axis, we plot $\log_{10}$ of each count:

$$\log_{10} 3000 = 3.477, \quad \log_{10} 300 = 2.477, \quad \log_{10} 30 = 1.477, \quad \log_{10} 3 = 0.477.$$

The heights are now $3.477, 2.477, 1.477, 0.477$ — equally spaced, one unit apart. Every bar is visible, and the constant gap of exactly 1 reveals that each bin holds one tenth of the previous, a tenfold decay that the linear plot concealed completely.

Formal Name — the mathematics

A **logarithmic axis** places a value v at position $\log v$. Equal distances then represent equal *ratios* rather than equal differences: the gap from 1 to 10 is the same length as from 10 to 100.
A distribution whose tail decays as a power law, $\mathbb{P}(|X| > x) \sim x^{-\alpha}$, becomes a straight line of slope $-\alpha$ when both axes are logarithmic, since

$$\ln \mathbb{P}(|X| > x) = -\alpha \ln x + \text{constant}.$$

Straightness on a log–log plot is the visual signature of a power law, and the slope reads off the tail index directly.

Where the Analogy Breaks

A log axis cannot display a count of zero, since $\ln 0$ is undefined — empty bins simply vanish rather than sitting on the axis. It also flatters the eye: differences that are genuinely enormous are compressed into a modest vertical gap, so a model that is off by a factor of ten can look close. The log axis makes the tail *visible*; it does not make disagreements there *small*.

Why It Matters Here — link to the thesis

The stylised-facts figure produced for every market in this project includes a log-scale distribution panel for exactly this reason, with values outside the plotted range clipped rather than dropped so that no observation silently disappears. The panel is where a generator that has matched the centre of the distribution but missed its tails is caught — and Chapter 9 shows that missing the tails is the characteristic failure of these models.

Chapter 3

Calculus I — Change

Before and After

Before this chapter you need: functions, domain and range (§1.3–§1.4), slope of a straight line (§1.5), quadratics and curvature (§1.6), the exponential and the logarithm (§2.1–§2.2). **After it you will understand:** what a derivative is and how to compute one, what a second derivative says about the shape of a curve, and what an integral accumulates. Chapter 4 lifts all of it to many variables, which is where machine learning actually lives.

Calculus is the mathematics of change. Everything in this thesis that *learns* does so by measuring how a number responds to a small nudge and then nudging in the direction that helps. This chapter builds that measurement for functions of one variable.

3.1 Limits, Informally

Situation — the everyday picture

A runner covers 100 metres in 10 seconds. Her average speed is 10 m/s, but she was not travelling at 10 m/s throughout — she started at zero. To find her speed at the exact instant $t = 4$ seconds, you might time her over the interval from 4 to 5 seconds. That is still an average. So you shrink the window: 4 to 4.1, then 4 to 4.01, then 4 to 4.001.

Problem — where it breaks or why it matters

You cannot shrink the window to nothing. Over an interval of exactly zero duration she covers exactly zero distance, and $0/0$ is not a number. Yet the sequence of averages over shrinking windows plainly settles down toward a particular value. We need language for "the value it settles toward" that does not require ever reaching a window of zero width.

Worked Arithmetic — concrete numbers

Take $f(x) = x^2$ and ask for the slope at $x = 3$.
Average slope from 3 to $3 + h$ is

$$\frac{f(3+h) - f(3)}{h} = \frac{(3+h)^2 - 9}{h} = \frac{9 + 6h + h^2 - 9}{h} = \frac{6h + h^2}{h} = 6 + h,$$

where the cancellation is legal for every $h \neq 0$.

h	Average slope $6 + h$
1	7
0.1	6.1
0.01	6.01
0.001	6.001
-0.001	5.999
-0.1	5.9

Approaching from either side, the values close in on 6. At $h = 0$ the expression $(6h + h^2)/h$ is $0/0$ and says nothing — but $6 + h$ at $h = 0$ is plainly 6. The limit is 6.

Formal Name — the mathematics

We write

$$\lim_{h \to 0} g(h) = L$$

to mean: the values $g(h)$ can be made as close to L as desired by taking h sufficiently close to (but not equal to) 0.

The limit describes the value g *approaches*; whether $g(0)$ exists, and what it equals if it does, is a separate question. In the example above $g(h) = (6h + h^2)/h$ is undefined at $h = 0$ yet has limit 6 there.

Where the Analogy Breaks

Not every function settles. Consider $g(h) = 1/h$ near zero: from the right it grows without bound, from the left it falls without bound, and there is no value it approaches. Others approach different values from each side and so have no limit either. The limit is a claim that must be checked, not a formality — and the same care resurfaces in Chapter 9, where the sample kurtosis of a heavy-tailed series has no limit at all, drifting upward without settling as n grows.

Why It Matters Here — link to the thesis

Every derivative in this thesis is a limit of this kind. The autograd machinery in PyTorch does not compute limits numerically — it applies the rules of §3.3, which were themselves derived from limits once and for all.

3.2 The Derivative as Instantaneous Slope

Situation — the everyday picture

Your car's speedometer reads 60 km/h. It is not telling you a distance divided by a duration over any interval you could name — it is reporting how fast the odometer is changing *right now*. The dashboard shows one number, and it is the limit of the previous section made physical.

Problem — where it breaks or why it matters

For a straight line, one slope describes the whole thing (§1.5). For anything that bends, the slope differs at every point, so "the slope" is not a number but a new *function* of position.

Building that function is the central construction of calculus.

Worked Arithmetic — concrete numbers

The two-point picture first. For $f(x) = x^2$, from $x = 1$ to $x = 4$: heights 1 and 16, change 15, run 3, ratio 5. That is the average slope over the interval — the slope of the straight line joining the two points, called a *secant*.
Shrink the interval toward $x = 1$:

From	To	Rise	Run	Slope
1	4	15	3	5
1	2	3	1	3
1	1.1	0.21	0.1	2.1
1	1.01	0.0201	0.01	2.01

Settling on 2. Repeating the algebra of §3.1 at a general point:

$$\frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h \xrightarrow[h\to 0]{} 2x.$$

So the derivative of x^2 is $2x$: at $x = 1$ it is 2, at $x = 3$ it is 6 (matching §3.1), at $x = 0$ it is 0 — the flat bottom of the bowl.

Sanity check against §1.6. There the average slopes over unit steps were $1, 3, 5, 7$. The derivative at the *midpoints* $0.5, 1.5, 2.5, 3.5$ is $2x = 1, 3, 5, 7$. Exactly. ✓

Formal Name — the mathematics

The **derivative** of f at x is

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h},$$

when the limit exists. Equivalent notations: $f'(x)$, $\dfrac{df}{dx}$, $\dfrac{d}{dx}f(x)$.
$f'(x)$ is the slope of the **tangent** — the straight line touching the curve at x and matching its direction there. A function with a derivative everywhere on an interval is **differentiable** there.

Where the Analogy Breaks

Not every function has a derivative everywhere. The absolute value $|x|$ has slope -1 to the left of zero and $+1$ to the right, and no single tangent at the corner: the limit from each side disagrees. This is not an exotic pathology. The ReLU activation used throughout deep learning is exactly this shape, and the gradient penalty of Chapter 13 involves $\big|\,\|\nabla D\| - 1\big|$, which has the same kink. Practical systems patch the corner by picking a convention; the mathematics genuinely has no answer there.

Why It Matters Here — link to the thesis

Training any model in this thesis means computing $\partial(\text{loss})/\partial(\text{weight})$ for every weight and stepping against it. The gradient-clipping threshold of 1.0 applied in all three generators is a bound on the size of exactly these derivatives.

3.3 The Rules of Differentiation

Situation — the everyday picture

Nobody computes limits by hand to differentiate. Once the limit has been evaluated for a handful of basic functions, and once a few rules for combining them are established, differentiation becomes bookkeeping — which is precisely why a computer can do it automatically.

Problem — where it breaks or why it matters

The functions we actually differentiate are built by adding, multiplying, dividing and composing simpler ones. Without combination rules, each new combination would require returning to the limit definition from scratch.

Worked Arithmetic — concrete numbers

Power rule: $\dfrac{d}{dx}x^n = nx^{n-1}$.
Check on x^3 at $x = 2$. The rule gives $3x^2 = 12$. Numerically, with $h = 0.001$:

$$\frac{(2.001)^3 - 2^3}{0.001} = \frac{8.012006001 - 8}{0.001} = 12.006,$$

converging to 12 as h shrinks. ✓

Sum rule: $(f+g)' = f' + g'$. For $f(x) = x^2 + x^3$, $f'(x) = 2x + 3x^2$. At $x = 2$: $4 + 12 = 16$.

Product rule: $(fg)' = f'g + fg'$.
Take $f = x^2$, $g = x^3$, so $fg = x^5$ and the power rule predicts $5x^4$. Via the product rule: $2x \cdot x^3 + x^2 \cdot 3x^2 = 2x^4 + 3x^4 = 5x^4$. ✓
Note that $(fg)' \neq f'g'$: here $f'g' = 2x \cdot 3x^2 = 6x^3$, which is wrong.

Chain rule: $\big(f(g(x))\big)' = f'(g(x)) \cdot g'(x)$.
Take $f(u) = u^2$ and $g(x) = 3x + 1$, so $f(g(x)) = (3x+1)^2$. The rule gives $2(3x+1) \cdot 3 = 6(3x+1) = 18x + 6$. Expanding first: $(3x+1)^2 = 9x^2 + 6x + 1$, whose derivative is $18x + 6$. ✓

Exponential and logarithm:

$$\frac{d}{dx}e^x = e^x, \qquad \frac{d}{dx}\ln x = \frac{1}{x} \quad (x > 0).$$

The first is the property promised in §2.1: e^x is its own derivative, the unique function (up to scaling) that is its own rate of change.

Formal Name — the mathematics

For differentiable f, g and constant c:

$$\begin{aligned} (cf)' &= cf' & (f \pm g)' &= f' \pm g' \\ (fg)' &= f'g + fg' & \left(\frac{f}{g}\right)' &= \frac{f'g - fg'}{g^2} \quad (g \neq 0) \\ (f \circ g)'(x) &= f'(g(x))\, g'(x) & \frac{d}{dx}x^n &= nx^{n-1} \end{aligned}$$

The chain rule is the one that matters most: it is the mathematical content of backpropagation.

Where the Analogy Breaks

The rules make differentiation look purely mechanical, and symbolically it is. But mechanical is not the same as numerically harmless. Applying the chain rule through fifty composed layers multiplies fifty derivatives together, and if each is around 0.5 the product is about 10^{-15} — the vanishing-gradient problem. The algebra is exact; the arithmetic that implements it is not.

Why It Matters Here — link to the thesis

The chain rule applied to a deep composition is backpropagation, and it is why Chapter 4's multivariable version is the load-bearing piece of this whole book. The vanishing-gradient issue above is the direct reason this project replaced sigmoid activations with tanh in the TimeGAN latent path: $\tanh'(0) = 1$ against $\sigma'(0) = 0.25$, a fourfold stronger signal per layer.

3.4 Second Derivatives and Curvature

Situation — the everyday picture

You are in a car. The speedometer says how fast you are going; the push you feel in your back says how fast *that* is changing. Speed is the first derivative of position, acceleration the second. They answer different questions, and you can have one without the other: at constant 100 km/h the speed is large and the acceleration is zero.

Problem — where it breaks or why it matters

The first derivative alone cannot tell a valley from a peak. At the bottom of a bowl and at the top of a hill the slope is zero in both cases. Distinguishing them requires knowing which way the curve bends.

Worked Arithmetic — concrete numbers

Take $f(x) = x^2$, so $f'(x) = 2x$ and $f''(x) = 2$.
At $x = 0$: $f'(0) = 0$, so it is a turning point. $f''(0) = 2 > 0$, meaning the slope is increasing as we pass through — from negative, to zero, to positive. That is a valley.
Check directly: $f(-1) = 1$, $f(0) = 0$, $f(1) = 1$. Lower in the middle. ✓

Now $g(x) = -x^2$, so $g'(x) = -2x$ and $g''(x) = -2$.
At $x = 0$: $g'(0) = 0$ again, but $g''(0) = -2 < 0$. The slope is decreasing through the point — positive, zero, negative. A peak.
Check: $g(-1) = -1$, $g(0) = 0$, $g(1) = -1$. Higher in the middle. ✓

The ambiguous case. $h(x) = x^3$ has $h'(0) = 0$ and $h''(0) = 0$. It is neither peak nor valley: $h(-1) = -1$ and $h(1) = 1$, so the function passes straight through while momentarily flattening. When the second derivative vanishes, the test is silent.

Formal Name — the mathematics

The **second derivative** $f''(x)$ is the derivative of f'. It measures **concavity**:

- $f''(x) > 0$: convex here (curving upward, holds water);
- $f''(x) < 0$: concave here (curving downward, sheds water);
- $f''(x) = 0$: no information from this test.

Second-derivative test. If $f'(a) = 0$ and $f''(a) > 0$, then a is a local minimum; if $f'(a) = 0$ and $f''(a) < 0$, a local maximum.

Where the Analogy Breaks

"Local" is doing heavy lifting. The test inspects an infinitesimal neighbourhood and certifies nothing beyond it: a function may have a local minimum at a and still take far lower values somewhere else entirely. Every optimisation result in this thesis is local in exactly this sense — nothing in the training procedure certifies a global optimum, and nothing in the reported numbers should be read as claiming one.

Why It Matters Here — link to the thesis

Convexity is what makes an optimisation problem easy, and its absence is what makes GAN training hard. A convex loss has one basin and any downhill walk finds it. A GAN's objective is not convex, and moreover the two networks are optimising against each other, so the surface each is descending shifts as the other learns. This is the structural reason for the instability Chapter 13 documents, and for the mode collapse the post-generation guard in this project is built to detect.

3.5 Integrals as Accumulated Area

Situation — the everyday picture

A tap fills a bucket. The flow rate varies — fast at first, then slowing. You know the rate at every instant and want the total volume after ten minutes. If the rate were constant you would multiply rate by time. It is not, so you cannot.

Problem — where it breaks or why it matters

Differentiation goes from total to rate. The reverse trip — rate to total — is needed just as often, and it is not simply the same operation run backwards in any obvious way. It requires its own construction.

Worked Arithmetic — concrete numbers

Constant rate first. A tap running at 2 litres/minute for 3 minutes delivers $2 \times 3 = 6$ litres. On a graph of rate against time this is a rectangle of height 2 and width 3 — area 6. Total is area under the rate curve.

Varying rate. Let the rate be $f(t) = 2t$ litres/minute, from $t = 0$ to $t = 3$. Approximate with three rectangles of width 1, using the rate at the right edge of each:

$$f(1) \cdot 1 + f(2) \cdot 1 + f(3) \cdot 1 = 2 + 4 + 6 = 12.$$

Too big — the rate is rising, so the right edge overstates each strip. Using left edges:

$$f(0) + f(1) + f(2) = 0 + 2 + 4 = 6,$$

too small. The truth is between 6 and 12.

Refine to six strips of width 0.5, right edges:

$$0.5(f(0.5) + f(1) + \cdots + f(3)) = 0.5(1 + 2 + 3 + 4 + 5 + 6) = 0.5 \times 21 = 10.5.$$

Left edges give $0.5(0+1+2+3+4+5) = 7.5$. The bracket has narrowed from $[6, 12]$ to $[7.5, 10.5]$.

Exactly. The region under $f(t) = 2t$ from 0 to 3 is a triangle of base 3 and height 6, so its area is $\frac{1}{2} \times 3 \times 6 = 9$ litres — and 9 sits inside both brackets. ✓

Formal Name — the mathematics

The **definite integral** of f from a to b,

$$\int_a^b f(x)\,dx,$$

is the limit of such rectangle sums as the widths shrink to zero. It equals the signed area between the graph and the horizontal axis — signed because regions below the axis count negatively.

The elongated S is a stretched "sum", and dx records the vanishing width of each strip.

Where the Analogy Breaks

Area is the right picture in one dimension and becomes a misleading one quickly. In the probability chapters an integral over a two-dimensional region is a volume, and the integrals defining Wasserstein distance are over spaces of *joint distributions* — objects with no spatial interpretation at all. What survives from the picture is the accounting: an integral totals up infinitely many infinitesimal contributions. That much always holds.

Why It Matters Here — link to the thesis

A probability density is a rate — probability per unit of return — so probabilities are integrals of it. The statement that a 5σ day has probability 5.73×10^{-7} under a Gaussian is the value of an integral over the two tails, and that single number is what yields the once-per-6,922-years figure that motivates this entire thesis.

3.6 The Fundamental Theorem

Situation — the everyday picture

Your car has an odometer and a speedometer. The speedometer reading is the rate of change of the odometer. Equally, the odometer reading is the accumulation of the speedometer over the journey. Two instruments, one relationship, read in opposite directions.

Problem — where it breaks or why it matters

Computing integrals by shrinking rectangles is exhausting and, for most functions, impossible by hand. If accumulation really is the inverse of differentiation, then every derivative already known becomes an integral known for free — and §3.3 supplied a whole table of them.

Worked Arithmetic — concrete numbers

Redo the tap by reversing differentiation. We need a function whose derivative is $2t$. From the power rule, $\frac{d}{dt}t^2 = 2t$, so $F(t) = t^2$ works.

Then

$$\int_0^3 2t\,dt = F(3) - F(0) = 9 - 0 = 9,$$

matching the triangle exactly, with no rectangles at all. ✓

Second example. $\int_1^4 2x\,dx = F(4) - F(1) = 16 - 1 = 15.$
Geometric check: the region is a trapezium with parallel sides $f(1) = 2$ and $f(4) = 8$, width 3. Area $= \frac{1}{2}(2+8) \times 3 = 15$. ✓

Third. $\int_1^e \frac{1}{x}\,dx = \ln(e) - \ln(1) = 1 - 0 = 1$, since $\frac{d}{dx} \ln x = 1/x$. The area under $1/x$ from 1 to e is exactly 1 — which is one way of saying what the number e is.

Formal Name — the mathematics

Fundamental Theorem of Calculus. If $F' = f$ on $[a, b]$ and f is continuous, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Conversely, $\frac{d}{dx} \int_a^x f(t)\,dt = f(x)$.
F is an **antiderivative** of f. Antiderivatives are not unique — any constant may be added, since constants differentiate to zero — but the constant cancels in $F(b) - F(a)$, so the definite integral is unaffected.

Where the Analogy Breaks

The theorem guarantees an antiderivative exists for any continuous f; it does not guarantee you can write one down. The Gaussian density $e^{-x^2/2}$ has no antiderivative expressible in elementary functions, which is exactly why the normal CDF Φ must be tabulated or computed numerically. The 5.73×10^{-7} quoted above is a numerical result, not a closed-form one.

Why It Matters Here — link to the thesis

The theorem links densities to probabilities throughout this thesis: the CDF is the integral of the density, and the density is the derivative of the CDF. Every quantile — and Value at Risk is precisely a quantile — is found by inverting that relationship.

Chapter 4

Calculus II — Many Variables

Before and After

Before this chapter you need: derivatives and the rules for computing them (§3.2–§3.3), second derivatives and concavity (§3.4), composition of functions (§1.7).
After it you will understand: what a partial derivative is, why the gradient points uphill, how the chain rule propagates through a deep composition — which is backpropagation — and what $\|\nabla D\|_2 = 1$ is asking for in the WGAN gradient penalty.

This is the load-bearing chapter of the book. Gradient descent, backpropagation and the gradient penalty are all direct applications of what follows. Everything here is the one-variable calculus of Chapter 3 applied one variable at a time, and then assembled.

A note on ordering: this chapter needs to collect several numbers into a single object and occasionally measure its length. It does so using only school geometry — Pythagoras — and defers the general theory of such objects to Chapter 5, which gives them their proper treatment.

4.1 Functions of Several Variables

Situation — the everyday picture

The temperature in a room depends on where you stand. Move toward the radiator and it rises; move toward the window and it falls. To state the temperature you must give two numbers — how far along, how far across — and you get back one number.

Problem — where it breaks or why it matters

A curve drawn on a page cannot represent this. Two inputs and one output need three dimensions, and the objects we actually optimise in this thesis have hundreds of thousands of inputs. Any intuition tied to pictures will fail; what survives is the arithmetic.

Worked Arithmetic — concrete numbers

Take $f(x, y) = x^2 + y^2$ — squared distance from the origin. Evaluate on a grid:

	$y=0$	$y=1$	$y=2$	$y=3$
$x=0$	0	1	4	9
$x=1$	1	2	5	10
$x=2$	4	5	8	13
$x=3$	9	10	13	18

Read across row $x = 2$: values $4, 5, 8, 13$, rising as y grows. Read down column $y = 1$: values $1, 2, 5, 10$, rising as x grows. The single lowest entry is 0 at the corner $(0, 0)$ — the bottom of a bowl, now in two input directions rather than one.

Notice the entries equal to 5: at $(1, 2)$ and at $(2, 1)$. Different points, same height. Joining all points of equal height gives a **contour**, exactly as on a map. Here the contours are circles about the origin.

Formal Name — the mathematics

A function of several variables assigns one output number to each combination of inputs, written $f : \mathbb{R}^n \to \mathbb{R}$. The notation $\mathbb{R}^n$ means ordered lists of n real numbers; $\mathbb{R}^2$ is pairs (x, y), $\mathbb{R}^3$ is triples, and so on.
A **level set** or **contour** is the collection of inputs sharing one output value: $\{(x, y) : f(x, y) = c\}$.

Where the Analogy Breaks

Two inputs can be pictured as a landscape; three already cannot, and the loss function of the TimeGAN generator in this project has 11,400 inputs — one per weight. Nothing visual survives. What does survive is that each input can be varied on its own while the rest are held fixed, and that is the entire content of the next section.

Why It Matters Here — link to the thesis

A neural network's loss is exactly such a function: inputs are the weights, output is a single number measuring how badly the network is doing. Training searches that space for a low point.

4.2 Partial Derivatives

Situation — the everyday picture

You are on a hillside at a crossroads. One road runs due east, another due north. Standing still, you can ask two separate questions: if I step east, do I go up or down, and how steeply? And the same for north. Two roads, two answers, one place.

Problem — where it breaks or why it matters

"The slope" of a surface is not a single number, because the answer depends on which way you face. But if we agree to ask only about the coordinate directions — one at a time, everything else frozen — each question reduces to the one-variable derivative already available.

Worked Arithmetic — concrete numbers

Take $f(x, y) = x^2 + y^2$ at the point $(3, 4)$, where $f = 9 + 16 = 25$.

Step east (x changes, y frozen at 4). The function becomes $g(x) = x^2 + 16$, a one-variable function. Its derivative is $2x$, which at $x = 3$ is 6.
Check numerically: $f(3.001, 4) = 9.006001 + 16 = 25.006001$, so $(25.006001 - 25)/0.001 = 6.001$, tending to 6. ✓

Step north (y changes, x frozen at 3). Now $h(y) = 9 + y^2$, derivative $2y$, which at $y = 4$ is 8.
Check: $f(3, 4.001) = 9 + 16.008001 = 25.008001$, giving $(25.008001 - 25)/0.001 = 8.001 \to 8$. ✓

The recipe. To differentiate with respect to one variable, treat every other variable as a constant. For $f(x, y) = x^2y^3$:

$$\frac{\partial f}{\partial x} = 2xy^3 \quad (y^3 \text{ is just a number}), \qquad \frac{\partial f}{\partial y} = 3x^2y^2 \quad (x^2 \text{ is just a number}).$$

At $(2, 1)$: $\partial f/\partial x = 2 \cdot 2 \cdot 1 = 4$ and $\partial f/\partial y = 3 \cdot 4 \cdot 1 = 12$.

Formal Name — the mathematics

The **partial derivative** of f with respect to x_i is

$$\frac{\partial f}{\partial x_i} = \lim_{h\to 0} \frac{f(x_1, \ldots, x_i + h, \ldots, x_n) - f(x_1, \ldots, x_n)}{h},$$

the ordinary derivative in the i-th coordinate with all others held fixed. The curved ∂ distinguishes it from the d of one-variable calculus and is a reminder that other variables exist and are being held still.

Where the Analogy Breaks

Partial derivatives only ever look along the coordinate axes. A ridge running diagonally can have zero slope due east and zero due north while dropping steeply to the southwest — both partials vanish and the point is still not a minimum. The coordinate directions are a convenient basis, not a complete description, and §4.3 is what repairs the gap.

Why It Matters Here — link to the thesis

$\partial(\text{loss})/\partial w$ for a single weight w is what an optimiser consumes. PyTorch's `backward()` computes exactly this for every weight in the network in one sweep.

4.3 The Gradient

Situation — the everyday picture

Back at the crossroads. You know the eastward slope is 6 and the northward slope is 8. You are not obliged to walk along a road — you may set off in any direction. Which one climbs fastest, and how steep is it?

Problem — where it breaks or why it matters

The partials give two numbers; the question asks for a direction and a steepness. Something must convert the pair into that, and the conversion had better be the one that actually identifies the steepest route, since gradient descent depends on going exactly opposite to it.

Worked Arithmetic — concrete numbers

Collect the partials into an ordered pair, written in brackets:

$$\nabla f(3,4) = [\,6,\ 8\,].$$

Claim: this pair points in the steepest uphill direction, and its length is the steepest slope.

Its length, by Pythagoras:

$$\sqrt{6^2+8^2} = \sqrt{36+64} = \sqrt{100} = 10.$$

Check that no direction beats 10. Moving a small distance h in a direction that is a fraction a east and b north (with $a^2+b^2=1$) changes f by approximately $h(6a+8b)$. Try several:

Direction (a,b)	$6a+8b$
Due east $(1,0)$	6
Due north $(0,1)$	8
Diagonal $(0.707, 0.707)$	$4.24+5.66=9.90$
$(0.6, 0.8)$	$3.6+6.4=10.0$
$(0.5, 0.866)$	$3.0+6.93=9.93$

The winner is $(0.6, 0.8)$, giving exactly 10 — and $(0.6, 0.8)$ is $[6,8]$ divided by its length 10. The gradient direction wins, and the best achievable rate equals the gradient's length. ✓

Downhill is the exact reverse, $[-6,-8]$, dropping at rate 10. That is the direction gradient descent steps in.

Formal Name — the mathematics

The **gradient** of $f:\mathbb{R}^n\to\mathbb{R}$ is the ordered list of its partial derivatives:

$$\nabla f = \left[\frac{\partial f}{\partial x_1},\ \frac{\partial f}{\partial x_2},\ \ldots,\ \frac{\partial f}{\partial x_n}\right].$$

Its **length** (Chapter 5 calls this the Euclidean or 2-norm) is

$$\|\nabla f\|_2 = \sqrt{\left(\frac{\partial f}{\partial x_1}\right)^2 + \cdots + \left(\frac{\partial f}{\partial x_n}\right)^2}.$$

Two facts, both verified numerically above: ∇f points in the direction of steepest increase, and $\|\nabla f\|_2$ is the rate of increase in that direction. Where f has a local maximum or minimum in the interior of its domain, $\nabla f = \mathbf{0}$ — every partial vanishes at once.

Where the Analogy Breaks

Steepest *here* is not fastest *overall*. The gradient is purely local: it describes an infinitesimal neighbourhood and knows nothing of the terrain beyond. Repeatedly following it can zig-zag down a narrow valley, taking many small steps where one well-chosen direction would have done. It is greedy, and greedy is not optimal — which is why Chapter 10 (Optimisation)'s momentum and adaptive methods exist at all.

Why It Matters Here — link to the thesis

Two direct uses. First, gradient descent: every model here is trained by repeatedly stepping along $-\nabla(\text{loss})$. Second, and more specifically, the WGAN-GP gradient penalty is a statement about $\|\nabla D\|_2$ — it adds a cost whenever the length of the critic's gradient departs from 1. With that definition in hand, $\|\nabla D\|_2 = 1$ now reads concretely: whichever direction you perturb the input, the critic's output should change at a rate of exactly one unit per unit of perturbation. Chapter 13 explains why that is the right thing to ask for.

4.4 The Chain Rule and Backpropagation

Situation — the everyday picture

The factory line of §1.7, now with a quality inspector at the end who scores each finished item. A dial on station one is nudged. The painted part changes a little; the boxed part changes as a result; the score changes as a result of that. How much did the score move per unit of dial?

Problem — where it breaks or why it matters

The dial and the score are separated by every station in between. Measuring the effect by physically turning the dial and re-running the line is possible for one dial, but a network has hundreds of thousands, and re-running once per dial is hopelessly expensive. We need the whole answer from one pass.

Worked Arithmetic — concrete numbers

A chain of three. Let

$$u = g(x) = x + 3, \qquad v = f(u) = u^2, \qquad L = k(v) = 2v.$$

At $x = 1$: $u = 4$, $v = 16$, $L = 32$.
Local rates: $du/dx = 1$, $dv/du = 2u = 8$, $dL/dv = 2$.
Chain them by multiplying:

$$\frac{dL}{dx} = \frac{dL}{dv} \cdot \frac{dv}{du} \cdot \frac{du}{dx} = 2 \times 8 \times 1 = 16.$$

Verify by substitution. $L = 2(x + 3)^2$, so $dL/dx = 4(x + 3)$, which at $x = 1$ is 16. ✓
Verify numerically. At $x = 1.001$: $u = 4.001$, $v = 16.008001$, $L = 32.016002$. Then $(32.016002 - 32)/0.001 = 16.002 \to 16$. ✓

Two paths. Now let one quantity feed two others which recombine:

$$u = 2x, \qquad v = x^2, \qquad L = u + v.$$

At $x = 3$: $u = 6$, $v = 9$, $L = 15$.
$\partial L/\partial u = 1$ and $\partial L/\partial v = 1$; $du/dx = 2$ and $dv/dx = 2x = 6$. Influence arrives along both routes and the contributions *add*:

$$\frac{dL}{dx} = 1 \times 2 + 1 \times 6 = 8.$$

Check: $L = 2x + x^2$, so $dL/dx = 2 + 2x = 8$ at $x = 3$. ✓

The two operations are the whole algorithm: multiply along a path, add across paths.

Formal Name — the mathematics

If L depends on x through intermediate quantities $u_1, \ldots, u_m$, then

$$\frac{\partial L}{\partial x} = \sum_{j=1}^{m} \frac{\partial L}{\partial u_j} \frac{\partial u_j}{\partial x}.$$

Backpropagation evaluates this from the output backwards. A *forward pass* computes and stores every intermediate value; a *backward pass* starts with $\partial L/\partial L = 1$ and walks toward the inputs, multiplying by each local derivative and summing where paths meet.
Its efficiency is the point: one backward pass yields $\partial L/\partial w$ for *every* weight simultaneously, at roughly the cost of one forward pass — not one pass per weight.

Where the Analogy Breaks

The multiplication along a path is also the algorithm's weakness. Through n layers the result is a product of n local derivatives; if these are systematically below 1 the product collapses toward zero, and if above 1 it explodes. Neither is a flaw in the chain rule — the algebra is exact — but both are real obstacles in floating-point arithmetic, and both are why activation functions, initialisation schemes and gradient clipping receive so much attention.

Why It Matters Here — link to the thesis

This is how every model in this thesis is trained. It is also why the activation choice in TimeGAN mattered so much: with sigmoid in the latent path, the local derivative is at most 0.25, so three layers multiply to at most 0.0156 and the learning signal reaching the early layers is negligible. Switching to tanh, whose derivative reaches 1 at the origin, is what moved the autoencoder's R^2 from roughly zero to +0.995. The gradient-norm clip of 1.0 used everywhere in this project is a direct guard against the exploding case.

4.5 The Jacobian

Situation — the everyday picture

A machine takes two dials and drives three gauges. Turning either dial moves potentially all three gauges. To describe the machine's responsiveness you need one number for each dial–gauge pairing: two dials times three gauges is six numbers, naturally laid out in a grid.

Problem — where it breaks or why it matters

The gradient of §4.3 was defined for functions returning a single number. A neural network layer returns many numbers at once, so its responsiveness is not one list but a whole table of them — and the notation must keep track of which entry means what.

Worked Arithmetic — concrete numbers

Take $F(x, y) = (x + y,\ xy,\ x^2)$: two inputs, three outputs.
Write the outputs as $F_1 = x + y$, $F_2 = xy$, $F_3 = x^2$ and differentiate each with respect to each input:

$$\begin{aligned} \partial F_1/\partial x &= 1, & \partial F_1/\partial y &= 1, \\ \partial F_2/\partial x &= y, & \partial F_2/\partial y &= x, \\ \partial F_3/\partial x &= 2x, & \partial F_3/\partial y &= 0. \end{aligned}$$

Arrange as a grid, rows indexed by output and columns by input, evaluated at $(x, y) = (2, 3)$:

$$J = \begin{bmatrix} 1 & 1 \\ 3 & 2 \\ 4 & 0 \end{bmatrix}.$$

Reading it. Row 2, column 1 is 3: nudging x moves F_2 at rate 3. Check: $F_2 = xy$, and at $(2.001, 3)$ it is 6.003, up from 6, so the rate is 3. ✓
Row 3, column 2 is 0: y has no effect on $F_3 = x^2$ at all. ✓

Formal Name — the mathematics

For $F : \mathbb{R}^n \to \mathbb{R}^m$ with components $F_1, \ldots, F_m$, the **Jacobian** is the $m \times n$ grid

$$J = \begin{bmatrix} \partial F_1/\partial x_1 & \cdots & \partial F_1/\partial x_n \\ \vdots & \ddots & \vdots \\ \partial F_m/\partial x_1 & \cdots & \partial F_m/\partial x_n \end{bmatrix}, \qquad J_{ij} = \frac{\partial F_i}{\partial x_j}.$$

When $m = 1$ the Jacobian has a single row and is the gradient. The chain rule for composed multi-output functions is the product of their Jacobians, in the grid-multiplication sense defined in §5.2.

Where the Analogy Breaks

Jacobians are rarely built explicitly in practice. For a layer with 1,000 inputs and 1,000 outputs the grid holds a million entries, and a network has many such layers. Automatic differentiation instead computes the *product* of a Jacobian with a given list of numbers, which needs far less memory and is all backpropagation ever actually requires.

Why It Matters Here — link to the thesis

Every layer of every network here has a Jacobian, and backpropagation is the chained product of them. The critic's gradient in WGAN-GP is a one-row Jacobian — the critic outputs a single score — which is why ∇D is a gradient and its Euclidean length is the quantity the penalty constrains.

4.6 The Hessian

Situation — the everyday picture

A valley may be a narrow gorge or a broad shallow bowl. Both have a flat floor, so the gradient is zero in each. They demand completely different step sizes: the gorge punishes a long stride by throwing you up the opposite wall, while the bowl makes short strides a waste of time.

Problem — where it breaks or why it matters

The second-derivative test of §3.4 distinguished peaks from valleys with one number. With many inputs there is curvature in every direction and, worse, cross-terms describing how the slope in one direction changes as you move in another. One number cannot carry that.

Worked Arithmetic — concrete numbers

Take $f(x, y) = x^2 + 3y^2$. First partials:

$$\frac{\partial f}{\partial x} = 2x, \qquad \frac{\partial f}{\partial y} = 6y.$$

Both vanish only at $(0, 0)$, so that is the sole candidate turning point.
Now differentiate each partial with respect to each variable:

$$\frac{\partial^2 f}{\partial x^2} = 2, \quad \frac{\partial^2 f}{\partial y^2} = 6, \quad \frac{\partial^2 f}{\partial x \partial y} = 0, \quad \frac{\partial^2 f}{\partial y \partial x} = 0.$$

As a grid:

$$H = \begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix}.$$

Reading it. Both diagonal entries are positive, so the surface curves upward in both coordinate directions: $(0, 0)$ is a minimum. The y direction curves three times as sharply as x, so the bowl is an elongated ellipse, steep across and shallow along.
Verify. $f(1, 0) = 1$ while $f(0, 1) = 3$. The same unit step costs three times as much in y. ✓

A saddle. For $g(x, y) = x^2 - y^2$ the gradient $[2x, -2y]$ also vanishes only at the origin, and

$$H = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}.$$

One positive, one negative: a minimum along x, a maximum along y. Neither peak nor valley but a saddle, and $g(1, 0) = 1$ against $g(0, 1) = -1$ confirms it. Such points are everywhere in high dimensions — with many directions available, it is far more likely that some curve up and others down than that all agree.

Formal Name — the mathematics

The **Hessian** of $f : \mathbb{R}^n \to \mathbb{R}$ is the $n \times n$ grid of second partial derivatives,

$$H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

For twice-continuously-differentiable f the order of differentiation does not matter, so $H_{ij} = H_{ji}$ and the grid is symmetric.

At a point where $\nabla f = \mathbf{0}$: if every direction curves upward the point is a local minimum; if every direction curves downward, a local maximum; if directions disagree, a saddle. Chapter 5 makes "every direction" precise via eigenvalues (§5.6).

Where the Analogy Breaks

The Hessian is the right diagnostic and the wrong computation. For a network with 100,000 weights it holds 10^{10} entries — tens of gigabytes — and must be recomputed at every step. No training procedure in this thesis forms one. Optimisers instead approximate curvature cheaply, which is exactly what Adam's per-parameter scaling amounts to (Chapter 10 (Optimisation)).

Why It Matters Here — link to the thesis

Curvature explains why a single global learning rate struggles: in a valley shaped like the $[2, 0; 0, 6]$ example, the step that suits one direction is wrong for the other. Adam — the optimiser used for every model in this project — addresses precisely this by maintaining a separate effective step size per parameter.

Chapter 5

Linear Algebra

Before and After

Before this chapter you need: functions and composition (§1.3, §1.7), linear functions and slope (§1.5), gradients and Jacobians (§4.3, §4.5).
After it you will understand: what a vector and a matrix are, why matrix multiplication is defined the way it is, what a norm measures, and what $\|\nabla D\|_2 = 1$ demands of the critic — the last piece needed before the gradient penalty of Chapter 13 can be read as a sentence rather than a formula.

Chapter 4 repeatedly collected numbers into lists and grids and promised a proper account later. This is that account. Neural networks are, at bottom, alternating stacks of matrix multiplications and simple non-linear functions, so the vocabulary here is the vocabulary in which every model in this thesis is written.

5.1 Scalars, Vectors and Matrices

Situation — the everyday picture

A shop tracks three products. Monday's sales are 4, 7 and 2 units. You could write three separate numbers, or you could write one row, $[4, 7, 2]$, and treat Monday as a single object. Add Tuesday, Wednesday and Thursday and you have a table — one row per day, one column per product.

Problem — where it breaks or why it matters

Handling each number separately means writing a separate line of algebra for each, and a network layer would need a thousand such lines. Treating the collection as one object lets a single symbol stand for the lot and a single operation act on all of it at once.

Worked Arithmetic — concrete numbers

Vectors. Monday $\mathbf{u} = [4, 7, 2]$, Tuesday $\mathbf{v} = [1, 0, 5]$.
Addition is entry by entry:

$$\mathbf{u} + \mathbf{v} = [4 + 1,\ 7 + 0,\ 2 + 5] = [5, 7, 7].$$

Total units sold across the two days, per product. The interpretation survives the operation, which is the sign that the operation is the right one.

Scaling multiplies every entry:

$$3\mathbf{u} = [12, 21, 6],$$

what three identical Mondays would produce.

The dot product. Suppose prices are $\mathbf{p} = [10, 2, 5]$ per unit. Monday's revenue is

$$\mathbf{u} \cdot \mathbf{p} = 4 \times 10 + 7 \times 2 + 2 \times 5 = 40 + 14 + 10 = 64.$$

Multiply matching entries, add the results. One number out of two lists.

Matrices. Stack the two days:

$$A = \begin{bmatrix} 4 & 7 & 2 \\ 1 & 0 & 5 \end{bmatrix},$$

a 2×3 matrix — 2 rows, 3 columns, always in that order. The entry $A_{23} = 5$ sits in row 2, column 3: Tuesday's sales of product 3.

Formal Name — the mathematics

A **scalar** is a single number. A **vector** is an ordered list of numbers; the number of entries is its **dimension**. A **matrix** is a rectangular array, of **size** $m \times n$ for m rows and n columns. For vectors of equal dimension, addition and scalar multiplication act entrywise. The **dot product** of $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i,$$

a scalar. Entry A_{ij} of a matrix is in row i, column j.

Where the Analogy Breaks

The spreadsheet picture makes a matrix look like passive storage. Most matrices in this thesis are not data at all — they are *actions*. A layer's weight matrix is a thing that transforms its input, and the next section defines multiplication precisely so that this reading is the correct one.

Why It Matters Here — link to the thesis

A window of 128 daily returns is a vector in $\mathbb{R}^{128}$; a batch of 64 such windows is a 64×128 matrix. The noise fed to the QuantGAN generator is a vector in $\mathbb{R}^{100}$. These are the literal objects the code moves around.

5.2 Matrix Multiplication as Composition

Situation — the everyday picture

Two currency conversions in sequence: pounds to euros, then euros to dollars. Each is a multiplication. Doing both is a single conversion from pounds to dollars, and its rate is the product of the two rates. The composite is itself a conversion of the same kind.

Problem — where it breaks or why it matters

Layers of a network apply in sequence, exactly like the conversions. If each layer is a matrix, the composition of two layers should again be a matrix — otherwise composing them would produce some new kind of object and the theory would not close. Matrix multiplication is *defined* so that it does close.

Worked Arithmetic — concrete numbers

Matrix times vector first. Let

$$A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

Each output entry is the dot product of a row of A with $\mathbf{x}$:

$$A\mathbf{x} = \begin{bmatrix} 2 \times 4 + 0 \times 5 \\ 1 \times 4 + 3 \times 5 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \end{bmatrix}.$$

Now compose. Let

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}.$$

Apply A then B to $\mathbf{x}$:

$$B(A\mathbf{x}) = B \begin{bmatrix} 8 \\ 19 \end{bmatrix} = \begin{bmatrix} 1 \times 8 + 1 \times 19 \\ 0 \times 8 + 2 \times 19 \end{bmatrix} = \begin{bmatrix} 27 \\ 38 \end{bmatrix}.$$

The composite matrix. BA has entry (i, j) equal to the dot product of row i of B with column j of A:

$$BA = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 \cdot 2 + 1 \cdot 1 & 1 \cdot 0 + 1 \cdot 3 \\ 0 \cdot 2 + 2 \cdot 1 & 0 \cdot 0 + 2 \cdot 3 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 2 & 6 \end{bmatrix}.$$

Test it on $\mathbf{x}$:

$$(BA)\mathbf{x} = \begin{bmatrix} 3 \times 4 + 3 \times 5 \\ 2 \times 4 + 6 \times 5 \end{bmatrix} = \begin{bmatrix} 27 \\ 38 \end{bmatrix}. \quad \checkmark$$

Same answer. One matrix now does the work of two.

Order matters. Reversing gives

$$AB = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 1 & 7 \end{bmatrix} \neq BA.$$

Just as painting-then-boxing differed from boxing-then-painting in §1.7.

Formal Name — the mathematics

For A of size $m \times k$ and B of size $k \times n$, the product AB has size $m \times n$ with

$$(AB)_{ij} = \sum_{l=1}^{k} A_{il} B_{lj}.$$

The inner dimensions must match: an $m \times k$ can only multiply a $k \times n$.

Matrix multiplication is **associative**, $(AB)C = A(BC)$, and **distributive** over addition, but **not commutative**: $AB \neq BA$ in general.

Where the Analogy Breaks

Composition of matrices captures composition of *linear* maps only. A network built purely from matrices would collapse: stack fifty of them and the product is a single matrix, so fifty layers would have exactly the expressive power of one. The non-linear activation between layers is what prevents the collapse, and it is the entire reason deep networks are deeper than shallow ones.

Why It Matters Here — link to the thesis

A dense layer computes $W\mathbf{x} + \mathbf{b}$ — a matrix multiplication and a shift. The GRU cells in TimeGAN and the convolutions in QuantGAN and FinGAN are structured matrix multiplications. The parameter counts reported for this project's models (9,744 for the TimeGAN embedder, 11,400 for its generator) are counts of entries in these matrices.

5.3 Transpose, Identity and Inverse

Situation — the everyday picture

Undoing things. Multiplying by 5 is undone by dividing by 5, and doing both leaves the number untouched. Is there a matrix that undoes another, and a matrix that does nothing at all?

Problem — where it breaks or why it matters

Backpropagation runs a network backwards, and solving a linear system means undoing a transformation. Both need an account of reversal — including an account of when reversal is impossible, which turns out to be the more informative case.

Worked Arithmetic — concrete numbers

Transpose: flip rows and columns.

$$A = \begin{bmatrix} 4 & 7 & 2 \\ 1 & 0 & 5 \end{bmatrix}, \qquad A^\top = \begin{bmatrix} 4 & 1 \\ 7 & 0 \\ 2 & 5 \end{bmatrix}.$$

A 2×3 becomes 3×2; entry (i, j) moves to (j, i).

Identity: ones on the diagonal, zeros elsewhere.

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad I \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \cdot 4 + 0 \cdot 5 \\ 0 \cdot 4 + 1 \cdot 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}. \quad \checkmark$$

Inverse. For $A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$, the inverse is

$$A^{-1} = \begin{bmatrix} 0.5 & 0 \\ -1/6 & 1/3 \end{bmatrix}.$$

Check:

$$A^{-1}A = \begin{bmatrix} 0.5 \cdot 2 + 0 \cdot 1 & 0.5 \cdot 0 + 0 \cdot 3 \\ -\frac{1}{6} \cdot 2 + \frac{1}{3} \cdot 1 & -\frac{1}{6} \cdot 0 + \frac{1}{3} \cdot 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{3} + \frac{1}{3} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I. \quad \checkmark$$

When it fails. Take $C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, whose second row is twice the first. Then

$$C \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{and} \quad C \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Two different inputs, identical output. No rule could recover which one you started from, so no inverse exists. Information was destroyed — the same failure mode as multiplying an equation by zero in §1.2.

Formal Name — the mathematics

The **transpose** $A^\top$ satisfies $(A^\top)_{ij} = A_{ji}$, and $(AB)^\top = B^\top A^\top$ — the order reverses.
The **identity** I_n has $I_{ii} = 1$ and $I_{ij} = 0$ for $i \neq j$; it satisfies $AI = IA = A$.
A square A is **invertible** if some A^{-1} has $AA^{-1} = A^{-1}A = I$. Such an A^{-1} is unique when it exists. A matrix with no inverse is **singular**, and A is singular exactly when some non-zero vector satisfies $A\mathbf{x} = \mathbf{0}$.

Where the Analogy Breaks

Invertibility is a yes/no question in theory and a matter of degree in practice. A matrix can be technically invertible while being so close to singular that computing A^{-1} amplifies rounding error catastrophically. Numerical work therefore speaks of *conditioning* — how nearly singular a matrix is — rather than treating invertibility as a clean binary.

Why It Matters Here — link to the thesis

The transpose appears throughout backpropagation: if a layer's forward pass multiplies by W, the corresponding backward pass multiplies by $W^\top$. Singularity is also the right way to understand mode collapse — a generator that maps many different noise inputs to the same output has destroyed information in exactly the way C above does, and no amount of downstream processing can recover the diversity that was lost.

5.4 The Determinant

Situation — the everyday picture

Take the unit square — corners at $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$, area 1 — and apply a matrix to each corner. You get a parallelogram. Its area tells you the factor by which this transformation inflates or shrinks regions.

Problem — where it breaks or why it matters

We need a single number answering "does this transformation destroy information?". Squashing a two-dimensional region flat onto a line reduces its area to zero, and that is exactly the situation in which the map cannot be undone.

Worked Arithmetic — concrete numbers

A stretch. $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ sends $(1,0) \mapsto (2,0)$ and $(0,1) \mapsto (0,3)$. The unit square becomes a 2×3 rectangle of area 6.
Formula for 2×2: $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$. Here $2 \times 3 - 0 \times 0 = 6$. ✓

A shear. $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ sends $(1,0) \mapsto (1,0)$ and $(0,1) \mapsto (1,1)$. The square becomes a parallelogram with base 1 and height 1 — area still 1. And $\det B = 1 \times 1 - 1 \times 0 = 1$. ✓A shear slides without changing area.

A collapse. $C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ from §5.3 sends $(1,0) \mapsto (1,2)$ and $(0,1) \mapsto (2,4)$. Both images lie on the same line through the origin — $(2,4)$ is exactly twice $(1,2)$. The square is flattened onto a line, area zero. And $\det C = 1 \times 4 - 2 \times 2 = 0$. ✓
The determinant being zero and the inverse failing to exist are the same fact.

Formal Name — the mathematics

The **determinant** $\det A$ of a square matrix is the signed factor by which it scales volume. For 2×2,

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

Key properties:

- $\det A = 0$ if and only if A is singular;
- $\det(AB) = \det A \cdot \det B$;
- a negative determinant means orientation is flipped, like a mirror.

Where the Analogy Breaks

The volume picture is exact but the computation does not scale. Evaluating a determinant by the schoolbook cofactor expansion costs on the order of $n!$ operations — for $n = 20$ that is 10^{18}, beyond reach. Practical algorithms use factorisations costing about n^3 instead. Never mistake a clean definition for a usable algorithm.

Why It Matters Here — link to the thesis

Determinants govern how probability densities transform under an invertible map, via the Jacobian determinant — the mechanism underpinning normalising flows, which Chapter 12 (Generative Models) contrasts with GANs. GANs deliberately avoid this machinery, which is what frees them from any invertibility requirement and also why they provide no likelihood.

5.5 Norms

Situation — the everyday picture

You are at a street corner and a friend is three blocks east and four blocks north. How far away is the friend? If you can fly, five blocks — straight line. If you must follow streets, seven — three plus four. Both are honest answers to "how far"; they differ because they measure different things.

Problem — where it breaks or why it matters

"How big is this vector?" has no single answer, and the gradient penalty in WGAN-GP asks precisely that question about ∇D. Before that constraint can be understood, the notion of size must be pinned down — including which one is meant.

Worked Arithmetic — concrete numbers

Take $\mathbf{v} = [3, 4]$.

Euclidean (2-norm) — straight line:

$$\|\mathbf{v}\|_2 = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

Manhattan (1-norm) — along the streets:

$$\|\mathbf{v}\|_1 = |3| + |4| = 7.$$

Maximum (∞-norm) — the largest single component:

$$\|\mathbf{v}\|_\infty = \max(|3|, |4|) = 4.$$

Three sizes for one vector: 5, 7, 4. Always state which.

A three-entry example. $\mathbf{w} = [1, -2, 2]$:

$$\|\mathbf{w}\|_2 = \sqrt{1 + 4 + 4} = \sqrt{9} = 3, \qquad \|\mathbf{w}\|_1 = 1 + 2 + 2 = 5, \qquad \|\mathbf{w}\|_\infty = 2.$$

Scaling. Doubling a vector doubles every norm: $\|2\mathbf{w}\|_2 = \sqrt{4 + 16 + 16} = \sqrt{36} = 6 = 2 \times 3$. ✓

Unit length. Dividing by the 2-norm gives length exactly 1: $\mathbf{v}/5 = [0.6, 0.8]$, and $\sqrt{0.36 + 0.64} = 1$. ✓This is the vector that won the steepest-direction contest in §4.3 — now recognisable as the gradient rescaled to unit length.

Formal Name — the mathematics

A **norm** $\|\cdot\|$ assigns a non-negative size to each vector, subject to three requirements:

1. $\|\mathbf{v}\| = 0$ only when $\mathbf{v} = \mathbf{0}$;
2. $\|c\mathbf{v}\| = |c|\,\|\mathbf{v}\|$ for any scalar c;
3. $\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$ (the triangle inequality: no detour is shorter than going direct).

The p-**norm** is

$$\|\mathbf{v}\|_p = \left(\sum_i |v_i|^p \right)^{1/p},$$

giving the Manhattan norm at $p = 1$, the Euclidean at $p = 2$, and the maximum norm in the limit $p \to \infty$. Unsubscripted $\|\cdot\|$ conventionally means the Euclidean norm.

Where the Analogy Breaks

The choice of norm is not cosmetic. In high dimensions the norms diverge sharply: for a vector of 10,000 entries each equal to 1, the 2-norm is 100, the 1-norm is 10,000, and the ∞-norm is 1. A constraint that is mild in one may be severe in another, so "the gradient is bounded" means nothing until the norm is named.

Why It Matters Here — link to the thesis

This is the last piece the gradient penalty needed. The WGAN-GP objective adds

$$\lambda\,\mathbb{E}\Big[\big(\|\nabla D(\hat{x})\|_2 - 1\big)^2\Big]$$

to the critic's loss, with $\lambda = 10$ throughout this project. Read plainly: compute the critic's gradient at a point, measure its Euclidean length, and penalise the square of its deviation from 1. Chapter 13 explains why 1 is the target — it enforces the Lipschitz condition that makes the critic's output a valid Wasserstein estimate. The gradient-norm clipping applied to every model here, `max_norm=1.0`, is a separate use of the same 2-norm: rescale the whole gradient whenever its Euclidean length exceeds 1.

5.6 Eigenvalues and Eigenvectors

Situation — the everyday picture

Stretch a rubber sheet: twice as wide, three times as tall. Most drawn arrows come out pointing somewhere new. But an arrow lying exactly horizontal stays horizontal, merely twice as long; a vertical one stays vertical, three times as long. Those two directions are special to this particular stretch.

Problem — where it breaks or why it matters

A general matrix mixes every direction into every other, which makes repeated application hard to reason about. If we can find the directions the matrix merely scales, its behaviour becomes transparent: along those directions it is nothing but multiplication by a number.

Worked Arithmetic — concrete numbers

Take $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$.

$$A\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad A\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix} = 3\begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Directions unchanged, lengths scaled by 2 and 3. Eigenvalues 2 and 3.

A generic direction is not preserved:

$$A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix},$$

which is not a multiple of $[1,1]$ — the arrow has rotated.

A less obvious case. $B = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$.

Try $[1,0]$: $B[1,0]^\top = [3,0]^\top = 3[1,0]^\top$. Eigenvalue 3. ✓
Try $[1,-1]$: $B[1,-1]^\top = [3-1,\ -2]^\top = [2,-2]^\top = 2[1,-1]^\top$. Eigenvalue 2. ✓

Note in both examples the eigenvalues are the diagonal entries — true for triangular matrices, not in general.

Formal Name — the mathematics

A non-zero vector $\mathbf{v}$ is an **eigenvector** of a square matrix A with **eigenvalue** λ if

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The direction is preserved; only the length changes, by the factor λ (reversing if $\lambda < 0$). Eigenvalues solve $\det(A - \lambda I) = 0$. A real matrix need not have real eigenvalues — a rotation has none, since it preserves no direction — but a *symmetric* real matrix always does.

Where the Analogy Breaks

Eigenvectors give a complete picture only when there are enough of them to span the space, which is not guaranteed. Some matrices are *defective* and cannot be diagonalised at all. The convenient picture — "every matrix is just scaling along some special axes" — is a property of a well-behaved subclass, not a universal truth.

Why It Matters Here — link to the thesis

The Hessian of §4.6 is symmetric, so its eigenvalues are real, and they make "curves upward in every direction" precise: all eigenvalues positive means a minimum, all negative a maximum, mixed signs a saddle. Their ratio is the condition number, which measures how elongated the valley is and therefore how badly a single global learning rate will perform — the quantitative version of the gorge-versus-bowl problem that motivates Adam.

5.7 Vector Spaces and Linear Transformations

Situation — the everyday picture

Recipes. Combine half a batch of one and two batches of another and you get a sensible recipe. Combine sound waves and you get a sound; combine forces and you get a force. In each case adding two objects and scaling one produce objects of the same kind. The collection is closed under those two operations.

Problem — where it breaks or why it matters

The same theorems keep reappearing for lists of numbers, for functions, for random variables. Rather than reproving each time, we identify what these settings share — and then prove things once, for anything that shares it.

Worked Arithmetic — concrete numbers

Is the set of vectors $[x, y]$ with $y = 2x$ a vector space?
Take $[1, 2]$ and $[3, 6]$, both on the line. Their sum is $[4, 8]$, and $8 = 2 \times 4$. ✓Still on the line.
Scale $[1, 2]$ by -5: $[-5, -10]$, and $-10 = 2 \times (-5)$. ✓
Zero: $[0, 0]$ satisfies $0 = 2 \times 0$. ✓
Closed under both operations and containing zero — yes.

Is the set with $y = 2x + 1$?
Take $[0, 1]$ and $[1, 3]$, both on the line. Their sum is $[1, 4]$, but $2 \times 1 + 1 = 3 \neq 4$. $\times$ Not closed. Nor does it contain $[0, 0]$, since $2 \times 0 + 1 = 1 \neq 0$.
The offset $+1$ is fatal, and this is exactly why $f(x) = mx + c$ with $c \neq 0$ is called *affine* rather than linear (§1.5).

Is a matrix map linear? For $A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$, $\mathbf{u} = [1, 0]$, $\mathbf{v} = [0, 1]$:

$$A(\mathbf{u} + \mathbf{v}) = A[1, 1]^\top = [2, 4]^\top,$$

$$A\mathbf{u} + A\mathbf{v} = [2, 1]^\top + [0, 3]^\top = [2, 4]^\top. \quad \checkmark$$

Formal Name — the mathematics

A **vector space** is a set closed under addition and scalar multiplication, containing a zero element, with each element having an additive inverse, and satisfying the usual associative, commutative and distributive laws.
A map T between vector spaces is a **linear transformation** if

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \quad \text{and} \quad T(c\mathbf{v}) = cT(\mathbf{v}).$$

Every linear transformation between finite-dimensional spaces is a matrix multiplication, once bases are fixed — which is why matrices and linear maps are so often treated as the same thing.
A **basis** is a minimal set of vectors from which every element can be built by addition and scaling; the number of them is the **dimension**.

Where the Analogy Breaks

Almost nothing interesting in this thesis is linear. Activation functions are deliberately non-linear, GAN losses are non-convex, and the map from noise to a synthetic return series is wildly non-linear by design. Linear algebra is not a description of these systems — it is the local language in which each small piece is written, and the thing calculus reduces them to one step at a time.

Why It Matters Here — link to the thesis

Every layer is a linear map followed by a non-linearity, and this alternation is what gives a deep network its expressive power: without the non-linearity the whole stack would collapse to a single matrix (§5.2). The set of returns a generator can produce is not a vector space — it is a curved subset of $\mathbb{R}^{128}$ — and Chapter 12 (Generative Models) takes up what that means for the manifold a generative model is trying to learn.

Chapter 6

Probability

Before and After

Before this chapter you need: fractions and percentages (§1.1), functions and their range (§1.3–§1.4), the exponential and logarithm (Chapter 2), integrals as accumulated area (§3.5), vectors and the dot product (§5.1).
After it you will understand: what a random variable is, how densities and probabilities relate, what expectation and variance actually average, why independence is the assumption market returns violate, and — the point the whole thesis turns on — why the Central Limit Theorem does not rescue you when the fourth moment does not exist.

Probability is the mathematics of uncertainty. This chapter builds it from counting outcomes up to the two limit theorems, and then shows precisely where those theorems stop applying to financial returns.

6.1 Sets and Events

Situation — the everyday picture

Roll one die. Six things can happen. You might care about "a six", or about "an even number", or about "more than four". Each of these is a collection of outcomes: $\{6\}$, $\{2, 4, 6\}$, $\{5, 6\}$. The question you ask picks out a collection.

Problem — where it breaks or why it matters

Before anything can be assigned a probability, we need to say clearly what the things being assigned probabilities *are*. "The chance it rains" is ambiguous until "it rains" is pinned to a definite collection of outcomes, and ambiguity here produces paradoxes later.

Worked Arithmetic — concrete numbers

Rolling one die, the complete list of outcomes is

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Let A = "even" = $\{2, 4, 6\}$ and B = "more than four" = $\{5, 6\}$.

Both happen (intersection): $A \cap B = \{6\}$ — one outcome.
Either happens (union): $A \cup B = \{2, 4, 5, 6\}$ — four outcomes.

A **does not happen** (complement): $A^c = \{1, 3, 5\}$ — three outcomes.

Counting check. $|A| = 3$, $|B| = 2$, $|A \cap B| = 1$, and

$$|A \cup B| = |A| + |B| - |A \cap B| = 3 + 2 - 1 = 4. \quad \checkmark$$

Subtracting the intersection is necessary because the outcome 6 lies in both sets and would otherwise be counted twice.

Formal Name — the mathematics

The **sample space** Ω is the set of all possible outcomes. An **event** is a subset of Ω. For events $A, B \subseteq \Omega$:

- $A \cap B$ (intersection): both occur;
- $A \cup B$ (union): at least one occurs;
- A^c (complement): A does not occur;
- $A \cap B = \emptyset$: the events are **disjoint** or **mutually exclusive** — they cannot both occur.

Where the Analogy Breaks

A die has six outcomes and every subset is a sensible event. For continuous quantities this breaks down: if Ω is the real line, some subsets are so pathological that no consistent probability can be assigned to them. Measure theory exists to exclude those, restricting attention to a well-behaved family of events. Nothing in this thesis meets such a set, but the restriction is why the formal definition below speaks of a chosen family rather than all subsets.

Why It Matters Here — link to the thesis

"A big move today" and "a big move yesterday" are the two events whose relationship measures volatility clustering. Making them precise — a return exceeding two standard deviations in absolute value — is what turns an intuition into something a generator can be scored against.

6.2 The Probability Axioms

Situation — the everyday picture

You are asked to bet on a horse race. Someone quotes you odds implying a 60% chance for one horse and a 70% chance for another, and there are five horses. You do not need to know anything about horses to know these numbers are broken: they already sum to more than certainty.

Problem — where it breaks or why it matters

Assigning numbers to events is only useful if the assignment is coherent. Three rules are enough to rule out every incoherent assignment, and everything else in probability follows from them.

Worked Arithmetic — concrete numbers

Fair die, each outcome probability 1/6.

Non-negative: every outcome has probability $1/6 > 0$. ✓
Total is one: $6 \times \frac{1}{6} = 1$. ✓
Disjoint events add: $A = \{2, 4, 6\}$ and $C = \{1, 3\}$ share nothing, so

$$\mathbb{P}(A \cup C) = \mathbb{P}(\{1, 2, 3, 4, 6\}) = \tfrac{5}{6},$$

and $\mathbb{P}(A) + \mathbb{P}(C) = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$. ✓
Overlapping events do not simply add. With $B = \{5, 6\}$:

$$\mathbb{P}(A) + \mathbb{P}(B) = \tfrac{3}{6} + \tfrac{2}{6} = \tfrac{5}{6},$$

but $A \cup B = \{2, 4, 5, 6\}$ has probability $\frac{4}{6}$. The excess $\frac{1}{6}$ is exactly $\mathbb{P}(A \cap B) = \mathbb{P}(\{6\})$, so

$$\mathbb{P}(A \cup B) = \tfrac{3}{6} + \tfrac{2}{6} - \tfrac{1}{6} = \tfrac{4}{6}. \quad ✓$$

Complement. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A) = 1 - \frac{3}{6} = \frac{3}{6}$, and $A^c = \{1, 3, 5\}$ indeed has three outcomes. ✓

Formal Name — the mathematics

A **probability measure** $\mathbb{P}$ assigns a number to each event subject to:

1. $\mathbb{P}(A) \geq 0$ for every event A;
2. $\mathbb{P}(\Omega) = 1$;
3. if $A_1, A_2, \ldots$ are pairwise disjoint, then $\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i)$.

These are the **Kolmogorov axioms**. Immediate consequences: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(A) \leq 1$, and the **inclusion–exclusion** rule

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Where the Analogy Breaks

The axioms say what a coherent assignment looks like. They say nothing about which assignment is *correct* for any real situation — that is a modelling question, answered by data or judgement, not by mathematics. A Gaussian model of returns satisfies every axiom perfectly and is still badly wrong about markets, as §6.12 shows.

Why It Matters Here — link to the thesis

The axioms are what make the shuffled-real control a valid baseline. Shuffling permutes outcomes without changing which values occur, so every probability depending only on the collection of values is untouched — which is precisely why the control scores a Wasserstein distance of exactly 0.000e+00 against real data while failing every ordering-sensitive test.

6.3 Conditional Probability

Situation — the everyday picture

A city has a 60% chance of rain on any given day. But if it rained yesterday, the chance today is 80%. Knowing yesterday's outcome changes the probability you assign to today's.

Problem — where it breaks or why it matters

Unconditional probabilities describe the long run and are silent about the present. A risk model that knows only that big moves happen 5% of the time, with no way to use the fact that yesterday was violent, is discarding most of what the data contains.

Worked Arithmetic — concrete numbers

Let A = "big market move today", B = "big market move yesterday".
MOEX, 20-year data: $\mathbb{P}(A) = 3.8\%$ while $\mathbb{P}(A \mid B) = 22.9\%$.

Ratio: $22.9/3.8 = 6.0$. Knowing yesterday was violent multiplies today's probability by roughly six.

Where the formula comes from. Conditioning on B restricts attention to the days when B happened, so B becomes the new sample space. Of those days, we ask what fraction also had A:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

On the die. $A = \{2, 4, 6\}$, $B = \{5, 6\}$:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(\{6\})}{\mathbb{P}(\{5, 6\})} = \frac{1/6}{2/6} = \frac{1}{2}.$$

Check by counting: told the roll exceeded four, it is 5 or 6, one of which is even. One in two. ✓

Note the asymmetry: $\mathbb{P}(B \mid A) = \frac{1/6}{3/6} = \frac{1}{3} \neq \frac{1}{2}$.

Formal Name — the mathematics

For $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Rearranging gives the **multiplication rule** $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\,\mathbb{P}(B)$.
Events A and B are **independent** when $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, equivalently $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Where the Analogy Breaks

Conditioning is not causation. MOEX's clustering after February 2022 is driven by a stream of news, not by the previous day's move causing the next. And the relationship is about *magnitude*, not direction: knowing yesterday was violent says a great deal about how large today's move will be and essentially nothing about its sign. That asymmetry is the whole content of volatility clustering.

Why It Matters Here — link to the thesis

The ratio $\mathbb{P}(A \mid B)/\mathbb{P}(A) = 6.0$ is a direct measurement of volatility clustering, and reproducing it is what separates a generator that has learned market dynamics from one that has only learned the histogram. The shuffled control destroys exactly this structure: after permutation $\mathbb{P}(A \mid B) \approx \mathbb{P}(A)$, the ratio collapses to 1, and the control fails every temporal metric while acing every fidelity one.

6.4 Bayes' Theorem

Situation — the everyday picture

A test for a rare disease is 99% accurate. You test positive. The intuitive reading — "99% chance I have it" — is wrong, and not slightly. If the disease affects one person in a thousand, the true figure is closer to 9%.

Problem — where it breaks or why it matters

People reliably confuse $\mathbb{P}(\text{positive} \mid \text{ill})$ with $\mathbb{P}(\text{ill} \mid \text{positive})$. The previous section already showed these differ; Bayes' theorem says exactly how, and the gap is largest precisely when the underlying event is rare — which is the regime every tail-risk question lives in.

Worked Arithmetic — concrete numbers

Take 100,000 people, one in a thousand ill, test 99% accurate in both directions.

	Ill	Healthy	Total
Test positive	99	999	1,098
Test negative	1	98,901	98,902
Total	100	99,900	100,000

Ill: 100 people, of whom 99% test positive → 99.
Healthy: 99,900 people, of whom 1% test positive → 999 false alarms.

Given a positive test:

$$\mathbb{P}(\text{ill} \mid +) = \frac{99}{99 + 999} = \frac{99}{1098} = 0.0902 = 9.0\%.$$

The false alarms outnumber the true positives ten to one, because there are a thousand times more healthy people to generate them. ✓

Via the formula:

$$\mathbb{P}(\text{ill} \mid +) = \frac{\mathbb{P}(+ \mid \text{ill})\mathbb{P}(\text{ill})}{\mathbb{P}(+)} = \frac{0.99 \times 0.001}{0.01098} = 0.0902. \quad \checkmark$$

Formal Name — the mathematics

Bayes' theorem. For events with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\,\mathbb{P}(A)}{\mathbb{P}(B)}.$$

The denominator expands by the **law of total probability**:

$$\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\mathbb{P}(A^c).$$

$\mathbb{P}(A)$ is the **prior**, $\mathbb{P}(A \mid B)$ the **posterior**, and $\mathbb{P}(B \mid A)$ the **likelihood**.

Where the Analogy Breaks

Bayes' theorem is an identity, not a philosophy. It tells you how to update given a prior; it does not tell you what prior to hold, and where the prior is contested the theorem settles nothing. The disease example is clean only because the base rate of one in a thousand was handed over as fact.

Why It Matters Here — link to the thesis

Base rates dominate rare-event inference, and every tail question in this thesis is a rare-event question. A test that flags a bad generator with 99% accuracy still produces mostly false alarms if bad generators are rare — which is the same arithmetic that makes multiple testing across 19 metrics (§7.9) so treacherous.

6.5 Random Variables

Situation — the everyday picture

Roll two dice and add them. The outcome of the experiment is a pair — say $(3, 4)$ — but what you care about is the single number 7. You have attached a number to each outcome.

Problem — where it breaks or why it matters

Events are collections of outcomes, and outcomes are often not numbers: heads or tails, a pair of dice, a market state. To do arithmetic — averages, spreads, distances — we need numbers, and we need a disciplined way of attaching them.

Worked Arithmetic — concrete numbers

Two dice, 36 equally likely outcomes. Let X be the sum.

X	2	3	4	5	6	7	8	9	10	11	12
Ways	1	2	3	4	5	6	5	4	3	2	1

The counts total $1+2+3+4+5+6+5+4+3+2+1 = 36$. ✓
$X = 7$ arises six ways — $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$ — so $\mathbb{P}(X = 7) = 6/36 = 1/6$, while $\mathbb{P}(X = 2) = 1/36$.

X is a *function*: it takes an outcome and returns a number, exactly the machine of §1.3. What makes it *random* is that its input is the result of a chance experiment, not the function itself.

Formal Name — the mathematics

A **random variable** is a function $X : \Omega \to \mathbb{R}$ assigning a real number to each outcome. It is **discrete** if its range is finite or countable, **continuous** if it takes values across an interval.

We write $\mathbb{P}(X = x)$ for the probability of the event $\{\omega \in \Omega : X(\omega) = x\}$, and $\mathbb{P}(X \leq x)$ similarly. Capital letters denote the random variable, lower case a particular value it might take.

Where the Analogy Breaks

"Random variable" is a poor name twice over: it is not a variable, and it is not random. It is an ordinary, fully determined function. All the randomness lives in which outcome the experiment produces — the function merely translates that outcome into a number.

Why It Matters Here — link to the thesis

The daily log return r_t is a random variable. So is every statistic computed from it: the Hill estimator, the sample kurtosis, the discriminative AUC. Treating these as random variables rather than fixed numbers is what makes it meaningful to ask about their sampling variability — the subject of Chapter 7.

6.6 PMF, PDF and CDF

Situation — the everyday picture

Two ways to describe rainfall. You could say how likely each amount is — "2 mm is common, 40 mm is rare". Or you could say how likely it is to be *at most* a given amount — "90% of days see 10 mm or less". Both describe the same weather.

Problem — where it breaks or why it matters

For discrete quantities, listing the probability of each value works. For continuous ones it collapses: the probability of a return being exactly $0.0123456\ldots$ is zero, and that is true of every single value, so the list carries no information. Continuous quantities need a different device.

Worked Arithmetic — concrete numbers

Discrete: the PMF. For the two-dice sum, $\mathbb{P}(X = 2) = 1/36$, $\mathbb{P}(X = 7) = 6/36$, and the values sum to 1.

Discrete: the CDF. Accumulate:

$$F(4) = \mathbb{P}(X \leq 4) = \tfrac{1}{36} + \tfrac{2}{36} + \tfrac{3}{36} = \tfrac{6}{36} = \tfrac{1}{6}.$$

And $F(12) = 1$, since every outcome is at most 12.

Continuous: why point probabilities vanish. Pick a real number uniformly from $[0, 1]$. The chance it lands in $[0.3, 0.4]$ is 0.1; in $[0.30, 0.31]$ is 0.01; in an interval of width w is w. Shrink w to zero and the probability goes to zero. Any single point has probability exactly zero, yet some point certainly occurs.

Continuous: the density. What survives is probability *per unit width.* For the uniform on $[0, 1]$ the density is $f(x) = 1$ throughout, and

$$\mathbb{P}(0.3 \leq X \leq 0.4) = \int_{0.3}^{0.4} 1\, dx = 0.4 - 0.3 = 0.1. \quad \checkmark$$

The density is a rate; the probability is its integral (§3.5). This is why a density may exceed 1 — a uniform on $[0, 0.5]$ has $f(x) = 2$ — without anything being wrong: it is a rate, not a probability.

Formal Name — the mathematics

For a discrete X, the **probability mass function** is $p(x) = \mathbb{P}(X = x)$, with $p(x) \geq 0$ and $\sum_x p(x) = 1$.
For a continuous X, the **probability density function** f satisfies

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)\,dx, \qquad f(x) \geq 0, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1.$$

In both cases the **cumulative distribution function** is

$$F(x) = \mathbb{P}(X \leq x),$$

non-decreasing, with $F(-\infty) = 0$ and $F(+\infty) = 1$. For continuous X the two are linked by the Fundamental Theorem of Calculus (§3.6): $F' = f$, and F is the integral of f.

Where the Analogy Breaks

A density is not a probability and cannot be compared across different quantities without care: change the units of x and f changes too, since it is measured per unit of x. The CDF has no such problem — it is a probability, and dimensionless. This is one reason distances between distributions in this thesis are built from CDFs and quantiles rather than densities.

Why It Matters Here — link to the thesis

The Kolmogorov–Smirnov statistic is the largest vertical gap between two CDFs. Value at Risk is an inverse CDF — the return below which only 5% of days fall. And the histograms in the stylised-facts figure are estimates of f; the log-scaled panel (§2.5) is the one that makes the tail of f legible.

6.7 Expectation

Situation — the everyday picture

A fairground game costs \$2. You win \$10 with probability 1/8 and nothing otherwise. Play once and you either win or lose. Play ten thousand times and the outcome is no longer in doubt: you will have lost money, and by a predictable amount.

Problem — where it breaks or why it matters

Single outcomes are unpredictable; long-run averages are not. We need the number that long-run averages converge to, computed from the distribution rather than from data.

Worked Arithmetic — concrete numbers

The game. Win \$10 with probability 1/8, \$0 with probability 7/8:

$$\mathbb{E}[\text{winnings}] = 10 \times \tfrac{1}{8} + 0 \times \tfrac{7}{8} = 1.25.$$

Cost \$2, so the expected profit is $1.25 - 2 = -0.75$ per play. Over 10,000 plays, expect to lose about \$7,500.

One die.

$$\mathbb{E}[X] = \tfrac{1}{6}(1+2+3+4+5+6) = \tfrac{21}{6} = 3.5.$$

Note that 3.5 is not a possible outcome. The expectation is a balance point, not a prediction.

Two dice. By the same weighting,

$$\mathbb{E}[X] = \tfrac{1}{36}(2\cdot 1 + 3\cdot 2 + \cdots + 12\cdot 1) = \tfrac{252}{36} = 7.$$

Which is $3.5 + 3.5$ — expectation adds. ✓

Linearity, checked. Let $Y = 2X + 1$ for one die. Directly: Y takes values $3, 5, 7, 9, 11, 13$, averaging $\frac{48}{6} = 8$. By the rule: $2\mathbb{E}[X] + 1 = 2(3.5) + 1 = 8$. ✓

Formal Name — the mathematics

The **expectation** of a random variable is

$$\mathbb{E}[X] = \sum_x x\,p(x) \quad \text{(discrete)}, \qquad \mathbb{E}[X] = \int_{-\infty}^{\infty} x\,f(x)\,dx \quad \text{(continuous)}.$$

It is **linear**: for constants a, b and any X, Y,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y],$$

which holds whether or not X and Y are independent.
For a function of X, $\mathbb{E}[g(X)] = \sum_x g(x)p(x)$. In general $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$.

Where the Analogy Breaks

Expectation requires the defining sum or integral to converge, and for heavy-tailed distributions it may not. If $\mathbb{P}(|X| > x) \sim x^{-\alpha}$ with $\alpha \leq 1$, then $\mathbb{E}[X]$ simply does not exist — not "is large" or "is hard to estimate", but is undefined. Our markets have $\hat{\alpha}$ between 2.54 and 3.21, so the mean and variance are fine, but this is a property to be checked rather than assumed, and §6.12 shows which moments do fail.

Why It Matters Here — link to the thesis

Every loss function minimised in this thesis is an expectation estimated by a batch average. The WGAN critic loss $\mathbb{E}[D(x_{\text{real}})] - \mathbb{E}[D(x_{\text{fake}})]$ and the gradient penalty $\mathbb{E}[(\|\nabla D\|_2 - 1)^2]$ are both expectations, approximated by averaging over a batch of 64 or 128 windows.

6.8 Variance and Higher Moments

Situation — the everyday picture

Two investments both average 5% a year. One returns between 4% and 6% every year; the other alternates +40% and −30%. The average is a wholly inadequate description of the difference.

Problem — where it breaks or why it matters

Location without spread is half a summary. And for financial returns even spread is not enough: the shape of the extremes matters more than their typical size, so we need a ladder of increasingly fine descriptions — and a clear account of when each rung exists.

Worked Arithmetic — concrete numbers

One die. $\mathbb{E}[X] = 3.5$. Deviations and their squares:

x	1	2	3	4	5	6
$x - 3.5$	-2.5	-1.5	-0.5	0.5	1.5	2.5
$(x-3.5)^2$	6.25	2.25	0.25	0.25	2.25	6.25

$$\mathrm{Var}(X) = \tfrac{1}{6}(6.25 + 2.25 + 0.25 + 0.25 + 2.25 + 6.25) = \tfrac{17.5}{6} = 2.9167,$$

so the standard deviation is $\sqrt{2.9167} = 1.708$.

The shortcut. $\mathbb{E}[X^2] = \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = \frac{91}{6} = 15.1667$, and

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 15.1667 - 12.25 = 2.9167. \quad \checkmark$$

Why squares, not absolute values. Squaring makes variance add for independent variables, which absolute deviation does not. For two independent dice, $\mathrm{Var}(\text{sum}) = 2.9167 + 2.9167 = 5.8333$.

Scaling. $\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$: doubling a return quadruples its variance but only doubles its standard deviation. This is why volatility is quoted as a standard deviation — it shares the units of the returns themselves.

Formal Name — the mathematics

The **variance** is

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$

and the **standard deviation** is $\sigma = \sqrt{\mathrm{Var}(X)}$.
The **k-th moment** is $\mathbb{E}[X^k]$. Standardised third and fourth moments give

$$\text{Skewness} = \frac{\mathbb{E}[(X-\mu)^3]}{\sigma^3}, \qquad \text{Kurtosis} = \frac{\mathbb{E}[(X-\mu)^4]}{\sigma^4},$$

with excess kurtosis defined as kurtosis minus 3, so that a normal distribution scores 0.
Properties: $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$, and $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ when X and Y are independent.

Where the Analogy Breaks

Higher moments exist only when their integrals converge. For a power-law tail $\mathbb{P}(|X| > x) \sim x^{-\alpha}$, the k-th moment is finite only if $k < \alpha$. The BRICS markets in this project have $\hat{\alpha}$ between 2.54 and 3.21:

- $k = 2$ (variance): finite everywhere, since $2 < 2.54$;
- $k = 3$ (skewness): infinite in four of five markets;
- $k = 4$ (kurtosis): *infinite in all five.*

The population kurtosis of these return series does not exist. A sample kurtosis can always be computed, but it is estimating nothing.

Why It Matters Here — link to the thesis

This is why `kurtosis_diff` and `skewness_diff` were moved out of the ranked metrics and into the descriptive-only group. Ranking models on an estimate of a quantity that is infinite is not a strict test — it is noise with a decimal point, and Chapter 9 shows the estimate diverging with sample size rather than converging.

6.9 Joint Distributions and Independence

Situation — the everyday picture

Height and weight, measured across a population. Knowing someone is tall changes what you expect their weight to be. Height and shoe colour, by contrast, tell you nothing about each other.

Problem — where it breaks or why it matters

Individual distributions cannot express a relationship. Two return series may have identical histograms and behave completely differently in combination — and the whole question of temporal structure is a question about the joint behaviour of r_t and r_{t-1}.

Worked Arithmetic — concrete numbers

Two dice. Let X be the first roll and S the sum.

Are the two rolls independent? $\mathbb{P}(\text{first} = 3) = 1/6$ and $\mathbb{P}(\text{second} = 5) = 1/6$; the pair $(3, 5)$ is one of 36 outcomes, so

$$\mathbb{P}(3,5) = \tfrac{1}{36} = \tfrac{1}{6} \times \tfrac{1}{6}. \quad \checkmark$$

The joint factorises, so they are independent.

Are X and S independent? $\mathbb{P}(X = 1) = 1/6$ and $\mathbb{P}(S = 12) = 1/36$. If independent, $\mathbb{P}(X = 1, S = 12)$ would be $\frac{1}{6} \times \frac{1}{36} = \frac{1}{216}$. But $S = 12$ requires both dice to be 6, so $X = 1$ and $S = 12$ cannot both happen:

$$\mathbb{P}(X = 1, S = 12) = 0 \neq \tfrac{1}{216}.$$

Not independent. ✓Knowing the sum tells you about the first roll.

Conditional distribution. Given $S = 4$, the possibilities are $(1, 3), (2, 2), (3, 1)$, so $X \in \{1, 2, 3\}$ each with probability $1/3$ — a completely different distribution from the unconditional uniform on $\{1, \ldots, 6\}$.

Formal Name — the mathematics

The **joint distribution** of X and Y gives $\mathbb{P}(X = x, Y = y)$, or in the continuous case a joint density $f(x, y)$ with

$$\mathbb{P}((X, Y) \in A) = \iint_A f(x, y)\, dx\, dy.$$

The **marginal** distribution of X recovers X alone by summing or integrating out Y.
X and Y are **independent** if the joint factorises for all values:

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y), \qquad \text{or} \qquad f(x, y) = f_X(x) f_Y(y).$$

Equivalently, the conditional distribution of X given Y does not depend on Y.

Where the Analogy Breaks

Marginals do not determine the joint. Two pairs of variables can have identical individual distributions and utterly different dependence — this is exactly the gap a copula is designed to describe. It is also the precise sense in which matching a histogram is not matching a process, and the reason the evaluation framework in this thesis needs a second family of metrics at all.

Why It Matters Here — link to the thesis

The shuffled-real control has *identical* marginals to the real data by construction and destroyed joint structure. That single fact is why an evaluation built only on marginal distances ranked it first: under the old unweighted 19-metric composite it scored an average rank of 1.24 against 2.47 for a genuine generator. The permutation-invariant metrics could not see the difference, because there is no difference for them to see.

6.10 Covariance and Correlation

Situation — the everyday picture

Ice cream sales and drowning deaths rise together over the year. Neither causes the other — both follow temperature — but the co-movement is real and measurable.

Problem — where it breaks or why it matters

Independence is a yes/no property. What is usually wanted is a graded measure: *how strongly* do two quantities move together, on a scale that permits comparison across quantities with different units?

Worked Arithmetic — concrete numbers

Four paired observations:

x	1	2	3	4
y	2	4	5	9

Means: $\bar{x} = 10/4 = 2.5$, $\bar{y} = 20/4 = 5$.

$x - \bar{x}$	-1.5	-0.5	0.5	1.5
$y - \bar{y}$	-3	-1	0	4
product	4.5	0.5	0	6

$$\mathrm{Cov}(x, y) = \tfrac{1}{4}(4.5 + 0.5 + 0 + 6) = \tfrac{11}{4} = 2.75.$$

Positive: the two rise together.

To correlation. Spreads: $\mathrm{Var}(x) = \frac{1}{4}(2.25 + 0.25 + 0.25 + 2.25) = 1.25$, so $\sigma_x = 1.118$; $\mathrm{Var}(y) = \frac{1}{4}(9 + 1 + 0 + 16) = 6.5$, so $\sigma_y = 2.550$.

$$\rho = \frac{2.75}{1.118 \times 2.550} = \frac{2.75}{2.851} = 0.965.$$

Close to the maximum of 1 — a strong, nearly linear relationship.

Units cancel. Measure y in different units, say doubled: covariance doubles to 5.5, but σ_y doubles too, so ρ is unchanged. ✓

Formal Name — the mathematics

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y],$$

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1].$$

$\rho = \pm 1$ exactly when Y is an increasing or decreasing linear function of X. If X and Y are independent then $\mathrm{Cov}(X, Y) = 0$.
The **autocorrelation** of a series at lag k is the correlation of the series with itself shifted by k:

$$\rho(k) = \frac{\mathrm{Cov}(r_t, r_{t-k})}{\mathrm{Var}(r_t)}.$$

Where the Analogy Breaks

Zero correlation does not imply independence. Let X be symmetric about zero and $Y = X^2$; then $\mathrm{Cov}(X, Y) = 0$ while Y is completely determined by X. Correlation detects *linear* association only.
This is exactly the situation for financial returns. The autocorrelation of r_t is near zero — returns are close to linearly unpredictable — while the autocorrelation of $|r_t|$ and r_t^2 is strongly positive and decays slowly. Returns are uncorrelated and emphatically not independent, which is the single most important structural fact in this thesis.

Why It Matters Here — link to the thesis

This is why the evaluation computes three separate autocorrelation metrics — `acf_returns_mae` on r_t, `acf_absolute_mae` on $|r_t|$, and `acf_squared_mae` on r_t^2. A generator that matches only the first has reproduced the absence of linear predictability while missing volatility clustering entirely, and only the second and third catch it.

6.11 The Law of Large Numbers

Situation — the everyday picture

Flip a fair coin ten times and you might see seven heads — unremarkable. Flip it ten thousand times and 7,000 heads would be extraordinary. The proportion settles toward one half, and it settles more firmly the longer you go.

Problem — where it breaks or why it matters

Expectation was defined from a distribution, and we asserted that long-run averages converge to it. That assertion needs to be a theorem, with stated conditions — because the conditions are exactly what financial data threatens.

Worked Arithmetic — concrete numbers

Rolling a fair die, $\mathbb{E}[X] = 3.5$. The standard deviation of the average of n rolls is $\sigma/\sqrt{n}$ with $\sigma = 1.708$:

n	sd of the average	typical spread
1	1.708	± 1.71
100	0.171	± 0.17
10,000	0.017	± 0.02

Each hundredfold increase in n shrinks the spread tenfold, since $\sqrt{100} = 10$. Convergence is real but slow: to halve the uncertainty you must quadruple the data.

The cost of that slowness. This project's per-market test set holds about 496 observations; pooling all five markets gives about 2,480. The pooled standard error is $\sqrt{2480/496} = \sqrt{5} = 2.24$ times smaller — which is why the downstream utility metrics are computed on the pooled set rather than per-market.

Formal Name — the mathematics

Law of Large Numbers. If $X_1, X_2, \ldots$ are independent with common distribution and $\mathbb{E}|X| < \infty$, then the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i \longrightarrow \mathbb{E}[X]$$

as $n \to \infty$. When the variance σ^2 is also finite, $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$, so the standard error is $\sigma/\sqrt{n}$.

Where the Analogy Breaks

The law requires the mean to exist. It says nothing about how fast convergence happens for any particular distribution, and for heavy-tailed data the answer can be "far more slowly than $1/\sqrt{n}$ suggests", because a single extreme observation can move the running average substantially even at large n. The guarantee is asymptotic; the sample you actually hold is finite.

Why It Matters Here — link to the thesis

Every metric in this thesis is an average over a finite sample and therefore carries a standard error. The discriminative AUC null is not exactly 0.5 but 0.506 ± 0.084 — measured across fifteen real-versus-real splits — and that spread is precisely the $\sigma/\sqrt{n}$ of this section. An observed AUC must be judged against that spread, not against the theoretical 0.5.

6.12 The Central Limit Theorem — and Where It Fails

Situation — the everyday picture

Roll a die 100 times and record the average. Repeat the whole experiment 10,000 times and plot the 10,000 averages. A bell curve appears — even though a single die is flat, with all six faces equally likely. The bell was not in the die; it came from the averaging.

Problem — where it breaks or why it matters

The Central Limit Theorem is the reason the normal distribution is everywhere, and it is routinely over-read as licensing Gaussian models for anything. Applied to daily market returns it makes predictions that are wrong by orders of magnitude, and understanding exactly which of its conditions fails — and which does not — is the intellectual core of this thesis.

Worked Arithmetic — concrete numbers

What the Gaussian predicts. For a normal distribution,

$$\mathbb{P}(|Z| > 5) = 2\Phi(-5) \approx 5.73 \times 10^{-7}.$$

At 252 trading days per year, the expected waiting time for one such day is

$$\frac{1}{252 \times 5.73 \times 10^{-7}} \approx 6{,}922 \text{ years.}$$

What the data shows. A single BRICS market contributes roughly 6,201 market-days — about 25 years. The Gaussian expects $6201 \times 5.73 \times 10^{-7} \approx 0.0036$ five-sigma days over that span: call it one day every three centuries. Eleven were observed.
The discrepancy is a factor of roughly $11/0.0036 \approx 3{,}000$. This is not a model that is slightly miscalibrated in the tails; it is a model whose tail predictions are meaningless.

Which condition failed? The CLT needs finite variance. Our markets have $\hat{\alpha} \in [2.54, 3.21] > 2$, so the variance *is* finite and the CLT *does* apply to averages of returns. The theorem is not violated.
What fails is the inference people draw from it. The CLT describes the distribution of the *average* of many returns. It says nothing whatever about the distribution of a *single* return, and a five-sigma day is a single observation, not an average. Modelling individual daily returns as Gaussian is not an application of the CLT — it is an unrelated assumption that happens to be false.

And the fourth moment. Even where the CLT applies, it applies to the mean. The analogous result for the sample variance requires a finite fourth moment, and with $\alpha < 4$ everywhere in our data that moment is infinite. So the sample kurtosis has no limiting normal distribution, no standard error, and no convergence — Chapter 9 shows it climbing from 2.53 at $n = 250$ to 11.84 at $n = 2{,}000$ on BOVESPA while the Hill estimator, which does not require finite moments, settles from 4.83 to 3.11.

Formal Name — the mathematics

Central Limit Theorem. Let $X_1, X_2, \ldots$ be independent with common distribution, mean μ and *finite* variance σ^2. Then

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \longrightarrow \mathcal{N}(0, 1)$$

in distribution as $n \to \infty$.
Three conditions carry the weight: independence, identical distribution, and finite variance. When the variance is infinite ($\alpha < 2$), averages converge instead to a **stable** distribution with heavy tails of their own, and the normal approximation never appears at any sample size.

Where the Analogy Breaks

The CLT is about sums and averages, not about individual observations, and this is the confusion it most often causes. It also assumes independence, which market returns violate through volatility clustering — versions of the theorem exist for dependent sequences, but they converge more slowly and require their own conditions.

So there are two distinct failures to keep apart. Applying the CLT to a single day's return is a misapplication of a correct theorem. Computing a sample kurtosis and treating it as an estimate is applying a theorem whose conditions genuinely do not hold.

Why It Matters Here — link to the thesis

This section is the motivation for the entire thesis. If daily returns were Gaussian, a closed-form risk model would suffice and no generative model would be needed. Because they are not — eleven five-sigma days where the model expects 0.0036 — we need generators that reproduce heavy tails and volatility clustering, and an evaluation framework honest enough to check whether they actually do. It is also why the tail index $\hat{\alpha}$, which needs no finite moments, is a ranked metric while kurtosis is not.

Chapter 7

Statistics

Before and After

Before this chapter you need: logarithms and the log rules (§2.3), derivatives and how to find a maximum (§3.3–§3.4), gradients (§4.3), random variables, expectation and variance (§6.5–§6.8), the law of large numbers and the CLT (§6.11–§6.12).
After it you will understand: why an estimate is itself a random quantity, what maximum likelihood maximises, why this project fits GARCH with a Gaussian likelihood it knows to be wrong, what a p-value does and does not mean, why a test on 125 days is nearly powerless, and what goes wrong when 19 metrics are tested at once.

Probability reasons from a known distribution to the behaviour of data. Statistics runs the arrow backwards: from data in hand to claims about the distribution that produced it. Every number reported in this thesis is a statistic, and every one carries uncertainty this chapter makes explicit.

7.1 Samples and Sampling Distributions

Situation — the everyday picture

You measure the temperature every day for a week: 15, 17, 14, 19, 16, 21, 13 degrees. You want the "typical" value and how much the values vary. Next week you repeat the exercise and get different numbers — and therefore a different average.

Problem — where it breaks or why it matters

The second week's average differs from the first, yet neither is wrong. The summary you compute is not a fixed property of the world but something that changes with the sample. Unless that variation is quantified, there is no way to tell a real difference from an accident of sampling.

Worked Arithmetic — concrete numbers

The week's data. $x = \{15, 17, 14, 19, 16, 21, 13\}$.

Mean: $\bar{x} = (15 + 17 + 14 + 19 + 16 + 21 + 13)/7 = 115/7 = 16.43$.
Deviations: $\{-1.43, 0.57, -2.43, 2.57, -0.43, 4.57, -3.43\}$.
Squared: $\{2.04, 0.33, 5.90, 6.60, 0.18, 20.88, 11.76\}$, totalling 47.69.
Sample variance: $s^2 = 47.69/6 = 7.95$, so $s = 2.82$°C.

Why divide by 6 and not 7. The deviations are taken from $\bar{x}$, which was computed from the same seven numbers. They therefore sum to zero by construction, so only six of them are free — fix any six and the seventh follows. Dividing by $n-1$ corrects for that, and without it the variance would be systematically too small.

The sampling distribution. If temperatures have true standard deviation $\sigma = 3$, then across repeated weeks the sample mean has standard deviation

$$\frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{7}} = 1.13.$$

So a second week averaging 17.5 instead of 16.4 is entirely unremarkable: the gap of 1.1 is one standard error.

Formal Name — the mathematics

A **sample** is n observations drawn from a population. A **statistic** is any function of the sample; being a function of random quantities, it is itself a random variable, and its distribution across repeated samples is its **sampling distribution**.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The **standard error** of the mean is $\sigma/\sqrt{n}$, estimated by $s/\sqrt{n}$. The $n-1$ divisor is **Bessel's correction**.

Where the Analogy Breaks

Standard deviation summarises spread well only for roughly symmetric distributions. Returns fall much further than they rise, so a single number understates downside risk. And the $\sigma/\sqrt{n}$ standard error is itself a CLT result requiring finite variance — available for returns ($\hat{\alpha} > 2$) but not for their fourth moment, which is why no standard error is ever quoted for a sample kurtosis in this thesis.

Why It Matters Here — link to the thesis

Daily volatility $\sigma = \mathrm{sd}(r_t)$ is the foundational scale in all five markets: MOEX 1.87%/day, BOVESPA 1.63%. Annualising multiplies by $\sqrt{252}$. Every reported metric is a statistic with a sampling distribution, which is why the pooled test set of about 2,480 observations is preferred to the per-market 496 wherever the metric permits pooling.

7.2 Estimators: Bias and Variance

Situation — the everyday picture

Two bathroom scales. One reads exactly 2 kg heavy every time — consistent but wrong. The other averages correctly across many weighings but scatters ±3 kg. Neither is obviously better; it depends what you need.

Problem — where it breaks or why it matters

"Accurate" bundles together two distinct failures — being systematically off, and being erratic — and they trade against each other. Separating them is what lets you choose an estimator deliberately rather than by habit.

Worked Arithmetic — concrete numbers

True weight 70 kg. Five weighings on each scale.

	readings					mean	spread
Scale A	72	72	72	72	72	72	0
Scale B	67	73	70	68	72	70	2.55

Scale A: bias $= 72 - 70 = +2$, variance $= 0$.
Scale B: bias $= 0$, standard deviation $= 2.55$.

Which is better? Compare mean squared error, MSE $= \text{bias}^2 + \text{variance}$:

$$\text{A: } 2^2 + 0 = 4, \qquad \text{B: } 0^2 + 6.5 = 6.5.$$

The biased scale wins — on this criterion, at this sample size. Averaging four readings from B, however, cuts its variance to $6.5/4 = 1.63$ and its MSE to 1.63, beating A. The right choice depends on how much data you have.

Bessel's correction as bias removal. On the seven temperatures, dividing by $n = 7$ gives $47.69/7 = 6.81$; by $n - 1 = 6$ gives 7.95. The first is biased low; the second is unbiased.

Formal Name — the mathematics

An **estimator** $\hat{\theta}$ is a statistic used to estimate a parameter θ. Its **bias** is $\mathbb{E}[\hat{\theta}] - \theta$, and it is **unbiased** when this is zero.

$$\text{MSE}(\hat{\theta}) = \mathbb{E}\big[(\hat{\theta} - \theta)^2\big] = \big(\text{bias}\big)^2 + \text{Var}(\hat{\theta}).$$

$\hat{\theta}$ is **consistent** if it converges to θ as $n \to \infty$. Consistency is about the limit; unbiasedness is about any fixed n. Neither implies the other.

Where the Analogy Breaks

Unbiasedness is a weaker virtue than its name suggests. It guarantees correctness on average over hypothetical repetitions, which is cold comfort when you have one sample. A biased estimator with much smaller variance is frequently the better instrument, and every regularised model in machine learning — including weight decay and early stopping — deliberately accepts bias to reduce variance.

Why It Matters Here — link to the thesis

The Hill estimator is biased at small samples yet consistent, and its bias shrinks as the tail sample grows. This is why Chapter 9 reports it converging from 4.83 at $n = 250$ to 3.11 at $n = 2{,}000$ on BOVESPA: the movement is bias disappearing, not instability. Sample kurtosis over the same range moves the opposite way — 2.53 to 11.84 — because it is not converging to anything at all.

7.3 Maximum Likelihood

Situation — the everyday picture

A coin lands heads 7 times in 10 flips. What is its bias? You would guess 0.7, and the reasoning behind that guess is: of all possible biases, 0.7 is the one that makes what actually happened most probable.

Problem — where it breaks or why it matters

That reasoning needs to become a procedure that works for any model, not just coins — one that turns "which parameter best explains this data" into a calculation.

Worked Arithmetic — concrete numbers

The coin. For bias p, the probability of a specific sequence with 7 heads in 10 flips is $p^7(1-p)^3$. Evaluate:

p	0.5	0.6	0.7	0.8	0.9
$p^7(1-p)^3$	0.000977	0.001791	**0.002224**	0.001678	0.000478

The largest value is at $p = 0.7$. ✓

By calculus. Maximise the logarithm instead — the log is increasing, so it has its maximum in the same place, and it turns the product into a sum (§2.3):

$$\ell(p) = 7\ln p + 3\ln(1-p).$$

Differentiate and set to zero, using $\frac{d}{dp}\ln p = 1/p$:

$$\ell'(p) = \frac{7}{p} - \frac{3}{1-p} = 0 \implies 7(1-p) = 3p \implies 7 = 10p \implies p = 0.7. \quad ✓$$

Confirm it is a maximum. $\ell''(p) = -7/p^2 - 3/(1-p)^2 < 0$ for every p in $(0,1)$, so the function is concave and 0.7 is its highest point (§3.4). ✓

Formal Name — the mathematics

Given data $x_1, \dots, x_n$ and a model with density $f(x \mid \theta)$, the **likelihood** treats the data as fixed and θ as the argument:

$$L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta), \qquad \ell(\theta) = \ln L(\theta) = \sum_{i=1}^{n} \ln f(x_i \mid \theta).$$

The **maximum likelihood estimator** is $\hat{\theta} = \arg\max_\theta \ell(\theta)$.
Under regularity conditions — including that the model is correctly specified — MLE is consistent, asymptotically unbiased, and asymptotically normal with the smallest achievable variance.

Where the Analogy Breaks

The likelihood is not a probability distribution over θ. It is a function of θ computed from fixed data, and it does not integrate to one; reading $L(0.7) = 0.0022$ as "the probability the bias is 0.7" is a category error. Turning the likelihood into a distribution over θ requires a

prior and Bayes' theorem (§6.4).
The optimality guarantees also assume the model is right. When it is not, the next section describes what survives.

Why It Matters Here — link to the thesis

Log-likelihood is maximised whenever a GARCH model is fitted in this project. The sum form is not a convenience but a necessity: a product of several thousand small densities underflows to zero in floating-point arithmetic, while the corresponding sum of logarithms is perfectly stable.

7.4 QMLE: Consistency Under Misspecification

Situation — the everyday picture

You measure a room with a tape that stretches slightly under tension. Every individual reading is a little off. But if the stretch is proportional, the *ratio* of two walls is still exactly right — the flawed instrument gets some quantities wrong while getting others right.

Problem — where it breaks or why it matters

Maximum likelihood's guarantees assume the model is correct. Real models never are. The practical question is not whether the model is wrong — it is — but which of its outputs survive being wrong, and this project's residual-kurtosis test depends entirely on the answer.

Worked Arithmetic — concrete numbers

The setup. GARCH$(1, 1)$ models the conditional variance:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \qquad \varepsilon_t = \sigma_t z_t,$$

where z_t is a standardised innovation. Fitting requires assuming a distribution for z_t.

The test we want to run. Real returns keep excess kurtosis *after* dividing out the time-varying volatility (Bollerslev 1987). That is stylised fact 7: fat tails are not fully explained by volatility clustering. We want to measure that leftover kurtosis in the standardised residuals $\hat{z}_t = \varepsilon_t/\hat{\sigma}_t$ and check whether a generator reproduces it.

Why not assume a Student-t? A t innovation would fit the data better — the returns really are heavy-tailed. But it fits better precisely by absorbing the excess kurtosis into the assumed innovation distribution. The residuals would then look t-shaped by construction, and measuring their kurtosis would recover the assumption rather than a property of the data. The test would be circular.

Why Gaussian works anyway. Maximising a Gaussian likelihood estimates (ω, α, β) consistently even when z_t is not Gaussian. The Gaussian score equations depend on the data only through first and second conditional moments, and those are correctly specified by the GARCH recursion whatever the shape of z_t. So:

- the variance path $\hat{\sigma}_t$ is estimated consistently — *right*;
- the standard errors from the Gaussian information matrix are — *wrong*, and need a sandwich correction;
- leftover kurtosis in $\hat{z}_t$ is — *genuine evidence*, because nothing in the fitting procedure put it there.

The deliberately wrong assumption is what makes the measurement honest.

Formal Name — the mathematics

Quasi-maximum likelihood estimation (QMLE) maximises a likelihood known to be misspecified. For GARCH, White (1982) and Bollerslev–Wooldridge (1992) establish that the Gaussian QMLE of the conditional-variance parameters is consistent and asymptotically normal under mild conditions, whatever the true innovation distribution.
The asymptotic variance is *not* the usual inverse information matrix but the **sandwich** form

$$A^{-1}BA^{-1},$$

where A is the expected Hessian and B the variance of the score. Under correct specification $A = B$ and this collapses to the standard result.

Where the Analogy Breaks

Consistency under misspecification is a guarantee about specific parameters, not a licence to ignore the model being wrong. Gaussian QMLE gets the variance dynamics right and the uncertainty quantification wrong. It also assumes the *variance equation* is correctly specified — if the true process has leverage effects or long-memory volatility that GARCH$(1, 1)$ cannot represent, no choice of innovation distribution repairs that.

Why It Matters Here — link to the thesis

This is the reasoning behind `resid_kurtosis_diff`, one of the seven ranked temporal metrics. The pipeline fits a constant-mean GARCH$(1, 1)$ by Gaussian QMLE, extracts $\hat{z}_t = r_t/\hat{\sigma}_t$, and compares the excess kurtosis of real and synthetic residuals. A generator that reproduces volatility clustering but not conditional fat tails will match the ACF metrics and fail this one — which is exactly the discrimination the metric exists to provide.

7.5 Hypothesis Testing

Situation — the everyday picture

A colleague claims they can tell tap water from bottled. You pour ten pairs and they get eight right. Impressive — or is it? Someone guessing would average five, and getting eight by luck alone is not extraordinary.

Problem — where it breaks or why it matters

Distinguishing a real effect from a lucky run needs a rule fixed in advance. Without one, any result can be narrated as impressive after the fact.

Worked Arithmetic — concrete numbers

Null hypothesis: they are guessing, so each trial is a fair coin, $p = 0.5$.
Probability of 8 or more correct out of 10 by chance:

$$\mathbb{P}(8) = \binom{10}{8}(0.5)^{10} = 45/1024 = 0.0439$$

$$\mathbb{P}(9) = \binom{10}{9}(0.5)^{10} = 10/1024 = 0.0098, \qquad \mathbb{P}(10) = 1/1024 = 0.0010$$

$$\mathbb{P}(\geq 8) = (45 + 10 + 1)/1024 = 56/1024 = 0.0547.$$

At the conventional 5% threshold, $0.0547 > 0.05$: do not reject. Eight out of ten is not quite enough.
Nine out of ten: $\mathbb{P}(\geq 9) = 11/1024 = 0.0107 < 0.05$. Reject.

The line between "unconvincing" and "convincing" falls between 8 and 9 correct — a single trial. That sharpness is an artefact of the threshold, not a feature of the evidence.

Formal Name — the mathematics

A test contrasts a **null hypothesis** H_0 with an **alternative** H_1. A **test statistic** is computed, and its distribution under H_0 determines whether the observed value is surprising. Two errors are possible:

- **Type I**: rejecting a true H_0. Probability α, chosen in advance, conventionally 0.05.
- **Type II**: failing to reject a false H_0. Probability β; the **power** is $1 - \beta$ (§7.8).

Failing to reject is not evidence for H_0. It means the data was insufficient to rule it out.

Where the Analogy Breaks

The 0.05 threshold is a convention with no mathematical standing — Fisher proposed it as a rough guide, not a law. Treating 0.049 and 0.051 as categorically different, when they differ by less than the sampling noise in either, is the single most common abuse of the framework.

Why It Matters Here — link to the thesis

The Kupiec test asks whether an observed VaR exceedance rate is distinguishable from the claimed 5%; Christoffersen asks whether exceedances cluster; the Anderson–Darling and Kolmogorov–Smirnov tests compare distributions. Each has a null, a statistic and a threshold — and §7.8 shows that on per-market samples several of them cannot reject anything.

7.6 What a p-Value Is and Is Not

Situation — the everyday picture

A study reports $p = 0.03$ and the write-up says “there is a 3% chance the result is due to chance”. That sentence is wrong, and the correct statement is narrower and less satisfying than most readers want.

Problem — where it breaks or why it matters

The p-value is the most misread number in quantitative work. Since several appear in this thesis — Kupiec, Christoffersen, ARCH, Anderson–Darling — what each one licenses needs stating precisely.

Worked Arithmetic — concrete numbers

Return to eight correct out of ten. The p-value is 0.0547.

What it means: *if* the person were guessing, the probability of seeing eight or more correct is 0.0547.

What it does not mean:

- Not the probability they are guessing. That is $\mathbb{P}(H_0 \mid \text{data})$, and getting it from $\mathbb{P}(\text{data} \mid H_0)$ requires a prior and Bayes' theorem (§6.4) — the two differ by exactly the factor that made a positive disease test only 9% informative.
- Not the probability the result is a fluke.
- Not a measure of effect size. With $n = 10{,}000$ a trivial edge such as 50.5% produces a tiny p; with $n = 10$ a large edge produces a big one.
- Not comparable across tests as a strength ranking.

A concrete demonstration of the last point. Someone correct on 51% of 100,000 trials gives $p \approx 0.00003$ — overwhelming significance for an edge nobody would care about. Someone correct on 90% of 10 trials gives $p = 0.0107$. The second is the far more impressive ability and the less significant result.

Formal Name — the mathematics

The ***p*-value** is the probability, computed under H_0, of observing a test statistic at least as extreme as the one obtained:

$$p = \mathbb{P}(T \geq t_{\text{obs}} \mid H_0).$$

It is a statement about data given a hypothesis, never about a hypothesis given data. Under H_0 the p-value is uniformly distributed on $[0, 1]$ — which is why one test in twenty falls below 0.05 when nothing whatever is going on.

Where the Analogy Breaks

Some p-values in this thesis are not even usable as continuous evidence. `scipy`'s **`anderson_ksamp`** clamps its result to the range $[0.001, 0.25]$ and warns whenever the true value falls outside. A reported 0.001 may mean 0.001 or 10^{-12}; the number has been censored. The **`arch_pvalue_diff`** metric was moved to descriptive-only partly for this reason — it saturates on real data, giving $|\Delta p| = 0.000000$ and no discrimination at all.

Why It Matters Here — link to the thesis

Kupiec and Christoffersen p-values are reported per model. They are read as "the data does not contradict correct coverage", never as "the model is correct" — and given the power calculations in the next section, on per-market samples the first statement is often true simply because almost nothing could contradict it.

7.7 Confidence Intervals

Situation — the everyday picture

A poll reports 52% support, "margin of error 3 points". The single number 52 is less informative than the range 49–55, because the range carries its own uncertainty with it.

Problem — where it breaks or why it matters

A point estimate invites false precision. Reporting $\hat{\alpha} = 3.11$ suggests a confidence that a sample of a few hundred tail observations cannot support, and an interval says so honestly.

Worked Arithmetic — concrete numbers

The seven temperatures: $\bar{x} = 16.43$, $s = 2.82$, $n = 7$.

Standard error: $s/\sqrt{n} = 2.82/\sqrt{7} = 1.066$.
95% interval (using ≈ 2 standard errors):

$$16.43 \pm 2 \times 1.066 = 16.43 \pm 2.13 = [14.30,\ 18.56].$$

The effect of more data. With the same s but $n = 700$:

$$\mathrm{SE} = 2.82/\sqrt{700} = 0.107, \qquad \text{interval} = 16.43 \pm 0.21.$$

A hundredfold increase in data narrows the interval tenfold — the $\sqrt{n}$ of §6.11 again.

The discriminative AUC null. Fifteen real-versus-real splits gave 0.506 ± 0.084. An observed AUC of 0.62 sits

$$z = \frac{0.62 - 0.506}{0.084} = 1.36$$

standard errors above the null centre, giving $p \approx 0.17$. Not significant — despite 0.62 looking comfortably above 0.5 to the naked eye.

Formal Name — the mathematics

A $(1-\alpha)$ **confidence interval** is constructed so that, across repeated samples, the interval contains the true parameter a proportion $1-\alpha$ of the time. For a mean with known σ:

$$\bar{x} \pm z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}, \qquad z_{0.025} = 1.96.$$

With σ estimated by s and small n, the t-distribution replaces the normal.

Where the Analogy Breaks

The confidence is in the procedure, not in the particular interval. Once computed, $[14.30, 18.56]$ either contains the true mean or it does not — there is no probability left. "95% probability the mean lies in this interval" is a Bayesian credible interval, which requires a prior and is a different object.
Note also that the interval above assumes the CLT applies. For a statistic with no finite variance — sample kurtosis on our data — the construction is unavailable in principle, not merely difficult.

Why It Matters Here — link to the thesis

The 0.506 ± 0.084 null is why the evaluation reports `discriminative_auc_absz`, the distance from the null centre measured in standard errors, rather than the raw AUC. Judging a generator against a theoretical 0.5 would flag ordinary sampling noise as a detectable difference.

7.8 Statistical Power

Situation — the everyday picture

You test a bathroom scale for a 200 g bias by weighing yourself twice. The test will not detect it — not because the bias is absent, but because two weighings cannot resolve 200 g against day-to-day variation. The instrument is too blunt for the question.

Problem — where it breaks or why it matters

A test that fails to reject tells you one of two things: the null is true, or your test was too weak to notice. These are entirely different conclusions, and only a power calculation separates them.

Worked Arithmetic — concrete numbers

Kupiec, this project's numbers. A VaR model claims 5% exceedances. Suppose the truth is 7%. Can the test detect the difference?

Test days n	Power at true $p = 7\%$	Setting
125	17%	5-year, per market
496	$\approx$ 56%	20-year, per market
2,480	$\approx$ 99%	20-year, pooled across 5 markets

Reading the top row. With 125 days, a genuinely broken model — exceeding its VaR 40% more often than advertised — goes undetected five times in six. A non-rejection carries essentially no information.

Why pooling helps so much. Exceedance counts scale with n while their standard deviation scales with $\sqrt{n}$, so the signal-to-noise ratio grows as $\sqrt{n}$. Going from 496 to 2,480 multiplies it by $\sqrt{5} = 2.24$, and that is enough to move power from a coin flip to near certainty.

The asymmetry this creates. At $n = 125$ a non-rejection is uninformative, while a rejection still means something — Type I error is controlled at 5% regardless of power. Low power damages only the negative result.

Formal Name — the mathematics

The **power** of a test is

$$1 - \beta = \mathbb{P}(\text{reject } H_0 \mid H_1 \text{ true}).$$

It rises with the sample size n, with the size of the true effect, and with the significance level α (at the cost of more false positives). It falls as the data becomes noisier.
Conventionally 80% power is treated as adequate. Below roughly 50%, a non-rejection should not be reported as support for the null.

Where the Analogy Breaks

Power is computed against a *specific* alternative. “The test has 56% power” is meaningless until the alternative is named — here, a true exceedance rate of 7% against a claimed 5%. Against a true rate of 6% the power is much lower; against 15%, much higher. Any power figure is an answer to a particular question.

Why It Matters Here — link to the thesis

This is the reason downstream utility is evaluated on the pooled test set of about 2,480 observations rather than per market. It is also why QLIKE is computed pooled: at per-market sample sizes the metric inverts, scoring Gaussian noise above real data. Reporting a per-market Kupiec non-rejection as evidence that a generator produces well-calibrated VaR would be a claim the data cannot support at 17% power.

7.9 Multiple Testing

Situation — the everyday picture

Twenty researchers each test a different useless drug. Each uses the 5% threshold. On average one of them finds a significant result and publishes it. The drug does nothing; the statistics were applied correctly; the finding is still spurious.

Problem — where it breaks or why it matters

This thesis computes many metrics per model per market. Every one is an opportunity for a chance result, and the probability that *at least one* misleads grows quickly with their number.

Worked Arithmetic — concrete numbers

Each test has a 5% false-positive rate under the null, so a 95% chance of behaving. For m independent tests, the probability that all behave is 0.95^m:

Tests m	$\mathbb{P}$(no false positive)	$\mathbb{P}$(at least one)
1	0.950	0.050
5	0.774	0.226
10	0.599	0.401
19	0.377	**0.623**

At the 19 metrics of the original composite, the chance of at least one spurious “significant” result is 62% — more likely than not, with every individual test correctly calibrated.

Bonferroni. To hold the overall rate at 5%, test each at $\alpha/m = 0.05/19 = 0.00263$. Then $0.99737^{19} = 0.951$, so the family-wise error rate is about 4.9%. ✓

The price. A threshold of 0.00263 instead of 0.05 demands far stronger evidence per test, which drives power down — and §7.8 already showed power was the binding constraint. Correcting for multiplicity and retaining power both require more data; there is no free version.

Formal Name — the mathematics

With m tests each at level α, the **family-wise error rate** is

$$\mathrm{FWER} = 1 - (1-\alpha)^m$$

under independence. The **Bonferroni correction** tests each at α/m, guaranteeing $\mathrm{FWER} \le \alpha$ whether or not the tests are independent.
Less conservative alternatives — Holm's step-down procedure, or Benjamini–Hochberg control of the false discovery rate — trade a weaker guarantee for more power.

Where the Analogy Breaks

Bonferroni assumes the worst case and is conservative when tests are correlated, which these emphatically are: `acf_absolute_mae` and `acf_squared_mae` measure nearly the same thing, and Wasserstein and energy distance both respond to the same distributional gaps. The effective number of independent tests is well below 19, so a strict Bonferroni over-corrects. Knowing the correction is too strong does not, however, license skipping it.

Why It Matters Here — link to the thesis

This is part of why the evaluation framework ranks models rather than accumulating significance tests. Ranks avoid the multiplicity problem: they make no distributional claim per metric and cannot generate false positives to correct for. The composite score is the mean of a fidelity rank and a temporal rank, and the taxonomy in Chapter 15 exists precisely so that seven permutation-invariant metrics cannot outvote seven ordering-sensitive ones — which is what let the shuffled control win the old unweighted 19-metric average at 1.24 against 2.47.

7.10 Linear Regression

Situation — the everyday picture

Plot exam score against hours studied. The points scatter but trend upward. You draw the straight line that best follows them, and use it to guess the score for someone who studied six hours.

Problem — where it breaks or why it matters

"Best" needs defining. Infinitely many lines pass near the points, and without a criterion the choice is arbitrary — and unrepeatable by anyone else.

Worked Arithmetic — concrete numbers

Four observations, hours x and score y:

x	1	2	3	4
y	2	4	5	9

These are the numbers from §6.10, where we found $\bar{x} = 2.5$, $\bar{y} = 5$, $\mathrm{Cov} = 2.75$ and $\mathrm{Var}(x) = 1.25$.

Slope:

$$\hat\beta = \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)} = \frac{2.75}{1.25} = 2.2.$$

Intercept: the line passes through $(\bar x, \bar y)$, so

$$\hat\alpha = \bar y - \hat\beta \bar x = 5 - 2.2 \times 2.5 = -0.5.$$

Fitted line: $\hat y = -0.5 + 2.2x$.

Fitted values and residuals:

x	1	2	3	4
y	2	4	5	9
$\hat y$	1.7	3.9	6.1	8.3
residual	0.3	0.1	−1.1	0.7

Residuals sum to $0.3 + 0.1 - 1.1 + 0.7 = 0$, as they must when an intercept is fitted. ✓
Sum of squared residuals: $0.09 + 0.01 + 1.21 + 0.49 = 1.80$.

R^2**.** Total squared deviation of y about its mean: $9 + 1 + 0 + 16 = 26$. So

$$R^2 = 1 - \frac{1.80}{26} = 0.931.$$

The line accounts for 93% of the variation in scores.

Formal Name — the mathematics

Simple linear regression models

$$y_i = \alpha + \beta x_i + \varepsilon_i,$$

and **ordinary least squares** chooses α, β to minimise $\sum_i (y_i - \alpha - \beta x_i)^2$. Setting both partial derivatives (§4.2) to zero gives

$$\hat\beta = \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)}, \qquad \hat\alpha = \bar y - \hat\beta \bar x.$$

The **coefficient of determination** is

$$R^2 = 1 - \frac{\sum_i (y_i - \hat y_i)^2}{\sum_i (y_i - \bar y)^2},$$

the fraction of variance explained. In matrix form with several predictors, $\hat{\boldsymbol\beta} = (X^\top X)^{-1} X^\top \mathbf{y}$ — which exists exactly when $X^\top X$ is invertible (§5.3).

Where the Analogy Breaks

R^2 never decreases when predictors are added, even meaningless ones, so it cannot be used to choose between models of different size. It also says nothing about whether a linear model was appropriate: data lying on a perfect parabola can produce a respectable R^2 from a straight line that is wrong everywhere. And regression measures association, not causation — the ice cream and drowning of §6.10 would fit beautifully.

Why It Matters Here — link to the thesis

R^2 is the diagnostic reported for the TimeGAN autoencoder: reconstruction R^2 above 0.8 is required before Phase 3 is allowed to proceed, because a lossy embedding means the adversarial stage trains on noise. The smoke run reached +0.995. Regression also underlies the ARCH test, which regresses squared residuals on their own lags, and the discriminative classifier, whose logistic form is regression with a non-linear link.

Chapter 8

Time Series Analysis

Before and After

Before this chapter you need: log returns (§2.4), random variables, expectation and variance (§6.5–§6.8), covariance and correlation (§6.10), independence (§6.9), QMLE (§7.4), regression and R^2 (§7.10).
After it you will understand: what a stochastic process is, why a random walk is non-stationary while its increments are not, how AR and MA models build dependence, what GARCH adds, and why two respected literatures report Hurst exponents of 0.6 and 0.1 for the same quantity.

A **time series** is a sequence of measurements indexed by time: $r_1, r_2, \ldots, r_T$. Market returns form a time series. This chapter builds the vocabulary for describing their structure, then applies it to the two stylised facts that matter most — near-zero return autocorrelation alongside strongly persistent volatility.

8.1 Stochastic Processes

Situation — the everyday picture

A thermometer outside your window records a temperature every hour, for ever. Any single hour's reading is a random variable. But you do not have one random variable — you have an endless, ordered family of them, and the ordering is the whole point.

Problem — where it breaks or why it matters

Chapter 6 described one random variable at a time, and §6.9 handled pairs. A market history is neither. It is thousands of random variables in a fixed order, where the relationship between neighbours is the object of study, and it needs its own vocabulary.

Worked Arithmetic — concrete numbers

One realisation versus the ensemble. Toss a coin five times, writing $+1$ for heads and -1 for tails: $+1, -1, -1, +1, +1$. That sequence is one *realisation* of the process. Toss again and you get a different one.
The process is the rule generating them; a realisation is one draw from it.

The awkward part. With markets we observe exactly one realisation. There is only one history of BOVESPA. Yet we want to speak of $\mathbb{E}[r_t]$ — an average over the ensemble of histories that might have occurred.

The bridge. Suppose the process is stationary and we average along our single path:

$$\bar{r} = \frac{1}{T} \sum_{t=1}^{T} r_t.$$

Under conditions given in §8.4, this time average converges to the ensemble average $\mathbb{E}[r_t]$. That property — **ergodicity** — is what licenses learning anything at all from one history.

Why it can fail. If a market's mean return shifted permanently partway through the sample, the time average would converge to a blend of two regimes and estimate neither. MOEX in February 2022 is exactly this worry made concrete.

Formal Name — the mathematics

A **stochastic process** is a collection of random variables $\{X_t\}_{t \in T}$ indexed by a set T, all defined on a common sample space. Here $T = \{1, 2, \ldots\}$ and the process is **discrete-time**. A single observed sequence is a **realisation** or **sample path**.
The process is **ergodic** (in the mean) if time averages along one path converge to the corresponding ensemble expectation.

Where the Analogy Breaks

Ergodicity is an assumption, not a theorem about markets, and it is not testable from a single path — which is the only kind we have. Every statistical statement in this thesis inherits that assumption. It is most suspect exactly where it matters most: during regime changes, when the process generating today's returns is arguably not the one that generated last decade's.

Why It Matters Here — link to the thesis

The train/validation/test split in this project is *temporal*, never shuffled: 80% earliest, then 10%, then 10% latest. Shuffling would let the model see the future, and it would also destroy exactly the ordering that makes this a process rather than a bag of numbers.

8.2 White Noise

Situation — the everyday picture

Static between radio stations. There is no tune, no rhythm, nothing in one instant that hints at the next. It is the auditory picture of "no structure".

Problem — where it breaks or why it matters

Before we can say a series *has* structure, we need the reference case that has none. Every model in this chapter is built by adding structure to this baseline, and every diagnostic asks whether what remains after modelling has returned to it.

Worked Arithmetic — concrete numbers

A white-noise sequence with $\sigma = 2$: mean zero, no correlation between any two distinct times.

t	1	2	3	4	5	6
X_t	1.4	−2.3	0.7	−0.2	3.1	−1.8

Lag-1 pairs: $(1.4, -2.3), (-2.3, 0.7), (0.7, -0.2), (-0.2, 3.1), (3.1, -1.8)$.
Knowing the first entry of a pair tells you nothing about the second. The theoretical autocorrelation is $\rho(k) = 0$ for every $k \geq 1$.

What "zero" looks like in a finite sample. A sample autocorrelation computed from T observations of white noise is not exactly zero; it scatters about zero with standard deviation roughly $1/\sqrt{T}$. At $T = 4{,}953$ — BOVESPA's length — that is

$$\frac{1}{\sqrt{4953}} = 0.0142,$$

so the usual $\pm 2/\sqrt{T}$ significance band is about ± 0.028.
Compare the measured $\hat{\rho}(r, 1) = +0.03$ from §8.5. It sits just at the edge of the band — which is the precise sense in which raw returns are "approximately unpredictable".

Formal Name — the mathematics

$\{\varepsilon_t\}$ is **white noise**, written $\varepsilon_t \sim \mathrm{WN}(0, \sigma^2)$, if

$$\mathbb{E}[\varepsilon_t] = 0, \qquad \mathrm{Var}(\varepsilon_t) = \sigma^2, \qquad \mathrm{Cov}(\varepsilon_t, \varepsilon_s) = 0 \text{ for } t \neq s.$$

It is **independent** white noise if the ε_t are mutually independent, and **Gaussian** white noise if additionally normal.
These are three distinct conditions of increasing strength.

Where the Analogy Breaks

Uncorrelated is not independent — the point already made in §6.10. Financial returns are close to white noise in the correlation sense and emphatically not independent: $|r_t|$ is strongly autocorrelated. A generator that produced genuine independent white noise would match the return ACF perfectly and fail every volatility-clustering metric, and the evaluation framework in this thesis is built to catch precisely that.

Why It Matters Here — link to the thesis

The $\pm 2/\sqrt{T}$ band is what the ACF panels in the stylised-facts figure draw as dashed lines. It is the reference against which "the returns are uncorrelated but their absolute values are not" becomes a measurement rather than an impression.

8.3 The Random Walk

Situation — the everyday picture

A drunk leaves a lamppost. Each second he steps one metre, left or right at random. After an hour, where is he? On average, back at the lamppost — but you would not bet on finding him there.

Problem — where it breaks or why it matters

The steps are white noise, yet the position is nothing like white noise: it wanders, it trends for long stretches, and its variance grows without limit. Understanding how a well-behaved increment produces a badly behaved level is exactly the relationship between returns and prices.

Worked Arithmetic — concrete numbers

Start at $S_0 = 0$, steps ± 1 with equal probability.
One realisation: steps $+1, +1, -1, +1, -1, -1, -1, +1$.

t	0	1	2	3	4	5	6	7	8
S_t	0	1	2	1	2	1	0	-1	0

Mean. $\mathbb{E}[S_t] = 0$ for every t — the steps are symmetric.
Variance. Steps are independent, so variances add (§6.8). Each step has variance 1, so

$$\operatorname{Var}(S_t) = t.$$

The standard deviation is $\sqrt{t}$: after 100 steps, typically 10 metres from the lamppost; after 10,000 steps, 100 metres.

Why this is not stationary. Condition 2 of stationarity demands constant variance. Here $\operatorname{Var}(S_t) = t$ grows for ever. The *level* fails; the *increments* $S_t - S_{t-1}$ are white noise and pass.

The financial translation. With $\ln P_t = \ln P_{t-1} + r_t$, the log price is a random walk whose increments are the log returns. Prices are non-stationary; returns are approximately stationary. This is the entire reason the thesis models r_t and never P_t.

Formal Name — the mathematics

A **random walk** is

$$S_t = S_{t-1} + \varepsilon_t = S_0 + \sum_{i=1}^{t} \varepsilon_i, \qquad \varepsilon_t \sim \mathrm{WN}(0, \sigma^2).$$

Then $\mathbb{E}[S_t] = S_0$ and $\operatorname{Var}(S_t) = t\sigma^2$, so the process is non-stationary. A **random walk with drift** adds a constant: $S_t = \mu + S_{t-1} + \varepsilon_t$, giving $\mathbb{E}[S_t] = S_0 + \mu t$.
A series that becomes stationary after differencing once is **integrated of order 1**, written $I(1)$.

Where the Analogy Breaks

The random walk is the textbook model of an efficient market and it gets the first moment right while getting the interesting part wrong. It has constant step variance, so it cannot produce volatility clustering, and with Gaussian steps it cannot produce heavy tails. Real returns are approximately uncorrelated — consistent with the walk — and strongly heteroskedastic and fat-tailed, which the walk forbids. It is the right null and a poor model.

Why It Matters Here — link to the thesis

Reconstructing a synthetic price path from generated returns is $P_t = P_0 \exp(\sum_i r_i)$ — accumulating a random walk. The $\sqrt{t}$ growth also explains why the post-generation guard checks $|\mathrm{ACF}(1)| < 0.3$ on the returns: a generator that emits a smooth wandering *path* instead of a return series shows up immediately as high first-order autocorrelation.

8.4 Stationarity

Situation — the everyday picture

The temperature in Barcelona fluctuates around 18°C year-round (ignoring seasons). The sea level has been rising 3 mm/year for decades. The temperature process is *stationary* — its statistical properties do not change over time. Sea level is *non-stationary* — its mean is drifting.

Problem — where it breaks or why it matters

If a series is non-stationary, any statistic computed from it depends on *when* it was measured, and is therefore useless for describing the future. Worse, standard inference breaks: two independent trending series will show a high correlation purely because both are going up.

Worked Arithmetic — concrete numbers

A non-stationary series. Prices $100, 102, 105, 103, 108, 112$.
First half mean: $(100 + 102 + 105)/3 = 102.3$. Second half: $(103 + 108 + 112)/3 = 107.7$. The mean has moved by 5.4.

Its log returns.

$$\ln(102/100) = 0.0198, \quad \ln(105/102) = 0.0290, \quad \ln(103/105) = -0.0192,$$

$$\ln(108/103) = 0.0474, \quad \ln(112/108) = 0.0364.$$

First half mean: $(0.0198 + 0.0290)/2 = 0.0244$. Second half: $(0.0474 + 0.0364)/2 = 0.0419$. Not identical — five observations cannot be — but both hover near zero rather than marching upward. Differencing removed the trend.

The distinction that matters. Stationarity does not mean constant. It means the *rules* are constant. A stationary series can be violent and erratic; what it cannot do is have those characteristics systematically change.

Formal Name — the mathematics

A process $\{X_t\}$ is **weakly stationary** if:

1. $\mathbb{E}[X_t] = \mu$, constant in t;
2. $\mathrm{Var}(X_t) = \sigma^2 < \infty$, constant in t;
3. $\mathrm{Cov}(X_t, X_{t+k}) = \gamma(k)$, depending on the lag k alone.

Strict stationarity demands that the entire joint distribution of any block be invariant to time shifts. Strict plus finite variance implies weak; weak does not imply strict.
Raw prices P_t are non-stationary. Log returns r_t are approximately weakly stationary.

Where the Analogy Breaks

Weak stationarity constrains only the first two moments, and returns satisfy it only *approximately*. Their higher moments shift with regime — the MOEX 2022 suspension being the clearest case in this dataset — and their conditional variance is explicitly time-varying, which is what GARCH models. The assumption is a working approximation adopted knowingly, not a verified property.

Why It Matters Here — link to the thesis

Every evaluation metric assumes it: Wasserstein, ACF-MAE, the Hurst exponent and the discriminative AUC are all computed as though the distribution of r_t is the same in train and test. The temporal 80/10/10 split without shuffling is the correct design under that assumption, and the walk-forward protocol exists to test how badly it is violated in practice.

8.5 The Autocorrelation Function

Situation — the everyday picture

You write down today's temperature and tomorrow's for 365 days and plot the pairs (x_t, x_{t+1}). Because temperature is persistent, the scatter slopes upward: the correlation is around +0.8. Do the same for today and next week and it is lower, maybe +0.4. After a month it is near zero.

Problem — where it breaks or why it matters

For market returns the correlation between r_t and r_{t+1} is near zero at every lag — markets are approximately unpredictable. But the correlation between $|r_t|$ and $|r_{t+1}|$, measuring how wild each day was, is clearly positive out to a hundred lags or more. One quantity, two completely different answers depending on the transform applied first.

Worked Arithmetic — concrete numbers

BOVESPA 20-year data, $n = 4{,}953$. Significance band $\pm 2/\sqrt{n} = \pm 0.028$ (§8.2).

Lag k	$\hat\rho(r,k)$	$\hat\rho(\lvert r\rvert,k)$	$\hat\rho(r^2,k)$
1	+0.03	+0.19	+0.17
5	−0.01	+0.14	+0.12
20	+0.01	+0.09	+0.08
50	≈ 0	+0.05	+0.04

Column 1 hovers inside the band — raw returns are essentially uncorrelated, so direction is unpredictable.
Columns 2 and 3 start well above the band and decay slowly. At lag 50 the value +0.05 is still nearly twice the band. Volatility is strongly predictable.

The decay rate matters. From lag 1 to lag 50, $\hat{\rho}(|r|)$ falls from 0.19 to 0.05 — a factor of about 4 across 50 lags. Exponential decay at rate ϕ would give $\phi^{50} = 0.05/0.19$, so $\phi = 0.974$: extremely slow. A hyperbolic decay k^{-d} fits comparably, and distinguishing the two is precisely the long-memory question of §8.11.

Formal Name — the mathematics

The **sample autocorrelation** at lag k is

$$\hat{\rho}(k) = \frac{\sum_{t=1}^{T-k}(x_t - \bar{x})(x_{t+k} - \bar{x})}{\sum_{t=1}^{T}(x_t - \bar{x})^2},$$

and the **autocorrelation function** is the sequence $\{\hat{\rho}(1), \hat{\rho}(2), \ldots\}$, with $\hat{\rho}(0) = 1$ always. The evaluation metric **ACF-MAE** compares real and synthetic ACF vectors:

$$\text{ACF-MAE} = \frac{1}{K}\sum_{k=1}^{K}\left|\hat{\rho}_{\text{real}}(k) - \hat{\rho}_{\text{syn}}(k)\right|, \qquad K = 20,$$

computed separately on r_t, $|r_t|$ and r_t^2.

Where the Analogy Breaks

ACF-MAE averages across lags, so it can hide a systematic error at one lag behind accuracy elsewhere. Its compensating virtue is that it never saturates: unlike the ARCH-LM p-value of §8.10, it stays informative on real fat-tailed data instead of collapsing to a constant.

Why It Matters Here — link to the thesis

ACF-MAE on $|r_t|$ and r_t^2 is the primary ranking signal for volatility clustering in the temporal family. The shuffled control scores 0.043 on raw returns and 0.076 on absolute returns — correctly penalised, because shuffling destroys exactly this structure while leaving every marginal quantity untouched.

8.6 Partial Autocorrelation

Situation — the everyday picture

Grandparent, parent, child heights all correlate. But the grandparent–child correlation is largely inherited *through* the parent. To find what the grandparent adds on its own, you must hold the parent fixed.

Problem — where it breaks or why it matters

The ACF at lag 2 conflates two things: a genuine direct link from $t-2$ to t, and the indirect route through $t-1$. Choosing how many lags a model needs requires separating them.

Worked Arithmetic — concrete numbers

Suppose a series follows $X_t = 0.6X_{t-1} + \varepsilon_t$ exactly — each value depends on the previous one only.

Its ACF. Correlation propagates down the chain:

$$\rho(1) = 0.6, \quad \rho(2) = 0.6^2 = 0.36, \quad \rho(3) = 0.6^3 = 0.216.$$

The ACF is non-zero at every lag, decaying geometrically — which might wrongly suggest that lags 2 and 3 matter directly.

Its PACF. Ask what lag 2 contributes *given* lag 1. The answer is nothing: once X_{t-1} is known, X_{t-2} is irrelevant by construction. So

$$\phi_{11} = 0.6, \qquad \phi_{22} = 0, \qquad \phi_{33} = 0.$$

The PACF cuts off sharply after lag 1 and announces the model order.

The diagnostic pattern. For an AR(p) process the PACF cuts off after lag p while the ACF decays gradually. For an MA(q) process it is the mirror image: the ACF cuts off after lag q and the PACF decays. Reading the two together identifies the model.

Formal Name — the mathematics

The **partial autocorrelation** at lag k, written ϕ_{kk}, is the correlation between X_t and X_{t-k} after linearly removing the effect of the intervening $X_{t-1}, \ldots, X_{t-k+1}$.
Equivalently, it is the coefficient on X_{t-k} in the regression

$$X_t = \phi_{k1}X_{t-1} + \cdots + \phi_{kk}X_{t-k} + \varepsilon_t.$$

Where the Analogy Breaks

The PACF removes *linear* effects only, and identifies model order only if the process really is linear. Financial returns are close to uncorrelated, so both their ACF and PACF are near zero at all lags — and neither reveals the volatility structure, which lives entirely in the transformed series $|r_t|$ and r_t^2. On returns, the classical identification toolkit is silent.

Why It Matters Here — link to the thesis

That silence is the argument for the models in this thesis. If the PACF of r_t identified a low-order linear model, a GAN would be unnecessary. The structure worth learning is non-linear and lives in the conditional variance, which is where GARCH and the generative models both operate.

8.7 AR, MA and ARMA Models

Situation — the everyday picture

Two ways to explain today's mood. Either it carries over from yesterday's mood (an echo of the state), or it is the accumulation of the last few days' events (an echo of the shocks). Both produce persistence, by different mechanisms.

Problem — where it breaks or why it matters

White noise has no memory; a random walk has infinite memory. Almost everything real lies between, and we need a family of models that spans the middle with a small number of parameters.

Worked Arithmetic — concrete numbers

AR(1): memory in the level. $X_t = 0.5X_{t-1} + \varepsilon_t$. Start $X_0 = 4$ with shocks $\varepsilon = 1, -2, 0.5$:

$$X_1 = 0.5(4) + 1 = 3, \quad X_2 = 0.5(3) - 2 = -0.5, \quad X_3 = 0.5(-0.5) + 0.5 = 0.25.$$

The starting value's influence halves each step: $4 \to 2 \to 1 \to 0.5$ in the absence of shocks. Stationary because $|0.5| < 1$; at coefficient 1 it becomes the random walk.

MA(1): memory in the shocks. $X_t = \varepsilon_t + 0.8\varepsilon_{t-1}$. With the same shocks and $\varepsilon_0 = 0$:

$$X_1 = 1 + 0.8(0) = 1, \quad X_2 = -2 + 0.8(1) = -1.2, \quad X_3 = 0.5 + 0.8(-2) = -1.1.$$

Each shock influences exactly two periods, then vanishes completely. The ACF is non-zero at lag 1 and exactly zero from lag 2 on — a hard cut-off.

Contrast. AR memory decays for ever but geometrically; MA memory is absolute but finite. ARMA combines both.

Formal Name — the mathematics

AR(p):

$$X_t = c + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t.$$

MA(q):

$$X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}.$$

ARMA(p, q) contains both sets of terms.
An AR(1) is stationary when $|\phi_1| < 1$; every MA(q) with finite coefficients is stationary. A stationary AR can be rewritten as an infinite MA and vice versa (Wold's theorem), so the choice between them is about parsimony, not expressive power.

Where the Analogy Breaks

ARMA models the conditional *mean* and assumes the variance is constant. For market returns the conditional mean is essentially zero and all the structure is in the variance, so an ARMA fit to r_t finds almost nothing — correctly, and uselessly. GARCH is precisely the ARMA idea transplanted onto σ_t^2, which is where the signal actually is.

Why It Matters Here — link to the thesis

This is why no ARMA model appears in the pipeline. The absence of linear structure in r_t is a stylised fact to be *reproduced*, not exploited, and `acf_returns_mae` scores a generator on whether its synthetic returns are as unpredictable as the real ones.

8.8 ARIMA and Differencing

Situation — the everyday picture

A thermometer reading rises all afternoon. Reporting "the average temperature" is unhelpful. Reporting "it rises about half a degree an hour" describes something stable, because you switched from the level to the change.

Problem — where it breaks or why it matters

ARMA requires stationarity, and much real data trends. Rather than abandon the machinery, we transform the data until the assumption holds — and then must be honest about what the transformation cost.

Worked Arithmetic — concrete numbers

Series: $10, 12, 15, 17, 20, 22$ — clearly trending, so non-stationary.

First difference: $2, 3, 2, 3, 2$. Mean 2.4, no trend. Stationary.

Second difference: $1, -1, 1, -1$. Also stationary, but the alternating sign is an artefact — differencing more than necessary injects negative correlation that was not in the data.

The financial case. Log prices $\ln P_t$ are $I(1)$: differencing once gives log returns, which are stationary. Differencing twice would be over-differencing and would introduce spurious MA structure.

Why differencing costs something. The first difference discards the level entirely. After modelling returns you know how the price moves but not where it is, which is why any synthetic price path in this thesis needs an externally supplied starting value P_0.

Formal Name — the mathematics

ARIMA(p, d, q) applies ARMA(p, q) to the series differenced d times. With $\Delta X_t = X_t - X_{t-1}$, the model is ARMA$(p, q)$ fitted to $\Delta^d X_t$.
A series requiring d differences to become stationary is $I(d)$. Log prices are $I(1)$; log returns are $I(0)$.
The **augmented Dickey–Fuller** test has a null of a unit root (non-stationarity), so rejecting it supports stationarity.

Where the Analogy Breaks

Differencing removes trends and amplifies noise, and repeated differencing can manufacture structure that was never present. Nor is it the only route to stationarity: a deterministic trend is better removed by regression, and using differences on such a series over-differences it. Choosing d badly is a real failure mode, not a formality.

Why It Matters Here — link to the thesis

Working in log returns *is* the $d = 1$ step, taken once at the data-loading stage and never revisited. Everything downstream — every model, metric and figure — operates on the $I(0)$ series.

8.9 GARCH: Modelling the Conditional Variance

Situation — the everyday picture

Think of the market as a mood thermometer. On quiet days it reads "calm" and barely moves; on anxious days "stormy", and moves a lot. The crucial point: today's mood is heavily influenced by yesterday's. If yesterday was stormy, today probably is too.

Problem — where it breaks or why it matters

A constant-variance model assigns the same probability to a 2% move on every day. Real markets have quiet stretches where 0.5% is a big day and turbulent stretches where 2% is routine. Such a model underestimates risk in the storm and overestimates it in the calm — simultaneously, and by large factors.

Worked Arithmetic — concrete numbers

GARCH$(1,1)$ fitted to BOVESPA, 20-year:

$$\omega = 0.000002, \qquad \alpha = 0.08, \qquad \beta = 0.91.$$

Long-run variance. Setting $\sigma_t^2 = \sigma_{t-1}^2 = \sigma_\infty^2$:

$$\sigma_\infty^2 = \frac{\omega}{1-\alpha-\beta} = \frac{0.000002}{1-0.99} = 0.0002,$$

so $\sigma_\infty = \sqrt{0.0002} = 1.41\%/\text{day}$. ✓
Persistence. $\alpha + \beta = 0.99$: shocks decay by only 1% per day.

One step after a 3% move ($\varepsilon_{t-1}^2 = 0.0009$, starting from the long-run level):

$$\sigma_t^2 = 0.000002 + 0.08(0.0009) + 0.91(0.0002) = 0.000256,$$

giving $\sigma_t = 1.60\%$ — 13% above the long-run level.

Half-life of a shock. Persistence 0.99 per day means a shock decays to half after n days where $0.99^n = 0.5$:

$$n = \frac{\ln 0.5}{\ln 0.99} = \frac{-0.693}{-0.01005} = 69 \text{ days}.$$

More than three trading months. That slow decay is volatility clustering expressed as a number.

Formal Name — the mathematics

GARCH$(1,1)$ (Bollerslev 1986):

$$\begin{aligned} r_t &= \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \qquad z_t \overset{iid}{\sim} F(0,1) \\ \sigma_t^2 &= \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2. \end{aligned}$$

Constraints: $\omega > 0$, $\alpha, \beta \geq 0$, and $\alpha + \beta < 1$ for covariance stationarity. Persistence is $\alpha + \beta$; the long-run variance is $\omega/(1-\alpha-\beta)$.
Fitting uses Gaussian QMLE (§7.4): $z_t \sim \mathcal{N}(0,1)$ is assumed even though it is false, because that keeps (ω, α, β) consistent while leaving genuine excess kurtosis visible in the residuals.

Where the Analogy Breaks

GARCH(1, 1) imposes a specific functional form and misses several documented features: the leverage effect (negative returns raise volatility more than positive ones of equal size), jumps, and volatility persistence that decays hyperbolically rather than geometrically. In this thesis GARCH is a diagnostic instrument, never a generative model.

Why It Matters Here — link to the thesis

GARCH does two jobs here. First, the conditional heavy-tail metric: fit by Gaussian QMLE, extract $\hat{z}_t = \varepsilon_t/\hat{\sigma}_t$, measure residual excess kurtosis, and compare real against synthetic (`resid_kurtosis_diff`). Second, downstream utility: fit on synthetic data, apply the parameters to filter real data, and backtest the resulting conditional VaR — the TSTR protocol of Chapter 16.

8.10 The ARCH Test

Situation — the everyday picture

You want a yes/no answer on whether a series has volatility clustering. The ARCH-LM test provides one: it checks whether past squared returns predict the current squared return.

Problem — where it breaks or why it matters

A test is only useful if it can come out either way on the data you have. This one cannot. On real fat-tailed returns it returns the same verdict for a genuine series and for a shuffled deck, and a test that always says the same thing measures nothing.

Worked Arithmetic — concrete numbers

The test regresses r_t^2 on its own p lags and forms $LM = nR^2$, which is $\chi^2(p)$ under the null of no ARCH.

Series	LM	p-value
Simulated GARCH, $n = 1{,}200$	56–125	≈ 0.0
Real BRICS	≈ 51	0.0
Real BRICS, *shuffled*	≈ 67	0.0

The problem in one line. The shuffled series has no volatility clustering whatever — shuffling destroyed it — yet the test rejects just as emphatically. The difference in p-values is $|\Delta p| = 0.000000$.

Why. The auxiliary regression is dominated by a handful of enormous squared returns. Shuffling leaves the *set* of squared returns untouched, so those extreme values are still present, still leveraging the regression, and still producing a large R^2. The statistic responds to the heavy tails, not to the ordering it was meant to detect.

Formal Name — the mathematics

Engle's (1982) **ARCH-LM** test fits

$$r_t^2 = \gamma_0 + \sum_{j=1}^{p} \gamma_j r_{t-j}^2 + \eta_t$$

and forms $LM = nR^2_{\text{aux}}$, distributed $\chi^2(p)$ under H_0: $\gamma_1 = \cdots = \gamma_p = 0$.

Where the Analogy Breaks

The test performs well in simulation — around 99% correct on simulated GARCH with Gaussian innovations — and fails on real data. That gap is a warning about simulation-based validation generally: a method verified only on data matching its own assumptions has been verified only against itself.

Why It Matters Here — link to the thesis

Both `arch_pvalue_diff` and `arch_stat_diff` sit in the descriptive group: computed and displayed for transparency, excluded from every composite rank. ACF-MAE on $|r_t|$ carries the volatility-clustering signal instead, because it degrades gracefully rather than saturating.

8.11 The Hurst Exponent and Long Memory

Situation — the everyday picture

The Nile's floods are not independent from year to year. A decade of high floods tends to be followed by another. Hurst, employed to size a dam, found that the range of cumulative flows grew faster with the record length than independence would allow.

Problem — where it breaks or why it matters

An ACF that decays slowly and one that decays geometrically look similar over twenty lags but imply very different behaviour over hundreds. Distinguishing them needs a statistic sensitive to the decay *rate* across scales, not to any single lag.

Worked Arithmetic — concrete numbers

The rescaled range R/S over a window of n observations scales as

$$\mathbb{E}[R_n/S_n] \approx c\, n^H.$$

Taking logarithms turns the exponent into a slope (§2.5):

$$\ln \mathbb{E}[R_n/S_n] = \ln c + H \ln n,$$

so H is estimated by regressing $\ln(R/S)$ on $\ln n$ across window sizes.

Reference values. $H = 0.5$ for independent data. For BRICS $|r_t|$, $H \approx 0.57$–0.65.

Does that beat chance? The estimator's standard deviation is about 0.022 at $n = 1{,}000$ and 0.004 at $n = 4{,}000$.
At 5-year data ($n \approx 995$): $\hat{H} = 0.60$ against 0.5 is $(0.60 - 0.50)/0.022 = 4.5$ standard errors — significant but with wide uncertainty.

At 20-year data ($n \approx 4{,}960$): $(0.60 - 0.50)/0.004 = 25$ standard errors. Unambiguous.

This is one of the concrete payoffs of extending the dataset from 5 to 20 years.

Formal Name — the mathematics

The **Hurst exponent** $H \in (0, 1)$ characterises long-range dependence:

- $H = 0.5$: no long-range dependence (random walk, Brownian motion);
- $H > 0.5$: persistent — long stretches above or below average;
- $H < 0.5$: anti-persistent — reversals more often than chance.

For $H > 0.5$ the autocorrelations decay hyperbolically as k^{2H-2} and are not summable, which is the formal content of **long memory**. Applied to $|r_t|$, $H > 0.5$ is stylised fact 4.

Where the Analogy Breaks

R/S is biased in small samples and confounded by short-range dependence and non-stationarity: a series with a trend, or with strong AR structure and no long memory at all, can still produce $\hat{H} > 0.5$. The next section takes this seriously, because the literature does not agree on what H even is for volatility.

Why It Matters Here — link to the thesis

`hurst_diff` $= |H_{\text{real}} - H_{\text{syn}}|$ belongs to the temporal family. The shuffled control scores about 0.052 — correctly penalised, since permutation destroys long memory while preserving every marginal quantity. A generator should reproduce $H \approx 0.6$ on the absolute returns of each market.

8.12 R/S versus DFA, and the Rough-Volatility Tension

Situation — the everyday picture

Two teams measure the same coastline. One reports it is smooth and gently curving; the other that it is jagged and irregular. Both are competent, both used defensible instruments — and both are right, because one walked it with a metre rule and the other with a kilometre wheel.

Problem — where it breaks or why it matters

The literature reports Hurst exponents for volatility of about 0.6 and about 0.1. These are not small disagreements to be split; they describe opposite behaviour — persistent versus violently anti-persistent. Anyone using H as an evaluation metric needs to know which claim their number belongs to.

Worked Arithmetic — concrete numbers

Two estimators.
R/S (Hurst 1951; Mandelbrot–Wallis 1969) uses the range of the cumulative deviations divided by the standard deviation. It assumes the series has no trend, and a trend inflates the range without inflating S, biasing $\hat{H}$ upward.
Detrended fluctuation analysis (Peng et al. 1994) fits and removes a polynomial within each window before measuring fluctuation size. It is therefore robust to trends that would fool

R/S.

Why they disagree. On a series with a linear trend and no long memory at all, R/S can report $\hat{H}$ well above 0.5 while DFA correctly returns ≈ 0.5. On BRICS volatility, both exceed 0.5, but R/S tends to give the larger figure. A reported H without its estimator named is not interpretable.

The larger tension. Two literatures:

Claim	Source	Measured on
$H > 0.5$ (long memory)	Ding–Granger–Engle (1993)	daily $\lvert r_t \rvert$, low frequency
$H \approx 0.1$ (rough)	Gatheral–Jaisson–Rosenbaum (2018)	intraday realised volatility

They are not measuring the same thing. Ding, Granger and Engle describe how slowly correlations in daily absolute returns decay over *long* horizons — months. Gatheral and co-authors describe how irregular the volatility path is over *short* ones — minutes. A process can be locally jagged and globally persistent, exactly as a coastline is.

And the roughness may be partly artefact. Cont and Das (2024) show that estimating H from discretely sampled realised volatility is biased downward: realised volatility is a noisy proxy for the latent path, and the estimation noise is itself close to uncorrelated, which drags the estimate toward roughness. Part of the gap between 0.6 and 0.1 is a genuine difference of scale; part is a measurement artefact of the high-frequency estimator.

Formal Name — the mathematics

Rescaled range. For a window of length n with mean $\bar{x}$, let $Z_k = \sum_{i=1}^{k}(x_i - \bar{x})$ be the cumulative deviation. Then

$$R_n = \max_k Z_k - \min_k Z_k, \qquad S_n = \mathrm{sd}(x_1, \ldots, x_n),$$

and H is the slope of $\ln(R_n/S_n)$ against $\ln n$.
DFA. Within each window, fit and subtract a polynomial trend from the integrated series, take the root-mean-square of the residuals as $F(n)$, and estimate H from $F(n) \sim n^H$.
Rough volatility posits $\log \sigma_t$ follows a fractional Brownian motion with $H \approx 0.1$, so the volatility path is far less regular than Brownian motion.

Where the Analogy Breaks

It is tempting to declare one literature wrong. Neither is. What is wrong is treating H as a single scale-free property of "volatility". The estimate depends on the estimator, the sampling frequency, the proxy used for latent volatility, and the horizon. Quoting a Hurst exponent without all four is quoting an incomplete measurement.

Why It Matters Here — link to the thesis

This project computes H by R/S on daily $\lvert r_t \rvert$ — squarely in the Ding–Granger–Engle setting, where $H \approx 0.6$ is the right expectation. `hurst_diff` is therefore a low-frequency persistence metric and makes no claim about path roughness. Nothing in the evaluation would detect a generator that matched daily persistence while producing an intraday path of the wrong regularity — which is a real limitation, and the honest place to record it is here rather than in the results.

8.13 Multivariate Time Series

Situation — the everyday picture

Five BRICS markets trade simultaneously. When one falls sharply, the others usually follow within a day. Modelling each in isolation discards that, and the discarded part is exactly what a portfolio owner cares about.

Problem — where it breaks or why it matters

This thesis trains one model per market and evaluates each separately. That is a deliberate scope decision, and stating what it excludes is part of stating what the results mean.

Worked Arithmetic — concrete numbers

Why cross-market structure matters. Suppose two markets each have daily volatility 1.5% and you hold half of each. If independent, the portfolio volatility is

$$\sqrt{0.5^2(1.5)^2 + 0.5^2(1.5)^2} = 1.5/\sqrt{2} = 1.06\%.$$

If perfectly correlated, it is 1.5%. At correlation 0.5:

$$\sqrt{0.25(2.25) + 0.25(2.25) + 2(0.25)(0.5)(2.25)} = \sqrt{1.6875} = 1.30\%.$$

The same two marginal distributions give portfolio risk anywhere from 1.06% to 1.50% — a 42% range — depending entirely on dependence.

The consequence for this thesis. Five separately trained generators, each perfectly reproducing its own market, would jointly reproduce nothing about co-movement. Every result here is a per-market result.

Formal Name — the mathematics

A **vector autoregression** VAR(p) models k series jointly:

$$\mathbf{X}_t = \mathbf{c} + A_1\mathbf{X}_{t-1} + \cdots + A_p\mathbf{X}_{t-p} + \boldsymbol{\varepsilon}_t,$$

where each A_i is a $k \times k$ matrix (§5.2) and $\boldsymbol{\varepsilon}_t$ has covariance matrix Σ. The off-diagonal entries of A_i carry cross-series lead–lag effects; those of Σ carry contemporaneous co-movement. Multivariate GARCH extends the same idea to a time-varying Σ_t, at the cost of a parameter count growing like k^2 or worse.

Where the Analogy Breaks

Multivariate models are expensive in exactly the way that matters: parameters grow quadratically in the number of series while data does not. And linear cross-correlation is the wrong tool for the case of interest, since markets couple far more strongly during crashes than in calm periods — tail dependence, which correlation cannot express. Section 9.7 takes that up.

Why It Matters Here — link to the thesis

This is a stated scope boundary. The pipeline trains and evaluates one model per market with no cross-market mixing — a deliberate design choice, since pooling markets with different volatility regimes would confound the comparison. The consequence is that nothing here evaluates joint behaviour, and any portfolio-level claim would be unsupported by these results.

Chapter 9

Heavy Tails and Extreme Values

Before and After

Before this chapter you need: logarithms and log–log plots (§2.3, §2.5), integrals (§3.5), moments and their existence (§6.8), the law of large numbers and the CLT (§6.11–§6.12), estimators and consistency (§7.2), joint distributions (§6.9).
After it you will understand: the single rule $\mathbb{E}|X|^k < \infty \iff k < \alpha$ that organises this entire chapter, how the Hill estimator is derived and why it survives where kurtosis does not, why one degenerate series in this project produced $\hat{\alpha} = 31{,}581$, and what copulas add for the multivariate case.

Almost everything in this chapter follows from one rule about which moments exist. Section 9.1 states it; the remaining sections are consequences.

9.1 Moment Existence: The Organising Principle

Situation — the everyday picture

You are averaging the wealth of people in a room. Add a hundred ordinary people and the average barely moves. Add one billionaire and it leaps. Add a second and it leaps again. The average never settles, because the next arrival can always outweigh everyone already counted.

Problem — where it breaks or why it matters

Standard statistics assumes averages settle. When the underlying distribution has a heavy enough tail they do not — not slowly, but never — and every downstream quantity built on that average inherits the failure. Knowing exactly which averages are safe is therefore the first question, not a technicality.

Worked Arithmetic — concrete numbers

Suppose $\mathbb{P}(|X| > x) = x^{-\alpha}$ for $x \geq 1$. The density is $f(x) = \alpha x^{-\alpha-1}$, and the k-th moment is

$$\mathbb{E}[|X|^k] = \int_1^\infty x^k \cdot \alpha x^{-\alpha-1}\, dx = \alpha \int_1^\infty x^{k-\alpha-1}\, dx.$$

The integral $\int_1^\infty x^p\, dx$ converges only when $p < -1$. Here $p = k - \alpha - 1$, so convergence requires

$$k - \alpha - 1 < -1 \iff k < \alpha.$$

Concretely, at $\alpha = 3$:

$$\mathbb{E}[|X|^2] = 3 \int_1^\infty x^{-2}\, dx = 3\Big[-x^{-1}\Big]_1^\infty = 3 \quad \text{(finite)},$$

$$\mathbb{E}[|X|^4] = 3 \int_1^\infty x^{0}\, dx = 3\Big[x\Big]_1^\infty = \infty \quad \text{(divergent)}.$$

Applied to this project. BRICS tail indices run $\hat{\alpha} \in [2.54, 3.21]$:

k	Quantity	Needs $\alpha >$	Status across BRICS
1	mean	1	finite everywhere
2	variance	2	finite everywhere
3	skewness	3	finite in 1 of 5 markets
4	kurtosis	4	**infinite everywhere**

Every remaining section of this chapter is a consequence of the last row.

Formal Name — the mathematics

A random variable has a **power-law** (regularly varying) tail with **tail index** $\alpha > 0$ if

$$\mathbb{P}(|X| > x) \sim L(x)\, x^{-\alpha} \quad \text{as } x \to \infty,$$

where L is slowly varying. The governing rule is

$$\boxed{\mathbb{E}[|X|^k] < \infty \iff k < \alpha}$$

Smaller α means a heavier tail and fewer finite moments.

Where the Analogy Breaks

“Infinite” here is not a large number. It is the statement that the defining integral diverges, so the quantity does not exist. A sample kurtosis can always be computed — the arithmetic never fails — but with $\alpha < 4$ it is not an estimate of anything, because there is nothing for it to converge to. This distinction is the whole of §9.3.

Why It Matters Here — link to the thesis

This rule is why `tail_index_diff` is a ranked fidelity metric while `kurtosis_diff` and `skewness_diff` were demoted to descriptive-only. Ranking generators on an estimate of an infinite quantity is not a strict test; it is noise wearing a decimal point.

9.2 Power Laws

Situation — the everyday picture

City sizes. New York has 8 million people; the next few have 4M, 3M, 2M. Halve the population threshold and roughly twice as many cities qualify. The same doubling holds whether you start at a million or ten thousand — the rule has no characteristic scale.

Problem — where it breaks or why it matters

The normal distribution has exponentially decaying tails, which makes extreme events not merely unlikely but effectively impossible. Power-law tails decay polynomially, which is slow enough to make extremes a routine feature of a few thousand observations rather than a once-per-civilisation event.

Worked Arithmetic — concrete numbers

NIFTY50, 20-year, $n = 4{,}954$.
The top 5% of $|r_t|$: 248 observations, from 1.97% up to 15.91%. The Hill estimate on that tail is $\hat{\alpha} \approx 2.8$, so $\mathbb{P}(|r| > x) \approx Cx^{-2.8}$.

Scale-free, checked. Doubling the threshold multiplies the exceedance probability by $2^{-2.8} = 0.144$ — the same factor whether you double from 2% to 4% or from 4% to 8%. Under a Gaussian the analogous factor collapses:

$$\frac{\mathbb{P}(|Z| > 4)}{\mathbb{P}(|Z| > 2)} = \frac{6.3 \times 10^{-5}}{4.6 \times 10^{-2}} = 0.0014, \qquad \frac{\mathbb{P}(|Z| > 8)}{\mathbb{P}(|Z| > 4)} = \frac{1.2 \times 10^{-15}}{6.3 \times 10^{-5}} = 2 \times 10^{-11}.$$

The Gaussian ratio falls by eight orders of magnitude between the two doublings; the power-law ratio does not move at all. That is the difference between exponential and polynomial decay.

Straight lines. Taking logarithms of $\mathbb{P}(|X| > x) \approx Cx^{-\alpha}$:

$$\ln \mathbb{P}(|X| > x) = \ln C - \alpha \ln x,$$

a straight line of slope $-\alpha$ on log–log axes (§2.5). Straightness there is the visual signature of a power law and the slope reads off α directly.

Formal Name — the mathematics

For a power-law tail with index α, the **survival function** is

$$\bar{F}(x) = \mathbb{P}(|X| > x) \sim L(x)x^{-\alpha},$$

and L slowly varying means $L(cx)/L(x) \to 1$ for every $c > 0$ — so L contributes no scaling of its own.
Typical values: financial returns $\alpha \approx 2$–4; city sizes and word frequencies $\alpha \approx 1$ (Zipf); a Gaussian has no finite α, its tail decaying faster than any power.

Where the Analogy Breaks

The tail index is an *asymptotic* property, describing behaviour as $x \to \infty$. Any finite sample — even 20 years of daily data — contains only a few hundred genuinely extreme observations, and the estimator must draw a line between the power-law tail and the ordinary body of the distribution. Where that line falls is a judgement call, and §9.4 shows how much the answer depends on it.

Why It Matters Here — link to the thesis

`tail_index_diff` $= |\hat{\alpha}_{\text{real}} - \hat{\alpha}_{\text{syn}}|$ is the primary heavy-tail metric in the fidelity family, replacing `kurtosis_diff`. At $n \approx 4{,}960$ the Hill estimator has a standard deviation near 0.035 — precise enough to separate generators that reproduce the tail from those that do not.

9.3 Why Kurtosis Is Unreliable Here

Situation — the everyday picture

You want a single number for how extreme a dataset is, and kurtosis is the obvious candidate. You compute it on 250 days of BOVESPA and get 2.53. You compute it on 2,000 days and get 11.84. Neither is wrong, and that is the problem.

Problem — where it breaks or why it matters

An estimator is supposed to settle as data accumulates. This one moves further with every additional observation, so there is no sample size at which the answer becomes trustworthy — and §9.1 already said why.

Worked Arithmetic — concrete numbers

BOVESPA, two sample sizes:

Sample n	Sample kurtosis	Hill $\hat{\alpha}$
250	2.53	4.83 (sd 1.253)
2,000	11.84	3.11 (sd 0.316)

Kurtosis multiplied by 4.7 as n grew eightfold. It is diverging.
Hill fell from 4.83 **to** 3.11 **and its standard deviation shrank by a factor of four.** It is converging, and the movement is bias disappearing (§7.2).

Why kurtosis diverges. The estimator is

$$\hat{\kappa}_n = \frac{n^{-1}\sum_i (x_i - \bar{x})^4}{\left(n^{-1}\sum_i (x_i - \bar{x})^2\right)^2}.$$

The numerator is a sample mean of X^4. The law of large numbers (§6.11) requires $\mathbb{E}[X^4]$ to be finite, and at $\alpha < 4$ it is not. So the numerator has nothing to converge to and grows with n; each new record extreme raises it again.

Why Hill does not. Hill averages *log spacings* of order statistics. Under a power law these are exponentially distributed (§9.4), and the exponential distribution has finite moments of every order. The law of large numbers applies to them regardless of α.

Formal Name — the mathematics

When $\alpha < 4$, the population kurtosis $\mathbb{E}[(X-\mu)^4]/\sigma^4$ is undefined. The sample kurtosis $\hat{\kappa}_n$ does not converge; it grows without bound, its numerator behaving like a sum in the domain of attraction of a stable law with index $\alpha/4 < 1$.
The Hill estimator, by contrast, is consistent and asymptotically normal for any $\alpha > 0$, provided the number of order statistics k satisfies $k \to \infty$ and $k/n \to 0$.

Where the Analogy Breaks

"MOEX sample kurtosis = 65.70" is not a measurement of the population kurtosis, which is infinite. It is an artefact of this particular sample, driven by a handful of observations — above all 24 February 2022 at −33.3%. A different 20-year window could give 30 or 150 with equal legitimacy. The Hill estimate of $\hat{\alpha} \approx 2.7$ for the same series is stable across windows.

Why It Matters Here — link to the thesis

This is why `kurtosis_diff` and `skewness_diff` moved from the seven ranked fidelity metrics to the seven descriptive ones. It is also why the length guard in `FinancialMetrics.__init__` warns when the compared series differ in length: a diverging statistic makes any kurtosis comparison between unequal-length series biased by length alone, independently of the data.

9.4 The Hill Estimator, Derived

Situation — the everyday picture

You want the slope of a straight line but only trust the right-hand portion of your data. So you discard the left, keep the largest few hundred points, and fit using those alone.

Problem — where it breaks or why it matters

The tail index is defined by behaviour as $x \to \infty$, and any real sample stops well short. We need an estimator that uses only the largest observations, where the power law plausibly holds, and ignores the body where it does not.

Worked Arithmetic — concrete numbers

Step 1: the key transformation. Suppose $\mathbb{P}(X > x) = (x/u)^{-\alpha}$ for $x \geq u$. Let $Y = \ln(X/u)$. Then for $y > 0$,

$$\mathbb{P}(Y > y) = \mathbb{P}(X > ue^y) = \left(e^y\right)^{-\alpha} = e^{-\alpha y}.$$

So Y is **exponential** with rate α. The logarithm has converted a power law into an exponential — and exponentials are easy.

Step 2: estimate the rate. An exponential with rate α has mean $1/\alpha$. By the law of large numbers, averaging the Y values estimates $1/\alpha$, so

$$\hat{\alpha} = \frac{1}{\bar{Y}} = \frac{k}{\sum_{i=1}^{k} \ln(X_{(n-i+1)}/X_{(n-k)})}.$$

That is the Hill estimator. It is maximum likelihood (§7.3) for the exponential rate.

Step 3: a numerical check. Take five tail observations with threshold $u = 2$: values $2.2, 2.7, 3.3, 4.5, 8.1$.

$$\ln(2.2/2) = 0.0953, \quad \ln(2.7/2) = 0.3001, \quad \ln(3.3/2) = 0.5008,$$

$$\ln(4.5/2) = 0.8109, \quad \ln(8.1/2) = 1.3971.$$

Sum $= 3.1042$, mean $= 0.6208$, so

$$\hat{\alpha} = 1/0.6208 = 1.61.$$

Why moments do not matter. Each $\ln(X_{(i)}/u)$ is exponential, which has finite moments of every order. Whatever α is — even below 1, where X has no mean at all — the log spacings are perfectly well behaved. This is exactly the property kurtosis lacks.

The bias–variance trade-off in k. Small k uses only the most extreme points, where the power law holds best: low bias, high variance. Large k reaches into the body where the power law fails: low variance, high bias. This project uses $k = \lfloor 0.05n \rfloor$, so 248 points at $n = 4{,}954$.

Formal Name — the mathematics

Order the sample $X_{(1)} \leq \cdots \leq X_{(n)}$. Using the top k, the **Hill estimator** is

$$\hat{\alpha}_{\text{Hill}} = \left[\frac{1}{k} \sum_{i=1}^{k} \ln \frac{X_{(n-i+1)}}{X_{(n-k)}} \right]^{-1}.$$

It is consistent and asymptotically normal, with $\sqrt{k}(\hat{\alpha} - \alpha) \to \mathcal{N}(0, \alpha^2)$, provided $k \to \infty$ and $k/n \to 0$. The standard error is therefore approximately $\hat{\alpha}/\sqrt{k}$.
At $\hat{\alpha} = 3.11$ and $k = 248$: $3.11/\sqrt{248} = 0.197$.

Where the Analogy Breaks

Hill assumes a power-law tail. Applied to data without one — a Gaussian, say — it returns a number regardless, and that number is meaningless rather than merely large. It is also sensitive to k: a Hill plot of $\hat{\alpha}$ against k should show a stable plateau, and if it does not, the power-law assumption is suspect. Reporting a single $\hat{\alpha}$ without inspecting that stability is reporting one point from a curve.

Why It Matters Here — link to the thesis

Hill supplies `tail_index_diff`, the primary heavy-tail fidelity metric. Its independence from moment existence is exactly why it replaced kurtosis: it remains a genuine estimator on data where the fourth moment does not exist. The next section shows what happens when its assumption is violated outright.

9.5 When Hill Fails: the 31,581 Case

Situation — the everyday picture

A tape measure works well until you try to measure something with no length. The instrument does not announce the mistake — it returns a number, and the number looks like a measurement.

Problem — where it breaks or why it matters

A collapsed generator produces a nearly constant series. It has no tail at all. The Hill estimator, asked for the index of a tail that does not exist, returns a finite number, and that number then enters a ranking table alongside genuine estimates as though it were comparable.

Worked Arithmetic — concrete numbers

What happened in this project. During a BOVESPA smoke test, TimeGAN collapsed and produced a series with standard deviation 0.0006 against a real 0.0103. Hill returned

$$\hat{\alpha} = 31{,}581.$$

Why. The estimator divides by the mean log spacing

$$\bar{Y} = \frac{1}{k} \sum_i \ln \frac{X_{(n-i+1)}}{X_{(n-k)}}.$$

On a near-constant series every order statistic is nearly equal, so each ratio is nearly 1 and each logarithm nearly 0.
Concretely, if the largest values are all within 0.003% of the threshold, then $\ln(1.00003) = 3 \times 10^{-5}$, and

$$\hat{\alpha} = \frac{1}{3 \times 10^{-5}} = 33{,}000.$$

Same order of magnitude as observed. The estimator is dividing by very nearly zero, and small denominators produce enormous quotients.

Why this is dangerous rather than merely wrong. The metric is $|\hat{\alpha}_{\text{real}} - \hat{\alpha}_{\text{syn}}|$. With $\hat{\alpha}_{\text{real}} \approx 3$, the collapsed model scores $|3 - 31581| = 31578$ — catastrophically bad, which is correct. But had the collapse been milder, giving $\hat{\alpha} = 3.0$ on a series with no tail whatsoever, it would have scored a *perfect* 0 on this metric while being degenerate. A number with no information content had entered a ranking.

The guard. Real equity indices sit between 2.4 and 4.2. The pipeline therefore returns `NaN` whenever

$$\hat{\alpha} > 20,$$

and `NaN` ranks last by `na_option='bottom'` — so a failed estimate is scored as the worst outcome rather than silently excluded. It also returns `NaN` when the threshold order statistic is non-positive or the mean log spacing is non-positive, both of which signal the same degeneracy.

Formal Name — the mathematics

The Hill estimator is undefined in the limit of a degenerate distribution. If X is constant, every log spacing is 0, $\bar{Y} = 0$, and $\hat{\alpha} = 1/0$ is undefined rather than infinite.
In floating-point arithmetic the spacings are small but non-zero, so the division succeeds and returns a large finite value carrying no information. Guarding requires an explicit range check, because no arithmetic error occurs.

Where the Analogy Breaks

The guard is a heuristic, not a theorem. A genuine distribution with $\alpha = 25$ — extremely thin-tailed but legitimate — would be rejected. That trade is deliberate: no equity index has such a tail, so within this domain the false-rejection rate is effectively zero, while the false-acceptance rate without the guard was demonstrably not.

Why It Matters Here — link to the thesis

This is one of three degeneracy guards protecting the evaluation. The other two are the post-generation check on mean, standard deviation and ACF(1), and the autoencoder R^2 threshold. Each exists because a collapsed model can otherwise score *well* on some metric by accident, and a benchmark that can be won by producing nothing is not a benchmark.

9.6 Extreme Value Theory

Situation — the everyday picture

A dam must survive the worst flood in a century, but you have only forty years of records. The largest flood you have seen is not the largest that can happen, and you must say something about a level you have never observed.

Problem — where it breaks or why it matters

The CLT (§6.12) describes averages, and averages are exactly what a risk manager does not care about. There is a parallel theory for maxima, and it says something surprisingly strong: whatever the underlying distribution, the maximum has only three possible limiting shapes.

Worked Arithmetic — concrete numbers

Maxima of uniform draws. Take n values uniform on $[0, 1]$. The maximum is below x only if all n are, so

$$\mathbb{P}(M_n \le x) = x^n.$$

At $n = 100$: $\mathbb{P}(M_{100} \le 0.95) = 0.95^{100} = 0.0059$. The maximum is almost certainly above 0.95, and as n grows it is pushed toward 1 — a *bounded* limit.

Maxima of exponential draws. With $\mathbb{P}(X > x) = e^{-x}$,

$$\mathbb{P}(M_n \le x) = \left(1 - e^{-x}\right)^n.$$

Here the maximum grows like $\ln n$ without bound, but only logarithmically — slowly.

Maxima of power-law draws. With $\mathbb{P}(X > x) = x^{-\alpha}$, the maximum grows like $n^{1/\alpha}$. At $\alpha = 3$ and n multiplied by 1,000, the typical maximum multiplies by $1000^{1/3} = 10$.

Three regimes — bounded, logarithmic, polynomial. The Fisher–Tippett theorem says these exhaust the possibilities.

Peaks over threshold. Rather than block maxima, fix a high threshold u and model the exceedances $X - u$. As u rises, that distribution approaches the generalised Pareto, and its shape parameter is $\xi = 1/\alpha$. At $\hat{\alpha} = 3.11$, $\xi = 0.32$ — firmly in the heavy-tailed regime, since $\xi > 0$.

Formal Name — the mathematics

Fisher–Tippett–Gnedenko. If suitably normalised maxima converge, the limit is a **generalised extreme value** distribution

$$G(x) = \exp\left\{-\left[1 + \xi\left(\frac{x-\mu}{\sigma}\right)\right]^{-1/\xi}\right\},$$

with three cases: $\xi = 0$ Gumbel (exponential-like tails), $\xi > 0$ Fréchet (power-law tails), $\xi < 0$ Weibull (bounded).

Pickands–Balkema–de Haan. For a high threshold u, exceedances follow approximately a **generalised Pareto** distribution

$$H(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}, \qquad y > 0,$$

with $\xi = 1/\alpha$ for power-law tails. Financial returns sit in the Fréchet domain.

Where the Analogy Breaks

EVT describes the limit as the threshold rises, and raising the threshold discards data. There is no escaping the trade: a higher threshold gives a better approximation on fewer points. EVT also assumes independent observations in its basic form, which volatility clustering violates — extremes arrive in bursts, so the effective sample of independent extremes is smaller than the count suggests.

Why It Matters Here — link to the thesis

The Hill estimator is the EVT tool this project actually uses, and it is the peaks-over-threshold approach with $k = \lfloor 0.05n \rfloor$ setting the threshold. The connection $\xi = 1/\alpha$ means that reporting $\hat{\alpha}$ between 2.54 and 3.21 is reporting ξ between 0.31 and 0.39: the same measurement in the vocabulary a risk manager would use.

9.7 Copulas and Tail Dependence

Situation — the everyday picture

Two swimmers, each an excellent swimmer on their own. Whether they can rescue each other depends on something their individual abilities do not capture: whether they get into trouble at the same time.

Problem — where it breaks or why it matters

Section 6.9 established that marginals do not determine the joint distribution, and §8.13 showed the portfolio consequences. What was missing is a device that separates the two questions cleanly — and, critically, one that can express dependence concentrated in the tails, which correlation cannot.

Worked Arithmetic — concrete numbers

Correlation misses the thing that matters. Consider two return series, each standard normal, in two scenarios:

- *Gaussian dependence* with correlation 0.5;
- *t-copula dependence*, also correlation 0.5, with 3 degrees of freedom.

Identical marginals, identical correlation. But the probability that both fall below their 1% quantile simultaneously differs sharply: for the Gaussian case it tends to zero as the quantile becomes more extreme, while for the t-copula it tends to a positive constant. Joint crashes are asymptotically impossible under one and routine under the other, with every number a correlation matrix records being identical.

The transformation. Apply each variable's own CDF: $U = F_X(X)$ and $V = F_Y(Y)$. Whatever the original distributions, U and V are uniform on $[0, 1]$ — so all marginal information has been stripped out and what remains in the joint distribution of (U, V) is pure dependence. That joint distribution is the copula.

Tail dependence, defined by a limit. The lower tail dependence coefficient is

$$\lambda_L = \lim_{q \to 0^+} \mathbb{P}(V \le q \mid U \le q).$$

The Gaussian copula has $\lambda_L = 0$ for every correlation below 1. The t-copula has $\lambda_L > 0$. This single number is what the 2008 credit crisis literature identified as missing from Gaussian-copula pricing models.

Formal Name — the mathematics

Sklar's theorem. Any joint CDF F with continuous marginals $F_1, \dots, F_d$ can be written uniquely as

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d)),$$

where the **copula** C is a joint CDF on $[0, 1]^d$ with uniform marginals. Marginals and dependence are thus fully separable: choose each independently.

Tail dependence:

$$\lambda_U = \lim_{q \to 1^-} \mathbb{P}(V > q \mid U > q), \qquad \lambda_L = \lim_{q \to 0^+} \mathbb{P}(V \le q \mid U \le q).$$

Where the Analogy Breaks

Copulas separate marginals from dependence, which is their strength and their trap. The separation is a mathematical fact, not a licence to pick the two independently and assume the result is realistic — and estimating tail dependence requires observations in the joint tail, which are by definition the scarcest data available. A copula fitted mostly on the body of the distribution tells you very little about λ_L.

Why It Matters Here — link to the thesis

This section is scope-setting rather than method. This thesis is univariate: one model per market, evaluated per market, with no cross-market dependence modelled or measured. Copulas are the natural device for the multivariate extension, and tail dependence would be the metric that matters — since the question a portfolio owner asks is whether the markets crash together, which nothing in the current evaluation addresses.

Chapter 10

Optimisation

Before and After

Before this chapter you need: quadratics and curvature (§1.6), derivatives and the second-derivative test (§3.2–§3.4), partial derivatives, the gradient and the Hessian (§4.2–§4.3, §4.6), norms (§5.5), eigenvalues (§5.6), expectation (§6.7), the law of large numbers (§6.11). **After it you will understand:** what gradient descent actually does, why mini-batching is not merely a memory convenience, how a learning rate interacts with curvature, and why a minimax game does not converge in the way a minimisation does — the structural reason GAN training is hard.

Every model in this thesis is produced by minimising something. This chapter takes the gradient of Chapter 4 and builds the algorithms that use it, ending with the two-player problem that the rest of the thesis lives inside.

10.1 Objective Functions

Situation — the everyday picture

You are packing a suitcase. You want everything to fit, nothing to break, and the weight under the airline limit. Before you can say whether one arrangement beats another, you have to decide what "better" means — and different answers give different packings.

Problem — where it breaks or why it matters

"Train the model" is not yet a computation. It becomes one only when there is a single number to make small, and the choice of that number determines everything the model will and will not learn.

Worked Arithmetic — concrete numbers

Squared error. Predictions $\hat{y} = (2, 5, 8)$ against truth $y = (3, 5, 6)$:

$$L = (2-3)^2 + (5-5)^2 + (8-6)^2 = 1 + 0 + 4 = 5.$$

Absolute error on the same data:

$$L = |2-3| + |5-5| + |8-6| = 1 + 0 + 2 = 3.$$

They rank differently. Compare two candidate predictors for $y = (3, 5, 6)$:

- Predictor A errs by $(0, 0, 3)$: squared $= 9$, absolute $= 3$.
- Predictor B errs by $(1, 1, 1)$: squared $= 3$, absolute $= 3$.

Squared error prefers B decisively; absolute error is indifferent. Squaring punishes one large mistake more than several small ones, so the objective encodes a judgement about which failures matter.

Why it matters here. On heavy-tailed returns, squared error is dominated by the rare enormous days — the same leverage that broke the ARCH-LM test in §8.10. An objective is never neutral.

Formal Name — the mathematics

An **objective function** (or **loss**, or **cost**) $L : \mathbb{R}^n \to \mathbb{R}$ maps parameters to a single number to be minimised. The problem

$$\theta^* = \arg\min_{\theta} L(\theta)$$

asks for the parameters attaining the smallest value. Maximisation is the same problem applied to $-L$.

L typically averages over data: $L(\theta) = \frac{1}{N}\sum_{i=1}^{N} \ell(\theta; x_i)$, an estimate of the expected loss $\mathbb{E}[\ell(\theta; X)]$.

Where the Analogy Breaks

The objective is a proxy for what you actually want, never the thing itself. This thesis wants generators that reproduce stylised facts; what gets minimised is an adversarial loss that mentions none of them. The gap between the two is why a separate evaluation framework exists at all — and why a model can improve its training loss while getting worse at the job.

Why It Matters Here — link to the thesis

The three models minimise different objectives — TimeGAN a sum of reconstruction, supervised and adversarial terms; QuantGAN and FinGAN a WGAN-GP critic loss. None of them optimises Wasserstein distance to the real marginal, ACF error, or tail index. The evaluation metrics are deliberately *not* the training objectives, which is what stops the comparison being circular.

10.2 Local versus Global Minima

Situation — the everyday picture

You are dropped somewhere in a mountain range in thick fog and told to find the lowest point. You can feel the slope underfoot. Walking downhill until the ground is level lands you at the bottom of *a* valley — with no way of knowing whether a deeper one lies over the ridge.

Problem — where it breaks or why it matters

Every practical algorithm is a fog-walker. It sees only local information, so it can certify only local optimality. Claims about global optima require either a special structure or an exhaustive search, and neither is available here.

Worked Arithmetic — concrete numbers

Take $f(x) = x^4 - 4x^2 + x$. Then

$$f'(x) = 4x^3 - 8x + 1.$$

Stationary points occur near $x \approx -1.44$, $x \approx 0.13$ and $x \approx 1.32$. Evaluate:

x	$f(x)$	Type
-1.44	-5.68	local minimum (the global one)
0.13	$+0.06$	local maximum
1.32	-3.52	local minimum

The consequence. Start the walk at $x = 2$ and the slope carries you to 1.32, where $f = -3.52$. You stop, correctly, at a point where $f'(x) = 0$ and $f''(x) > 0$. And you are 2.16 above the true minimum, with nothing in the local information to tell you so.

Starting point decides the outcome. From $x = -2$ the same algorithm reaches -1.44 and the global minimum. Same objective, same rule, different answer — determined entirely by initialisation.

Formal Name — the mathematics

θ^* is a **local minimum** if $L(\theta^*) \leq L(\theta)$ for all θ in some neighbourhood of θ^*, and a **global minimum** if the inequality holds for all θ in the domain.
First-order condition: $\nabla L(\theta^*) = \mathbf{0}$. Second-order sufficient condition: the Hessian is positive definite — every eigenvalue positive (§5.6) — which is the many-variable version of $f''(a) > 0$. A point with $\nabla L = \mathbf{0}$ and mixed-sign eigenvalues is a **saddle**.

Where the Analogy Breaks

In high dimensions the intuition from one variable misleads. For a random function of n variables, a stationary point is a local minimum only if all n eigenvalues happen to be positive, which becomes vanishingly unlikely as n grows. Saddles, not spurious minima, are the dominant obstacle — and a saddle has a downhill direction, so an algorithm that is not exactly stationary can eventually escape. That is part of why training very large models works better than the one-variable picture predicts.

Why It Matters Here — link to the thesis

Every result in this thesis is a local optimum, and nothing in the procedure certifies more. This is also why seeds are reported per seed and never averaged away: different seeds mean different initialisations, hence different valleys, and the spread across seeds is a genuine part of the finding rather than noise to be smoothed.

10.3 Convexity

Situation — the everyday picture

A smooth bowl has one lowest point, and a marble released anywhere inside reaches it. An egg carton has many, and where the marble stops depends on where you dropped it. Whether a problem is easy or hard is largely the question of which shape you have.

Problem — where it breaks or why it matters

The previous section showed local search cannot certify a global answer. There is one important exception, and knowing whether you are inside it tells you how much to trust the result.

Worked Arithmetic — concrete numbers

The chord test. A function is convex if the straight line joining any two points on its graph never dips below the graph.
For $f(x) = x^2$, take $x = 1$ and $x = 3$, with $f = 1$ and 9. The midpoint of the chord is at height

$$\tfrac{1}{2}(1 + 9) = 5,$$

while the function at the midpoint $x = 2$ is $f(2) = 4$. Since $4 \le 5$, the chord lies above. ✓

A non-convex case. $g(x) = x^4 - 4x^2 + x$ from §10.2, at $x = -1.44$ and $x = 1.32$: heights -5.68 and -3.52, chord midpoint $\frac{1}{2}(-5.68 - 3.52) = -4.60$. The midpoint is $x = -0.06$, where

$$g(-0.06) = 0.000013 - 0.0144 - 0.06 = -0.074.$$

Here $-0.074 > -4.60$: the function lies *above* the chord. Not convex. ✓

Via the second derivative. $f''(x) = 2 > 0$ everywhere for x^2 — convex. For g, $g''(x) = 12x^2 - 8$, which is negative for $|x| < \sqrt{2/3} = 0.816$ — so g is concave in that band and convex outside. Mixed sign, not convex. ✓

Formal Name — the mathematics

f is **convex** on a convex set if for all x, y and $t \in [0, 1]$

$$f(tx + (1-t)y) \le tf(x) + (1-t)f(y).$$

For twice-differentiable f this is equivalent to the Hessian being positive semi-definite everywhere.
The essential consequence: **for a convex function every local minimum is global**, and the set of minimisers is convex.

Where the Analogy Breaks

Convexity is the exception in this field. Squared error in a *linear* model is convex; put the same loss behind a neural network and it is not, because the composition of a convex function with a non-linear map need not be convex. Every objective minimised in this thesis is non-convex, so no global guarantee is available anywhere in it.

Why It Matters Here — link to the thesis

The reconstruction loss in TimeGAN's Phase 1 is quadratic in the reconstruction error, and its bowl shape is why an R^2 of $+0.995$ is meaningful — the autoencoder really did approach the bottom of a well-behaved objective. Phase 3 is adversarial and has no such structure, which is exactly the difference §10.8 describes.

10.4 Gradient Descent

Situation — the everyday picture

The fog-walker again, now with a method: feel which way is steepest downhill, take a step that way, and repeat. It requires nothing but local slope, which is precisely what is available.

Problem — where it breaks or why it matters

Section 4.3 established that $-\nabla L$ is the steepest downhill direction. Turning that into an algorithm requires deciding how far to step, and knowing when the procedure can be trusted to make progress.

Worked Arithmetic — concrete numbers

Minimise $f(x) = x^2$ from $x_0 = 4$, with $f'(x) = 2x$ and step size $\eta = 0.1$. The rule is $x \leftarrow x - \eta f'(x)$:

Step	x	$f'(x)$	next x
0	4.000	8.000	3.200
1	3.200	6.400	2.560
2	2.560	5.120	2.048
3	2.048	4.096	1.638

Each step multiplies x by $1 - 2\eta = 0.8$. After n steps $x_n = 4(0.8)^n$, converging to 0 geometrically. At $n = 30$: $4(0.8)^{30} = 0.005$.

Two variables. Minimise $f(x, y) = x^2 + 3y^2$ from $(2, 2)$ with $\eta = 0.1$. Here $\nabla f = [2x, 6y]$:

$$x : 2 \to 2 - 0.1(4) = 1.6, \qquad y : 2 \to 2 - 0.1(12) = 0.8.$$

The y coordinate moves much faster, because the surface is steeper in y — the elongated bowl of §4.6, with Hessian $\mathrm{diag}(2, 6)$.
Next step: $x \to 1.28$, $y \to 0.32$. The x direction shrinks by 0.8 per step, y by 0.4. One global step size, two very different effective rates.

Formal Name — the mathematics

Gradient descent iterates

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t),$$

with **learning rate** $\eta > 0$.
For an L-smooth convex objective — gradient Lipschitz with constant L — convergence is guaranteed when $\eta < 2/L$, at rate $O(1/t)$. For non-convex objectives the guarantee weakens to convergence toward a stationary point, with no claim about which one.

Where the Analogy Breaks

Gradient descent is greedy: it takes the locally steepest direction, which is rarely the direction pointing at the minimum. In the elongated bowl above it zig-zags, making rapid progress across the narrow direction and crawling along the shallow one. The number governing how bad this gets is the ratio of largest to smallest Hessian eigenvalue — the **condition number** — which here is $6/2 = 3$, mild. Real loss surfaces reach condition numbers in the thousands.

Why It Matters Here — link to the thesis

Every parameter in every model in this thesis is updated by a variant of this rule. The gradient-norm clip of `max_norm` $= 1.0$ applied in all three models rescales ∇L whenever $\|\nabla L\|_2 > 1$, bounding the step length regardless of how steep the surface becomes.

10.5 The Learning Rate

Situation — the everyday picture

Descending a staircase in the dark. Tiny shuffles get you down safely but take all night. Enormous strides are fast until one overshoots the step and you land further down than intended — or fall.

Problem — where it breaks or why it matters

The step size is the single most consequential setting in the algorithm, and it has no universally right value: too small wastes computation, too large diverges, and the boundary between them depends on curvature that changes as you move.

Worked Arithmetic — concrete numbers

Minimise $f(x) = x^2$ from $x_0 = 4$. Each step multiplies x by $(1 - 2\eta)$.

η	factor $\lvert 1-2\eta \rvert$	behaviour
0.01	0.98	converges, slowly
0.1	0.80	converges well
0.5	0.00	reaches the minimum in one step
0.9	0.80	converges, oscillating in sign
1.0	1.00	oscillates for ever between 4 and -4
1.5	2.00	**diverges**: $4 \to -8 \to 16 \to -32$

Check the divergent case. At $\eta = 1.5$ and $x = 4$:

$$x \leftarrow 4 - 1.5(8) = 4 - 12 = -8,$$

then $-8 - 1.5(-16) = -8 + 24 = 16$. Growing by a factor of 2 each step, away from the minimum. ✓

The stability threshold. Convergence needs $|1 - 2\eta| < 1$, so $0 < \eta < 1$. In general the bound is $\eta < 2/\lambda_{\max}$ where $\lambda_{\max}$ is the largest Hessian eigenvalue. Here $f'' = 2$, giving $\eta < 1$. ✓

The conditioning trap. For $f(x, y) = x^2 + 3y^2$, stability demands $\eta < 2/6 = 0.333$ from the y direction, while the x direction would happily take η up to 1. The steepest direction caps the rate for every direction, so the shallow one is starved.

Formal Name — the mathematics

The learning rate η must satisfy $\eta < 2/\lambda_{\max}(H)$ for stability on a quadratic. Common schedules reduce it over training:

- **step decay**: multiply by a factor at fixed intervals;
- **cosine annealing**: decay smoothly to near zero;
- **warmup**: start small and increase, to avoid early instability.

Adam (Kingma & Ba 2015) instead maintains per-parameter step sizes from running estimates of the gradient's first and second moments, which adapts to differing curvature without forming the Hessian (§4.6).

Where the Analogy Breaks

There is no learning rate that is right throughout training. Early on the surface is steep and small steps are safe; later it flattens and larger steps would help. Adaptive methods address this per parameter, and they are heuristics with no convergence guarantee on non-convex problems — they work well in practice, which is a different claim from working provably.

Why It Matters Here — link to the thesis

This project uses Adam throughout: 10^{-3} for TimeGAN, and 10^{-4} for both generator and critic in QuantGAN and FinGAN. The lower adversarial rate is standard for WGAN-GP, where a critic that moves too fast destabilises the game of §10.8. Adam's per-parameter scaling is the practical answer to the conditioning trap above.

10.6 Stochastic Gradient Descent and Batching

Situation — the everyday picture

You want the average height of a city's population. You could measure everyone — exact, and absurdly slow. Or measure a hundred people, get a good estimate in a minute, and repeat with a fresh hundred whenever you need another. If you only need the direction of a trend, the sample is enough.

Problem — where it breaks or why it matters

The true gradient averages over the whole dataset, and computing it for one update is wasteful when a small sample gives nearly the same direction. But the sample introduces noise, and the size of the sample controls a trade-off that this project got badly wrong once.

Worked Arithmetic — concrete numbers

The estimate and its noise. With N training windows, the full gradient is $\frac{1}{N}\sum_{i=1}^{N}\nabla\ell_i$. A batch of B drawn at random gives an unbiased estimate whose standard error scales as $1/\sqrt{B}$ (§6.11).
Quadrupling the batch halves the gradient noise — and quadruples the cost per step. Four small steps usually beat one large one.

What went wrong in this project. TimeGAN's training loop drew *one* batch per epoch instead of iterating over all of them.

With $n_{\text{windows}} = 3{,}835$ and `batch_size` $= 128$:

$$n_{\text{batches}} = \left\lfloor \frac{3835}{128} \right\rfloor = 29.$$

So each "epoch" performed 1 gradient step where it should have performed 29 — a 29× shortfall. Set against QuantGAN, which iterated properly over roughly 59 batches per epoch, TimeGAN was receiving a small fraction of the training its configuration appeared to specify.

Why it was invisible. Both models reported "epochs", and the epoch counts looked comparable. The unit was the same word and meant different things.

The fix, and then the real fix. Adding the inner batch loop brought TimeGAN to 29 steps per epoch. But the deeper problem remained: an epoch is $\lfloor n/B \rfloor$ steps, so identical epoch counts still mean different step counts whenever the models or folds differ in size. The project therefore specifies budgets in **gradient steps** directly, deriving

$$n_{\text{epochs}} = \left\lceil \frac{\text{steps}}{n_{\text{batches}}} \right\rceil$$

at run time. With a 9,000-step budget and 29 batches per epoch that is 311 epochs — a number nobody would have chosen by hand, and exactly the number required for parity.

Formal Name — the mathematics

Stochastic gradient descent replaces the full gradient with a mini-batch estimate:

$$\theta_{t+1} = \theta_t - \eta \, \frac{1}{B} \sum_{i \in \mathcal{B}_t} \nabla \ell_i(\theta_t),$$

where $\mathcal{B}_t$ is a randomly drawn batch of size B. The estimate is unbiased: $\mathbb{E}[\hat{g}] = \nabla L$. An **epoch** is one pass through the dataset, comprising $\lfloor N/B \rfloor$ steps. Convergence for SGD requires a decaying learning rate, $\sum_t \eta_t = \infty$ and $\sum_t \eta_t^2 < \infty$, though constant rates are used in practice.

Where the Analogy Breaks

Gradient noise is not purely a cost. It helps escape saddle points and sharp narrow minima, and models trained with small batches often generalise better than those trained with very large ones. The noise is part of why the method works, not only an approximation error to be minimised — so "use the largest batch that fits" is bad advice.

Why It Matters Here — link to the thesis

The epoch-versus-step distinction is the single most consequential lesson in this project's training code. Reporting a budget in epochs conceals a dependence on dataset size and batch size; reporting it in gradient steps does not. The parity assertion in the pipeline now enforces equal *generator updates* across all three models and refuses to run otherwise — because a benchmark in which one model silently trains 29× less than another measures the bug, not the models.

10.7 Constraints and Lagrange Multipliers

Situation — the everyday picture

Fence off the largest possible rectangular field with 100 metres of fencing. You are not free to make it arbitrarily large — the fencing is fixed, and the answer lives on that boundary rather than at an unconstrained optimum.

Problem — where it breaks or why it matters

Setting the gradient to zero finds unconstrained optima. When the answer is forced onto a constraint surface, the gradient there is generally *not* zero, and a different condition is needed.

Worked Arithmetic — concrete numbers

Maximise area $A = xy$ subject to perimeter $2x + 2y = 100$, i.e. $x + y = 50$.

By substitution. $y = 50 - x$, so $A(x) = x(50 - x) = 50x - x^2$. Then $A'(x) = 50 - 2x = 0$ gives $x = 25$, $y = 25$, $A = 625$. A square. And $A'' = -2 < 0$ confirms a maximum. ✓

By Lagrange. Form

$$\mathcal{L}(x, y, \lambda) = xy - \lambda(x + y - 50).$$

Setting the partial derivatives to zero:

$$\frac{\partial \mathcal{L}}{\partial x} = y - \lambda = 0, \qquad \frac{\partial \mathcal{L}}{\partial y} = x - \lambda = 0, \qquad \frac{\partial \mathcal{L}}{\partial \lambda} = -(x + y - 50) = 0.$$

The first two give $x = y = \lambda$; the third gives $2\lambda = 50$, so $\lambda = 25$ and $x = y = 25$. ✓Same answer.

What λ means. It is the rate at which the optimum improves as the constraint is relaxed. With 102 metres of fence the optimum is $25.5^2 = 650.25$, up 25.25 from 625 for 2 extra metres — about 12.6 per metre, and $\lambda/2 = 12.5$ per metre of fence since perimeter uses two metres per unit of $x + y$. ✓The multiplier is a shadow price.

Formal Name — the mathematics

To optimise $f(\mathbf{x})$ subject to $g(\mathbf{x}) = 0$, form

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda\, g(\mathbf{x})$$

and solve $\nabla_{\mathbf{x}} \mathcal{L} = \mathbf{0}$, $g(\mathbf{x}) = 0$. Geometrically this requires $\nabla f = \lambda \nabla g$: the gradients are parallel, so no movement along the constraint surface improves f.
For inequality constraints $g(\mathbf{x}) \leq 0$ the **KKT conditions** apply, adding $\lambda \geq 0$ and complementary slackness $\lambda g = 0$ — the multiplier is non-zero only where the constraint binds.

Where the Analogy Breaks

Exact constraint enforcement is rare in deep learning, because projecting onto a constraint surface after every step is expensive and often ill-defined. The usual substitute is a **penalty**: add a term that grows when the constraint is violated, and accept approximate satisfaction. That is a soft version of the same idea with a fixed multiplier chosen by hand rather than solved for.

Why It Matters Here — link to the thesis

The WGAN-GP gradient penalty is precisely such a soft constraint. The critic should satisfy $\|\nabla D\|_2 = 1$; rather than enforce it exactly, the objective adds $\lambda\,\mathbb{E}[(\|\nabla D\|_2 - 1)^2]$ with $\lambda = 10$ fixed. The constraint is encouraged, never guaranteed — a distinction Chapter 13 takes seriously.

10.8 Minimax Problems

Situation — the everyday picture

Rock–paper–scissors. Whatever you settle on, your opponent has a best reply, and against that reply you would rather have chosen differently. There is no pair of fixed choices both players are content with, so no amount of thinking converges to one.

Problem — where it breaks or why it matters

Everything so far assumed a single objective going downhill. A GAN has two networks with opposed objectives, and the surface each descends is reshaped by the other's movement. The convergence story from §10.4 does not transfer, and understanding why is the point of this section.

Worked Arithmetic — concrete numbers

The simplest adversarial objective. Let

$$L(x, y) = xy,$$

with one player minimising over x and the other maximising over y. The equilibrium is $(0, 0)$: neither can improve unilaterally.
Simultaneous gradient steps with rate η:

$$x \leftarrow x - \eta y, \qquad y \leftarrow y + \eta x.$$

Start at $(1, 0)$ with $\eta = 0.1$:

Step	x	y	$\sqrt{x^2 + y^2}$
0	1.000	0.000	1.000
1	1.000	0.100	1.005
2	0.990	0.200	1.010
3	0.970	0.299	1.015
4	0.940	0.396	1.020

The distance from equilibrium grows. It should be shrinking. Compute it exactly: after one joint step,

$$x'^2 + y'^2 = (x - \eta y)^2 + (y + \eta x)^2 = (1 + \eta^2)(x^2 + y^2).$$

The radius multiplies by $\sqrt{1 + \eta^2} > 1$ at every step, *for any positive* η. The players spiral outward for ever.

This is not a tuning failure. No learning rate fixes it — smaller η only slows the divergence. The problem is structural: each player descends correctly with respect to a target the other is moving.

Contrast with minimisation. For $f(x) = x^2$ the same simultaneous update gave a factor of 0.8 per step (§10.4), monotonically decreasing. Minimisation has a Lyapunov function — the objective itself, which falls every step. A minimax game has none.

Formal Name — the mathematics

A **minimax problem** seeks

$$\min_{\theta} \max_{\phi} V(\theta, \phi).$$

A **Nash equilibrium** is a pair (θ^*, ϕ^*) at which neither player can improve alone:

$$V(\theta^*, \phi^*) \leq V(\theta, \phi^*)\ \forall\theta, \qquad V(\theta^*, \phi^*) \geq V(\theta^*, \phi)\ \forall\phi.$$

For convex–concave V an equilibrium exists and averaged iterates converge. For the non-convex, non-concave V of a GAN, neither existence nor convergence is guaranteed, and simultaneous gradient steps can cycle or diverge as above.

Where the Analogy Breaks

The word "converged" does not mean for a GAN what it means for a regression. GAN losses characteristically oscillate without settling, and a stable loss can indicate collapse rather than success — a generator producing one output makes the critic's job easy and both losses flat. Loss curves are close to uninformative as a stopping criterion, which is why this project judges the models by external evaluation and explicit degeneracy guards instead.

Why It Matters Here — link to the thesis

This is the structural reason GAN training is hard, and the reason the pipeline carries the safeguards it does: a post-generation check on mean, standard deviation and ACF(1); a fixed gradient-step budget rather than training to convergence; and a full evaluation framework external to the training objective. Chapter 13 shows what the failures look like in practice, including a TimeGAN collapse to standard deviation 0.0002 against a real 0.0103.

Chapter 11

Neural Networks

Before and After

Before this chapter you need: composition of functions (§1.7), the chain rule (§3.3, §4.4), gradients and Jacobians (§4.3, §4.5), matrices and matrix multiplication (§5.1–§5.2), gradient descent and batching (§10.4, §10.6).
After it you will understand: what a neuron computes, why the choice between sigmoid and tanh changed this project's results by more than any other single edit, how backpropagation is the chain rule run backwards, and the two architectures the models here are built from — gated recurrent cells and dilated causal convolutions.

A neural network is a composition of simple functions with adjustable coefficients. Nothing in this chapter is conceptually new; it is Chapter 4's chain rule and Chapter 5's matrices, assembled and given names.

11.1 A Neuron

Situation — the everyday picture

A loan officer weighs several facts — income, debts, employment history — giving each a different importance, adds them up, and issues a yes or no. The weighting is the judgement; the threshold turns a score into a decision.

Problem — where it breaks or why it matters

We need the smallest unit that can be composed into something expressive and adjusted by gradient descent. It must combine many inputs into one number, and it must be differentiable so §10.4 applies.

Worked Arithmetic — concrete numbers

Inputs $\mathbf{x} = [2, 3, 1]$, weights $\mathbf{w} = [0.5, -1, 2]$, bias $b = 0.4$.

Weighted sum. This is the dot product of §5.1:

$$z = \mathbf{w} \cdot \mathbf{x} + b = 0.5(2) + (-1)(3) + 2(1) + 0.4 = 1 - 3 + 2 + 0.4 = 0.4.$$

Activation. Apply tanh:

$$a = \tanh(0.4) = 0.3799.$$

What the weights encode. The second input has weight -1: raising it lowers the output. The third has weight 2: it matters twice as much per unit as the first. Learning means adjusting these numbers.

Why the bias is needed. Without b, an all-zero input forces $z = 0$ and the neuron cannot represent a non-zero default. The bias is the intercept c of §1.5, and it is what makes the map affine rather than strictly linear (§5.7).

Formal Name — the mathematics

A **neuron** computes

$$a = \sigma(\mathbf{w} \cdot \mathbf{x} + b) = \sigma\left(\sum_i w_i x_i + b\right),$$

where $\mathbf{w}$ are **weights**, b is the **bias**, and σ is a non-linear **activation function**.
A **layer** applies many neurons to the same input, which is one matrix multiplication:

$$\mathbf{a} = \sigma(W\mathbf{x} + \mathbf{b}),$$

with σ applied entrywise. For n inputs and m neurons, W is $m \times n$.

Where the Analogy Breaks

The biological analogy is thin and worth dropping early. Real neurons fire discrete spikes, have complex temporal dynamics, and do not learn by backpropagation — there is no known mechanism for propagating an error signal backwards through a brain. The name is historical. What is being described is a parameterised differentiable function, and nothing about its usefulness depends on the biology.

Why It Matters Here — link to the thesis

Every parameter counted in this project is a weight or a bias in such a unit. The TimeGAN embedder's 9,744 parameters and its generator's 11,400 are counts of exactly these numbers, and gradient descent adjusts every one of them at each step.

11.2 Activation Functions and Their Derivatives

Situation — the everyday picture

A stack of currency conversions is still a single conversion (§5.2). Stack fifty and you have one exchange rate. Something has to break the linearity, or depth buys nothing at all.

Problem — where it breaks or why it matters

The non-linearity must do two jobs at once: make composition expressive, and pass a usable gradient backwards. These pull against each other, and getting the balance wrong silently cripples training — as it did here.

Worked Arithmetic — concrete numbers

The three functions and their derivatives.

$$\sigma(x) = \frac{1}{1+e^{-x}}, \qquad \sigma'(x) = \sigma(x)(1-\sigma(x)),$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \qquad \tanh'(x) = 1 - \tanh^2(x),$$

$$\mathrm{ReLU}(x) = \max(0, x), \qquad \mathrm{ReLU}'(x) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases}$$

At the centre of each range. Sigmoid outputs 0.5 when $x = 0$, and there

$$\sigma' = 0.5 \times 0.5 = 0.25.$$

tanh outputs 0 when $x = 0$, and there

$$\tanh' = 1 - 0^2 = 1.00.$$

A factor of four, per layer. Backpropagation multiplies these along the chain (§4.4). Through three layers, at the most favourable point of each:

Layers	sigmoid	tanh	ratio
1	0.25	1.0	4
2	0.0625	1.0	16
3	0.0156	1.0	64

Through a three-layer stack the sigmoid passes at most 1.6% of the gradient signal that tanh passes — and 0.25 is its *best* case. Away from the centre it is far worse: at $x = 3$, $\sigma(3) = 0.953$ and $\sigma' = 0.953(0.047) = 0.045$.

Saturation. Both sigmoid and tanh flatten at their extremes, where the derivative approaches zero and learning stalls. ReLU has derivative exactly 1 for all positive inputs and so does not attenuate at all — at the cost of returning exactly 0 for negative inputs, where the unit can stop learning permanently (a “dead” unit).

Formal Name — the mathematics

An **activation function** is a non-linear scalar function applied entrywise. Ranges and uses:

- σ: range $(0, 1)$, for probabilities and gates;
- tanh: range $(-1, 1)$, zero-centred, for hidden and bounded outputs;
- ReLU: range $[0, \infty)$, cheap and non-saturating, the default in deep feedforward and convolutional stacks.

Zero-centred outputs matter: a layer whose activations are all positive produces gradients that share a sign across a whole weight vector, which forces the optimiser to move in zig-zags.

Where the Analogy Breaks

No activation is best everywhere. tanh’s stronger central gradient still saturates; ReLU never saturates on the positive side but discards all negative information and can die. The right choice depends on where in the network the unit sits and what range its output must occupy — which is why the models here use different activations in different places rather than one throughout.

Why It Matters Here — link to the thesis

This is the single highest-impact change made to this project's models. TimeGAN's embedder, generator and supervisor originally used sigmoid on the latent path. The diagnostic showed the latent representation stuck near its centre — mean 0.5069, standard deviation 0.04 — and the autoencoder reconstruction R^2 was approximately zero, meaning Phase 3 was training on noise. Replacing sigmoid with tanh on those three latent paths, for exactly the 4× per-layer reason above, moved the reconstruction R^2 to +0.995.
The recovery network also uses tanh, but for a different reason: it emits *data*, and the data was normalised by $\tanh(z/3)$ into $(-1, 1)$. Its activation must match the range of the thing it is reconstructing, which the round-trip assertion in the code checks to a tolerance of 10^{-3}.

11.3 Forward Propagation

Situation — the everyday picture

The factory line of §1.7: raw material in at one end, finished product out at the other, each station transforming what it receives.

Problem — where it breaks or why it matters

With the unit and the non-linearity defined, we need to run a whole network and keep track of what each stage produced — because backpropagation will need those intermediate values again.

Worked Arithmetic — concrete numbers

A two-layer network: 2 inputs, 2 hidden units with tanh, 1 output, linear.

$$W^{(1)} = \begin{bmatrix} 1 & -1 \\ 0.5 & 2 \end{bmatrix}, \quad \mathbf{b}^{(1)} = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix}, \quad W^{(2)} = \begin{bmatrix} 1 & -2 \end{bmatrix}, \quad b^{(2)} = 0.1.$$

Input $\mathbf{x} = [1, 2]$.

Layer 1, pre-activation:

$$\mathbf{z}^{(1)} = W^{(1)}\mathbf{x} + \mathbf{b}^{(1)} = \begin{bmatrix} 1(1) + (-1)(2) \\ 0.5(1) + 2(2) \end{bmatrix} + \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} = \begin{bmatrix} -1 \\ 5.0 \end{bmatrix}.$$

Layer 1, activation:

$$\mathbf{a}^{(1)} = \tanh \begin{bmatrix} -1 \\ 5.0 \end{bmatrix} = \begin{bmatrix} -0.7616 \\ 0.9999 \end{bmatrix}.$$

Note the second unit is saturated: $\tanh(5) = 0.9999$, where $\tanh' = 1 - 0.9999^2 = 0.0002$. Almost no gradient will pass through it.

Layer 2:

$$z^{(2)} = 1(-0.7616) + (-2)(0.9999) + 0.1 = -0.7616 - 1.9998 + 0.1 = -2.6614.$$

Output -2.6614.

Formal Name — the mathematics

For a network of L layers, **forward propagation** computes

$$\mathbf{a}^{(0)} = \mathbf{x}, \qquad \mathbf{z}^{(l)} = W^{(l)}\mathbf{a}^{(l-1)} + \mathbf{b}^{(l)}, \qquad \mathbf{a}^{(l)} = \sigma^{(l)}(\mathbf{z}^{(l)}),$$

for $l = 1, \ldots, L$, with $\mathbf{a}^{(L)}$ the output.
Every $\mathbf{z}^{(l)}$ and $\mathbf{a}^{(l)}$ must be retained: the backward pass needs them, which is why training uses far more memory than inference.

Where the Analogy Breaks

The layer-by-layer picture suggests each stage is independently meaningful. It is not. A trained layer's weights are fitted assuming exactly the distribution the previous layer sends it, and feeding raw input to a middle layer produces nonsense. The stations are separable in notation only — the point already made in §1.7, and the reason a saturated unit like the second one above corrupts everything downstream of it.

Why It Matters Here — link to the thesis

The saturated unit in the worked example is not hypothetical. The diagnostic block added to TimeGAN's Phase 1 reports exactly this: the fraction of latent activations with $|h| > 0.98$. Above 50% saturation, tanh is clipping the gradient and the layer has effectively stopped learning.

11.4 Computational Graphs

Situation — the everyday picture

A recipe written as a flowchart: chop, then fry, then season, then plate. Drawing the dependencies makes it obvious which steps must precede which, and which could run in parallel.

Problem — where it breaks or why it matters

Backpropagation must visit every operation in reverse order and know exactly which quantities feed which. Doing this by hand for a network with dozens of layers is error-prone; representing the computation as a graph makes it mechanical.

Worked Arithmetic — concrete numbers

Take $L = (wx + b - y)^2$ with $w = 2$, $x = 3$, $b = 1$, $y = 5$.

Forward, as a sequence of elementary operations:

Node	Operation	Value
u	$w \times x$	6
v	$u + b$	7
e	$v - y$	2
L	e^2	4

Local derivatives, one per edge:

$$\frac{\partial L}{\partial e} = 2e = 4, \quad \frac{\partial e}{\partial v} = 1, \quad \frac{\partial v}{\partial u} = 1, \quad \frac{\partial u}{\partial w} = x = 3, \quad \frac{\partial v}{\partial b} = 1.$$

Chain them.

$$\frac{\partial L}{\partial w} = 4 \times 1 \times 1 \times 3 = 12, \qquad \frac{\partial L}{\partial b} = 4 \times 1 \times 1 = 4.$$

Verify directly. $L(w) = (3w + 1 - 5)^2 = (3w - 4)^2$, so $dL/dw = 2(3w - 4)(3) = 6(3w - 4)$.
At $w = 2$: $6(2) = 12$. ✓
And $L(b) = (6 + b - 5)^2 = (b + 1)^2$, so $dL/db = 2(b + 1) = 4$ at $b = 1$. ✓

Each edge carries one simple derivative; the algorithm just multiplies along paths and adds across them, exactly as §4.4 described.

Formal Name — the mathematics

A **computational graph** is a directed acyclic graph whose nodes are variables and operations and whose edges are data dependencies. The forward pass evaluates nodes in topological order.
Automatic differentiation in *reverse mode* traverses the graph backwards from the output, accumulating $\partial L/\partial(\cdot)$ at each node by the chain rule. One reverse pass yields the derivative of one output with respect to *all* inputs, at roughly the cost of the forward pass — which is exactly the right trade when there are many parameters and one loss.

Where the Analogy Breaks

Reverse mode is efficient because there is one output and many inputs. With many outputs and few inputs, forward mode is cheaper. Neither is universally better; the asymmetry of the training problem — millions of parameters, one scalar loss — is what makes reverse mode the right choice here.

Why It Matters Here — link to the thesis

PyTorch builds this graph as the forward pass executes and `backward()` walks it in reverse. Nothing in this project computes a derivative by hand. The detach operations in the TimeGAN training code exist precisely to cut edges in this graph, preventing gradients from flowing into components that a given phase is not supposed to update.

11.5 Backpropagation

Situation — the everyday picture

A production error appears at the end of the line. To assign responsibility you walk backwards station by station, asking at each: given the fault that arrived here, how much of it did this station add?

Problem — where it breaks or why it matters

The chain rule gives the derivative for one parameter. A network has hundreds of thousands, and computing them one at a time — one forward pass per parameter — would make training impossible. The whole set must come from a single traversal.

Worked Arithmetic — concrete numbers

Use the two-layer network of §11.3 with target $y = -2$. The output was $z^{(2)} = -2.6614$, so with squared loss

$$L = \left(z^{(2)} - y\right)^2 = (-0.6614)^2 = 0.4374.$$

Step 1, at the output.

$$\frac{\partial L}{\partial z^{(2)}} = 2(-0.6614) = -1.3228.$$

Step 2, second-layer weights. Since $z^{(2)} = W^{(2)}\mathbf{a}^{(1)} + b^{(2)}$,

$$\frac{\partial L}{\partial W^{(2)}} = \frac{\partial L}{\partial z^{(2)}}\,\mathbf{a}^{(1)\top} = -1.3228\,[-0.7616,\ 0.9999] = [1.0075,\ -1.3227].$$

Step 3, back through $W^{(2)}$. Multiply by the transpose (§5.3):

$$\frac{\partial L}{\partial \mathbf{a}^{(1)}} = W^{(2)\top}\frac{\partial L}{\partial z^{(2)}} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}(-1.3228) = \begin{bmatrix} -1.3228 \\ 2.6456 \end{bmatrix}.$$

Step 4, through the activation. Multiply entrywise by $\tanh'(z^{(1)}) = 1 - \tanh^2$:

$$\tanh'(-1) = 1 - 0.7616^2 = 0.4200, \qquad \tanh'(5) = 1 - 0.9999^2 = 0.0002.$$

$$\frac{\partial L}{\partial \mathbf{z}^{(1)}} = \begin{bmatrix} -1.3228 \times 0.4200 \\ 2.6456 \times 0.0002 \end{bmatrix} = \begin{bmatrix} -0.5556 \\ 0.0005 \end{bmatrix}.$$

Read the second entry. The saturated unit receives a gradient of 0.0005 — essentially nothing. Its weights will barely move, however wrong they are. This is the vanishing gradient, visible in four lines of arithmetic, and it is the same mechanism that the sigmoid-to-tanh change in §11.2 addressed at a whole-network scale.

Formal Name — the mathematics

Backpropagation computes, for $l = L$ down to 1:

$$\boldsymbol{\delta}^{(L)} = \nabla_{\mathbf{a}^{(L)}} L \odot \sigma'(\mathbf{z}^{(L)}), \qquad \boldsymbol{\delta}^{(l)} = \left(W^{(l+1)\top}\boldsymbol{\delta}^{(l+1)}\right) \odot \sigma'(\mathbf{z}^{(l)}),$$

where $\odot$ is entrywise multiplication. The parameter gradients are then

$$\frac{\partial L}{\partial W^{(l)}} = \boldsymbol{\delta}^{(l)}\mathbf{a}^{(l-1)\top}, \qquad \frac{\partial L}{\partial \mathbf{b}^{(l)}} = \boldsymbol{\delta}^{(l)}.$$

Cost: one forward and one backward pass, independent of the parameter count.

Where the Analogy Breaks

The repeated multiplication by $W^\top$ and σ' is what makes deep networks hard to train. If the factors are systematically below 1 the signal vanishes; above 1 it explodes. Neither is a defect of the chain rule, which is exact — both are properties of the arithmetic that implements it, and both are why initialisation schemes, normalisation layers and gradient clipping exist.

Why It Matters Here — link to the thesis

The `max_norm` $= 1.0$ gradient clip in all three models is the guard against the exploding case: whenever $\|\nabla L\|_2$ exceeds 1, the whole gradient is rescaled. The tanh change was the guard against the vanishing case. Both are responses to the same product structure.

11.6 Training a Network

Situation — the everyday picture

Practising a piece of music. Play a phrase, hear what went wrong, adjust, repeat — thousands of times. No single repetition matters; the accumulation does.

Problem — where it breaks or why it matters

Forward propagation, backpropagation and gradient descent are three separate mechanisms. Training is the loop that combines them, and the loop has parameters of its own that determine whether anything is learned.

Worked Arithmetic — concrete numbers

One update, end to end. From §11.5, $\partial L/\partial W^{(2)} = [1.0075,\ -1.3227]$ and $W^{(2)} = [1,\ -2]$. With $\eta = 0.1$:

$$W^{(2)} \leftarrow [1,\ -2] - 0.1[1.0075,\ -1.3227] = [0.8993,\ -1.8677].$$

Did the loss fall? Recompute the forward pass with the new weights, holding layer 1 fixed:

$$z^{(2)} = 0.8993(-0.7616) + (-1.8677)(0.9999) + 0.1 = -0.6849 - 1.8675 + 0.1 = -2.4524.$$

New loss: $(-2.4524 + 2)^2 = 0.2047$, down from 0.4374. ✓More than halved in one step.

The loop.

1. Draw a mini-batch of B examples.
2. Forward pass, storing intermediates.
3. Compute the loss.
4. Backward pass for all parameter gradients.
5. Clip the gradient if $\|\nabla L\|_2 > 1$.
6. Update every parameter.
7. Repeat for the budgeted number of steps.

Budget, concretely. This project fixes 9,000 generator updates. With TimeGAN's 29 batches per epoch (§10.6), that is $\lceil 9000/29 \rceil = 311$ epochs.

Formal Name — the mathematics

The training loop repeats:

$$\theta \leftarrow \theta - \eta\,\mathrm{clip}\left(\frac{1}{B}\sum_{i\in\mathcal{B}} \nabla_\theta \ell_i,\ 1.0\right).$$

Weights are initialised randomly — not at zero, which would make every neuron in a layer compute the same thing and receive the same gradient for ever, leaving them identical.

Standard schemes scale the initial variance by the layer width so that activations neither shrink nor blow up with depth.

Where the Analogy Breaks

"Train until convergence" is not available here. GAN losses oscillate (§10.8) and a flat loss can signal collapse rather than success, so there is no reliable stopping signal in the loss itself. Training runs for a fixed budget and is judged externally.

Why It Matters Here — link to the thesis

The fixed-step budget is a methodological requirement, not a convenience. Because the three models have different batch sizes and different architectures, equal epoch counts would give unequal gradient steps — the 29× discrepancy of §10.6. The parity check in the pipeline asserts equal generator updates and refuses to run otherwise.

11.7 Recurrent Networks

Situation — the everyday picture

Reading a sentence. You do not process each word in isolation — you carry the meaning built so far and update it with each new word. The carried state is what lets the ending make sense.

Problem — where it breaks or why it matters

A feedforward network takes a fixed-size input and has no memory. A return series is a sequence of arbitrary length in which order is the whole point, and it needs an architecture that processes it step by step while retaining something.

Worked Arithmetic — concrete numbers

A minimal recurrent unit: $h_t = \tanh(w_h h_{t-1} + w_x x_t + b)$ with $w_h = 0.8$, $w_x = 0.5$, $b = 0$, $h_0 = 0$.
Inputs $x = (1, 0, -1)$:

$$h_1 = \tanh(0.8(0) + 0.5(1)) = \tanh(0.5) = 0.4621,$$
$$h_2 = \tanh(0.8(0.4621) + 0) = \tanh(0.3697) = 0.3537,$$
$$h_3 = \tanh(0.8(0.3537) + 0.5(-1)) = \tanh(-0.2170) = -0.2137.$$

Memory is there. At $t = 2$ the input is 0, yet $h_2 = 0.3537 \neq 0$: the state carries information from $t = 1$.

And it fades. With no further input, the state would evolve as $h \leftarrow \tanh(0.8h)$, shrinking each step: $0.3537 \to 0.2755 \to 0.2169$. The influence of x_1 decays geometrically.

Why that is a problem. Unrolling over T steps and backpropagating multiplies T factors of roughly $w_h \tanh'$. With $w_h = 0.8$ and $\tanh' \leq 1$, over 128 steps — this project's sequence length — the product is at most

$$0.8^{128} = 4 \times 10^{-13}.$$

A dependency spanning the full window transmits essentially no gradient. This is the vanishing gradient of §11.5 in its most acute form.

Formal Name — the mathematics

A **recurrent neural network** maintains a hidden state:

$$\mathbf{h}_t = \sigma\big(W_h \mathbf{h}_{t-1} + W_x \mathbf{x}_t + \mathbf{b}\big),$$

with the same weights reused at every step — so the parameter count is independent of sequence length.
Training uses **backpropagation through time**: unroll the recurrence into a T-layer feedforward network sharing weights, and backpropagate through it.

Where the Analogy Breaks

Weight sharing makes RNNs compact and makes them deep in a peculiar way: a 128-step sequence is a 128-layer network in which every layer is identical. The repeated multiplication by the *same* matrix means the gradient's fate is governed by that matrix's largest eigenvalue (§5.6) — below 1 and it vanishes, above 1 and it explodes. There is no comfortable middle.

Why It Matters Here — link to the thesis

TimeGAN is built from recurrent cells. Its embedder, recovery, generator, supervisor and discriminator are all GRU stacks with hidden dimension 24 and three layers, operating on sequences of length 128 — precisely the regime where the plain RNN above would fail.

11.8 LSTM and GRU

Situation — the everyday picture

Taking notes in a lecture. You do not transcribe everything and you do not discard everything — you decide what to write down, what to keep from earlier, and what is now irrelevant. Selective retention is what makes the notes useful an hour later.

Problem — where it breaks or why it matters

The plain RNN forces its state through a multiplication at every step, so information decays geometrically whether or not it should. What is needed is a path along which information can travel unchanged, plus a learned decision about when to change it.

Worked Arithmetic — concrete numbers

A GRU has an update gate z_t deciding how much of the old state to keep:

$$\mathbf{h}_t = (1 - z_t) \odot \mathbf{h}_{t-1} + z_t \odot \tilde{\mathbf{h}}_t.$$

Gate closed. With $z_t = 0$, $\mathbf{h}_t = \mathbf{h}_{t-1}$ exactly. The state passes through untouched — no multiplication by a weight matrix, no decay.
Gate open. With $z_t = 1$, $\mathbf{h}_t = \tilde{\mathbf{h}}_t$: the old state is discarded entirely.
Partly open. With $z_t = 0.1$, 90% of the old state survives each step. Over 128 steps that is $0.9^{128} = 1.4 \times 10^{-6}$ — still decaying, but compare the plain RNN's $0.8^{128} = 4 \times 10^{-13}$. Seven orders of magnitude better, and z_t is *learned*, so the network can close the gate further where long memory is needed.

Why the additive form matters. The state update is a weighted sum rather than a matrix product. Differentiating $(1-z)h_{t-1} + z\tilde{h}_t$ with respect to h_{t-1} gives $(1-z)$ — a scalar the network controls, not a fixed eigenvalue it is stuck with.

Formal Name — the mathematics

The **GRU** (Cho et al. 2014) computes

$$
\begin{aligned}
\mathbf{z}_t &= \sigma(W_z\mathbf{x}_t + U_z\mathbf{h}_{t-1}) && \text{update gate}\\
\mathbf{r}_t &= \sigma(W_r\mathbf{x}_t + U_r\mathbf{h}_{t-1}) && \text{reset gate}\\
\tilde{\mathbf{h}}_t &= \tanh(W\mathbf{x}_t + U(\mathbf{r}_t \odot \mathbf{h}_{t-1})) && \text{candidate}\\
\mathbf{h}_t &= (1-\mathbf{z}_t)\odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t && \text{new state}
\end{aligned}
$$

The **LSTM** (Hochreiter & Schmidhuber 1997) keeps a separate cell state with three gates — input, forget and output — and roughly a third more parameters for comparable performance on most tasks.
Note that the gates use σ deliberately: a gate must lie in $(0,1)$ to act as a proportion, which is the one place sigmoid's range is exactly right.

Where the Analogy Breaks

Gating mitigates vanishing gradients; it does not eliminate them. Very long dependencies remain hard, and the sequential structure prevents parallelisation across time — step t cannot start until $t-1$ finishes. Both limitations are why convolutional and attention-based architectures displaced recurrent ones for long sequences.

Why It Matters Here — link to the thesis

TimeGAN uses GRU stacks throughout. The activation distinction of §11.2 applies precisely here: sigmoid is correct *inside* the gates, where a proportion in $(0,1)$ is wanted, and wrong on the latent output path, where the $4\times$ gradient penalty applies with no compensating benefit. The fix in this project changed the latter and left the former alone.

11.9 Temporal Convolutions, Dilation and Receptive Fields

Situation — the everyday picture

Skimming a document for a theme. You do not read every word in order; you glance at every tenth line and still catch the structure. Sampling at intervals covers far more ground for the same effort.

Problem — where it breaks or why it matters

Recurrent models are sequential and slow, and their memory decays. Convolutions process all positions in parallel, but a stack of ordinary convolutions sees only a narrow window unless it is made impractically deep. Dilation resolves this.

Worked Arithmetic — concrete numbers

Receptive field of a plain stack. A convolution with kernel size $k=3$ sees 3 positions. Stacking two extends the view by $(k-1)=2$ each time, so L layers see $1+L(k-1)$ positions.

To cover 128 days would need $L = 127/2 \approx 64$ layers.

With dilation. A dilated convolution skips $d-1$ positions between taps, so it spans $(k-1)d+1$ while using the same k weights. Doubling d each layer makes the receptive field grow geometrically.

This project's QuantGAN generator. Three temporal blocks, each with *two* causal convolutions of kernel size 3, and dilations $1, 2, 4$:

Block	Dilation	Extension per block	Cumulative field
–	–	–	1
0	1	$2 \times 2 \times 1 = 4$	5
1	2	$2 \times 2 \times 2 = 8$	13
2	4	$2 \times 2 \times 4 = 16$	29

Each block contributes $2 \times (k-1) \times d = 4d$. Total:

$$1 + 4(1+2+4) = 1 + 28 = 29.$$

The generator's output at each position depends on the previous **29** time steps — about six trading weeks — using only three blocks.

Causality. A causal convolution pads only on the left, so position t sees $t, t-1, t-2, \ldots$ and never $t+1$. Without this the generator could use the future to produce the present, which for a time-series model is fatal.

Formal Name — the mathematics

A **dilated causal convolution** with kernel size k and dilation d computes

$$y_t = \sum_{i=0}^{k-1} w_i \, x_{t-i\cdot d},$$

using only indices at or before t.
For a network of L blocks with c convolutions each and dilations d_l, the **receptive field** is

$$R = 1 + c(k-1) \sum_{l} d_l.$$

With $d_l = 2^l$ the sum is geometric, so R grows exponentially in depth while the parameter count grows linearly.

Where the Analogy Breaks

A large receptive field is an upper bound on what the model *can* see, not evidence of what it does see. Dilation also leaves gaps: a single dilated layer samples a sparse subset of positions, and only the stacking of different dilations fills them in. And a fixed receptive field is a hard horizon — 29 days here — beyond which no dependence can be represented at all, which sits awkwardly against the long-memory findings of §8.11, where volatility autocorrelation remains measurable past lag 50.

Why It Matters Here — link to the thesis

This is QuantGAN's mechanism. Its generator and discriminator are temporal convolutional networks with dilations $1, 2, 4$ over channel widths $[64, 64, 32]$, giving the 29-day receptive field computed above. That is a concrete, checkable architectural limit, and it is the honest answer to why a model might reproduce short-horizon volatility clustering while missing the slow decay that `hurst_diff` measures.

Chapter 12

Generative Models

Before and After

Before this chapter you need: logarithms (§2.3), integrals (§3.5), norms (§5.5), densities and expectation (§6.6–§6.7), maximum likelihood (§7.3), gradient descent and minimax (§10.4, §10.8), neural networks (Chapter 11).
After it you will understand: what a generative model is trying to learn, what entropy and KL divergence measure, why Jensen–Shannon divergence gives no usable gradient when the two distributions do not overlap, and how Kantorovich–Rubinstein duality turns Wasserstein distance into something a neural network can compute — which is where $\|\nabla D\|_2 = 1$ finally comes from.

A generative model learns to produce new samples resembling its training data. This chapter builds the vocabulary for saying how close two distributions are, shows why the obvious choice fails, and derives the alternative the models in this thesis actually use.

12.1 Distributions over Data

Situation — the everyday picture

Someone shows you ten thousand photographs of faces. You could not write down a formula for "face", yet you would recognise an eleventh instantly, and you could sketch something face-like yourself. You have absorbed a distribution without ever representing it explicitly.

Problem — where it breaks or why it matters

The set of plausible return series is a vanishingly small corner of all possible number sequences. To generate new ones we must characterise that corner — and writing it down directly is hopeless, so it must be learned.

Worked Arithmetic — concrete numbers

How small is the corner? A window of 128 daily returns is a point in $\mathbb{R}^{128}$ (§5.1). Discretise crudely into 10 buckets per day. The space of sequences has

$$10^{128}$$

elements — more than the number of atoms in the observable universe, by roughly 10^{48}. BOVESPA's 20-year history supplies about 4,953 windows. The training data covers a fraction of the space equal to

$$\frac{4953}{10^{128}} \approx 5 \times 10^{-125}.$$

Yet the task is not hopeless, because the plausible sequences are not scattered uniformly. They concentrate: near-zero mean, standard deviation around 1.6%, volatility clustered, tails heavy. Those constraints carve out a low-dimensional surface inside $\mathbb{R}^{128}$, and it is that surface a generative model learns.

A concrete constraint. Requiring $|\mathrm{ACF}(1)| < 0.03$ alone eliminates the overwhelming majority of sequences in the space — a random sequence has no reason to satisfy it. Each stylised fact removes another vast portion.

Formal Name — the mathematics

A **generative model** approximates an unknown data distribution p_{data} over a space $\mathcal{X}$, either by

- providing a density $p_\theta(x)$ that can be evaluated, or
- providing a procedure that draws samples $x \sim p_\theta$.

These are different capabilities. GANs give the second without the first: they sample but cannot report a likelihood.
The **manifold hypothesis** holds that real data concentrates near a low-dimensional surface embedded in the high-dimensional space.

Where the Analogy Breaks

"Learning the distribution" overstates what any of these models do. They learn to produce samples that pass the tests we apply. A model can match every metric in this thesis and still differ from p_{data} in ways no metric probes — which is exactly why the evaluation framework is built around a taxonomy of what each metric can and cannot see, rather than around a single score.

Why It Matters Here — link to the thesis

The manifold picture explains why the shuffled-real control is so awkward. It lies *exactly* on the correct marginal distribution while sitting nowhere near the correct manifold, because the manifold is defined by ordering constraints that permutation destroys. Any distance that ignores ordering rates it perfect.

12.2 Latent Variables and Sampling

Situation — the everyday picture

A pipe organ. A handful of stops and one keyboard produce an enormous variety of sounds. The controls are few; the outputs are many and richly structured, because the mechanism between them does the work.

Problem — where it breaks or why it matters

We need to generate points on a complicated surface in $\mathbb{R}^{128}$. Sampling directly from a complicated surface is hard. Sampling from something simple and transforming it is easy —

provided the transformation can be learned.

Worked Arithmetic — concrete numbers

The recipe. Draw $\mathbf{z}$ from a simple distribution, then apply a learned map G:

$$\mathbf{z} \sim \mathcal{N}(0, I) \in \mathbb{R}^{100}, \qquad \mathbf{x} = G(\mathbf{z}) \in \mathbb{R}^{128}.$$

Transformation reshapes distributions. Take z uniform on $[0, 1]$ and $x = z^2$. Uniform draws $0.1, 0.5, 0.9$ map to $0.01, 0.25, 0.81$. The inputs are evenly spread; the outputs bunch near zero. A simple squaring has produced a non-uniform distribution.

Check the density. For $x = z^2$ with z uniform, $\mathbb{P}(x \le a) = \mathbb{P}(z \le \sqrt{a}) = \sqrt{a}$, so the density is $\frac{d}{da}\sqrt{a} = \frac{1}{2\sqrt{a}}$ — unbounded near 0, far from uniform. ✓ A trivial map has produced a non-trivial distribution; a neural network can produce a very complicated one.

Dimensions in this project. QuantGAN and FinGAN draw $\mathbf{z} \in \mathbb{R}^{100}$; TimeGAN uses a 24-dimensional noise vector matched to its hidden dimension. In every case the latent space is far smaller than the 128-dimensional output, which is the manifold hypothesis built into the architecture.

Formal Name — the mathematics

A **latent variable model** specifies a prior $p(\mathbf{z})$ — usually $\mathcal{N}(0, I)$ — and a map $G_\theta : \mathcal{Z} \to \mathcal{X}$. Sampling is: draw $\mathbf{z} \sim p(\mathbf{z})$, return $G_\theta(\mathbf{z})$.
The induced density on $\mathcal{X}$ is

$$p_\theta(\mathbf{x}) = \int p(\mathbf{z})\, \delta\big(\mathbf{x} - G_\theta(\mathbf{z})\big)\, d\mathbf{z},$$

which is generally intractable. When G is invertible with a computable Jacobian determinant (§5.4), the change-of-variables formula gives the density explicitly — the basis of normalising flows. GANs impose no such restriction and forgo the density.

Where the Analogy Breaks

A low-dimensional latent space is a modelling assumption with teeth. If the true data manifold has higher dimension than $\mathcal{Z}$, the generator cannot cover it whatever its training — some genuine outputs are simply unreachable. Too large a latent space wastes capacity and can make training unstable. Neither failure is visible in the loss.

Why It Matters Here — link to the thesis

The generator is a function from noise to a return series, and sampling is a single forward pass. This is why generating a synthetic dataset costs almost nothing once training is done, and why a collapsed generator is so dangerous: if G maps every $\mathbf{z}$ to nearly the same output, sampling a million series yields a million near-copies, and only a diversity check will notice.

12.3 Maximum Likelihood as a Training Objective

Situation — the everyday picture

Section 7.3 found a coin's bias by choosing the value that made the observed flips most probable. The same principle should extend to fitting a generator: choose parameters making the observed data most probable.

Problem — where it breaks or why it matters

The principle extends; the computation does not. For a latent-variable generator the likelihood involves an integral over all latent values that could have produced a given output, and that integral has no closed form.

Worked Arithmetic — concrete numbers

Where it works. Fit $\mathcal{N}(\mu, \sigma^2)$ to three observations $2, 4, 6$. The log-likelihood is

$$\ell(\mu, \sigma) = -\frac{3}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_i (x_i - \mu)^2.$$

Differentiating in μ and setting to zero gives $\hat{\mu} = \bar{x} = 4$; differentiating in σ gives

$$\hat{\sigma}^2 = \tfrac{1}{3}\left[(2-4)^2 + (4-4)^2 + (6-4)^2\right] = \tfrac{8}{3} = 2.667.$$

Closed form, done.

Where it fails. For $\mathbf{x} = G_\theta(\mathbf{z})$ the density at an observed $\mathbf{x}$ requires

$$p_\theta(\mathbf{x}) = \int p(\mathbf{z})\, \delta\big(\mathbf{x} - G_\theta(\mathbf{z})\big)\, d\mathbf{z},$$

an integral over $\mathbb{R}^{100}$. There is no formula, and numerical integration in 100 dimensions is out of reach.

Worse than intractable. If the generator's output lies exactly on a 100-dimensional surface inside $\mathbb{R}^{128}$, then any real data point off that surface has density *exactly zero*, so $\ln p_\theta(\mathbf{x}) = -\infty$. The objective is not merely hard to compute; it is infinite for almost every real observation. Section 12.7 shows this is the same difficulty in another guise.

Formal Name — the mathematics

Maximum likelihood for a generative model maximises

$$\frac{1}{N}\sum_{i=1}^{N} \ln p_\theta(x_i),$$

which as $N \to \infty$ approaches $\mathbb{E}_{p_{\text{data}}}[\ln p_\theta(x)]$ — equivalent to minimising $D_{\text{KL}}(p_{\text{data}} \,\|\, p_\theta)$ (§12.6).
Tractable for: explicit densities, autoregressive models, normalising flows. Intractable for: implicit generators such as GANs, which is why they optimise a divergence estimated adversarially instead.

Where the Analogy Breaks

Maximum likelihood is not automatically the right target even where it is computable. Minimising $D_{\mathrm{KL}}(p_{\mathrm{data}} \parallel p_\theta)$ penalises assigning low probability to real data, but barely penalises assigning high probability to nonsense. Models trained this way tend to over-spread, hedging across the whole space — which for return generation means producing implausible series rather than admitting ignorance.

Why It Matters Here — link to the thesis

This is the structural reason the models here are adversarial. A likelihood-based generator would need a tractable density, which for the sequence structure at hand would mean strong architectural restrictions. GANs trade the likelihood away for freedom of architecture — and the price is that no model in this thesis can report how probable a given real series is under it, so evaluation must be entirely sample-based.

12.4 Entropy

Situation — the everyday picture

Two weather forecasters. One says "rain" every day in a climate where it rains half the time; the other reports genuine daily odds. Over a year, how many yes/no questions would you need to pin down the actual weather? That number measures how uncertain the weather is.

Problem — where it breaks or why it matters

Before distributions can be compared, we need a way to quantify the uncertainty in a single one — and it should have the property that a predictable distribution scores low and an unpredictable one scores high.

Worked Arithmetic — concrete numbers

A fair coin. Two outcomes, each 1/2:

$$H = -\left[\tfrac{1}{2}\log_2\tfrac{1}{2} + \tfrac{1}{2}\log_2\tfrac{1}{2}\right] = -\left[\tfrac{1}{2}(-1) + \tfrac{1}{2}(-1)\right] = 1 \text{ bit}.$$

One yes/no question suffices, and one is needed.

A biased coin, $p = 0.9$.

$$H = -[0.9\log_2 0.9 + 0.1\log_2 0.1] = -[0.9(-0.152) + 0.1(-3.322)] = 0.137 + 0.332 = 0.469 \text{ bits}.$$

Less than half. The outcome is usually heads, so less information arrives per flip.

A certain coin, $p = 1$. $H = -1\log_2 1 = 0$. No uncertainty, no information.

A fair die. Six outcomes each 1/6:

$$H = \log_2 6 = 2.585 \text{ bits}.$$

The pattern. Entropy is maximised by the uniform distribution and minimised by a certain one. For n equally likely outcomes it is exactly $\log_2 n$.

Formal Name — the mathematics

The **Shannon entropy** of a discrete distribution is

$$H(p) = -\sum_x p(x) \log p(x),$$

in bits with $\log_2$, in nats with ln. For a continuous density the **differential entropy** is

$$h(p) = -\int p(x) \ln p(x)\, dx.$$

Entropy is non-negative in the discrete case, maximised by the uniform distribution over a finite set, and equal to zero only for a point mass.

Where the Analogy Breaks

Differential entropy does not behave like the discrete kind. It can be negative — a narrow Gaussian has negative differential entropy — and it changes under a change of units, since rescaling x shifts h by a constant. It is not "the amount of information" in any absolute sense. Differences of entropies, which is what every divergence below uses, are better behaved.

Why It Matters Here — link to the thesis

Entropy is the building block for cross-entropy and KL divergence, which is how the original GAN objective is expressed. It also gives the right language for mode collapse: a generator producing one output has near-zero entropy, while the data it was meant to imitate has high entropy. That gap is what the diversity guards in this project detect.

12.5 Cross-Entropy

Situation — the everyday picture

You design a code for weather reports, assuming rain and sun are equally likely, so each gets a one-bit symbol. The real climate is rainy 90% of the time. Your code works, but wastes space — you gave a common event as many bits as a rare one.

Problem — where it breaks or why it matters

We need to measure the cost of using the *wrong* distribution when the truth is something else. That single quantity turns out to be the standard loss function for classification, and it is what the original GAN discriminator minimises.

Worked Arithmetic — concrete numbers

Truth $p = (0.9, 0.1)$; assumed model $q = (0.5, 0.5)$.

Cost under the correct code — the entropy:

$$H(p) = 0.469 \text{ bits (from §12.4)}.$$

Cost under the wrong code — the cross-entropy:

$$H(p, q) = -\big[0.9 \log_2 0.5 + 0.1 \log_2 0.5\big] = -\log_2 0.5 = 1 \text{ bit}.$$

The waste: $1 - 0.469 = 0.531$ bits per report. That excess is the KL divergence of §12.6.

A better model. With $q = (0.8, 0.2)$:

$$H(p,q) = -\big[0.9 \log_2 0.8 + 0.1 \log_2 0.2\big] = -[0.9(-0.322)+0.1(-2.322)] = 0.290+0.232 = 0.522.$$

Waste now $0.522 - 0.469 = 0.053$ bits. Closer model, smaller excess. ✓

As a classifier loss. With a true label $y \in \{0,1\}$ and predicted probability $\hat{y}$, binary cross-entropy is

$$\ell = -\big[y \ln \hat{y} + (1-y)\ln(1-\hat{y})\big].$$

At $y = 1$ and $\hat{y} = 0.9$: $\ell = -\ln 0.9 = 0.105$. At $\hat{y} = 0.5$: $\ell = 0.693$. At $\hat{y} = 0.01$: $\ell = 4.605$. Confident and wrong is punished heavily; the loss diverges as $\hat{y} \to 0$.

Formal Name — the mathematics

The **cross-entropy** of q relative to p is

$$H(p,q) = -\sum_x p(x) \log q(x),$$

the expected coding cost of using q when the truth is p. It satisfies $H(p,q) \geq H(p)$, with equality only when $q = p$ (Gibbs' inequality).
Binary cross-entropy is the two-outcome case and is the standard classification loss.

Where the Analogy Breaks

Cross-entropy is not symmetric: $H(p,q) \neq H(q,p)$. It is a measure of the cost of using q to describe p, and swapping the roles is a different question. It is also unbounded — a single confident, wrong prediction contributes an arbitrarily large loss, which makes it sensitive to mislabelled data.

Why It Matters Here — link to the thesis

Binary cross-entropy is the discriminator's loss in the original GAN formulation of Chapter 13. Its unboundedness is part of the instability story: when the discriminator becomes confident, the generator's gradient can vanish or explode, which is one of the motivations for the Wasserstein alternative.

12.6 KL Divergence

Situation — the everyday picture

The wasted bits from the previous section — the excess cost of the wrong code — deserve a name of their own, because that excess is exactly a measure of how far apart two distributions are.

Problem — where it breaks or why it matters

We want a number that is zero when two distributions agree and grows as they diverge. The coding waste supplies one, and it is the most widely used such quantity — but it has a defect that turns out to matter enormously here.

Worked Arithmetic — concrete numbers

From the previous section. $p = (0.9, 0.1)$, $q = (0.5, 0.5)$:

$$D_{\mathrm{KL}}(p \parallel q) = H(p, q) - H(p) = 1 - 0.469 = 0.531 \text{ bits.}$$

Directly:

$$D_{\mathrm{KL}}(p \parallel q) = 0.9 \log_2 \frac{0.9}{0.5} + 0.1 \log_2 \frac{0.1}{0.5} = 0.9(0.848) + 0.1(-2.322) = 0.763 - 0.232 = 0.531. \quad \checkmark$$

Asymmetry, checked.

$$D_{\mathrm{KL}}(q \parallel p) = 0.5 \log_2 \frac{0.5}{0.9} + 0.5 \log_2 \frac{0.5}{0.1} = 0.5(-0.848) + 0.5(2.322) = -0.424 + 1.161 = 0.737.$$

$0.531 \neq 0.737$. The two directions give different answers, so KL is not a distance.

The fatal case. Let q assign probability zero where p does not: $p = (0.5, 0.5)$, $q = (1, 0)$. Then

$$D_{\mathrm{KL}}(p \parallel q) = 0.5 \log_2 \frac{0.5}{1} + 0.5 \log_2 \frac{0.5}{0} = \infty.$$

Any region the model rules out but the data occupies makes KL infinite — not large, infinite. And an infinite loss has no usable gradient.

Formal Name — the mathematics

The **Kullback–Leibler divergence** is

$$D_{\mathrm{KL}}(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \qquad \text{or} \qquad \int p(x) \log \frac{p(x)}{q(x)}\, dx.$$

Properties: $D_{\mathrm{KL}} \geq 0$, with equality only when $p = q$; $D_{\mathrm{KL}}(p \parallel q) \neq D_{\mathrm{KL}}(q \parallel p)$; it violates the triangle inequality; and it is $+\infty$ whenever $q(x) = 0$ at some x with $p(x) > 0$.

Where the Analogy Breaks

The two directions encode opposite preferences. Minimising $D_{\mathrm{KL}}(p_{\mathrm{data}} \parallel p_\theta)$ punishes the model for ignoring any region the data occupies, producing over-spread, everything-is-possible models. Minimising the reverse punishes putting mass where the data has none, producing models that lock onto one mode and ignore the rest. Mode-seeking versus mode-covering is a choice of direction, not a tuning parameter.

Why It Matters Here — link to the thesis

The infinity is the crux. A generator's output lies on a low-dimensional surface in $\mathbb{R}^{128}$ (§12.1), so early in training the model and the data occupy essentially disjoint regions — and KL is infinite everywhere. It reports that the model is wrong without indicating any direction of improvement, which is precisely what a gradient must supply.

12.7 Jensen–Shannon Divergence and Disjoint Support

Situation — the everyday picture

Two islands with no bridge. Asked how to get from one to the other, an unhelpful guide says only "you cannot". True, and useless — you wanted a direction to start walking, not a verdict.

Problem — where it breaks or why it matters

JS divergence fixes KL's asymmetry and its infinities, and is what the original GAN objective implicitly minimises. It nonetheless fails in the same situation, for a related reason: it becomes *constant*, and a constant has zero gradient.

Worked Arithmetic — concrete numbers

Two point masses that do not overlap. Let p put all its mass at 0 and q all its mass at $\theta \neq 0$. Their mixture $m = \frac{1}{2}(p+q)$ puts $1/2$ at each location.

$$D_{\mathrm{KL}}(p \parallel m) = 1 \cdot \log_2 \frac{1}{1/2} = 1,$$

and likewise $D_{\mathrm{KL}}(q \parallel m) = 1$. So

$$\mathrm{JSD}(p,q) = \tfrac{1}{2}(1) + \tfrac{1}{2}(1) = 1 \text{ bit}.$$

Now move them closer. Set $\theta = 100$, then $\theta = 10$, then $\theta = 0.001$. As long as the supports do not overlap, the calculation is identical:

θ	JSD	W_1
100	1	100
10	1	10
1	1	1
0.001	1	0.001
0	0	0

Read the JSD column. It is 1 everywhere and then drops discontinuously to 0. Its derivative with respect to θ is zero everywhere it is defined. Gradient descent on JSD gets no signal at all — the objective is flat, so the generator has no indication that $\theta = 0.001$ is nearly right and $\theta = 100$ is badly wrong.

Read the W_1 column. It equals $|\theta|$: smooth, non-zero derivative everywhere, pointing directly at the answer. This single comparison is the entire argument for Wasserstein GANs.

Formal Name — the mathematics

The **Jensen–Shannon divergence** is

$$\mathrm{JSD}(p,q) = \tfrac{1}{2} D_{\mathrm{KL}}(p \parallel m) + \tfrac{1}{2} D_{\mathrm{KL}}(q \parallel m), \qquad m = \tfrac{1}{2}(p+q).$$

It is symmetric, always finite, bounded above by 1 bit, and $\sqrt{\mathrm{JSD}}$ is a genuine metric. Its defect: when p and q have disjoint support, $\mathrm{JSD} = \log 2$ regardless of how far apart they are, so its gradient with respect to the generator's parameters is zero.

Where the Analogy Breaks

It is tempting to dismiss disjoint support as a pathological edge case. It is the normal situation. A generator mapping $\mathbb{R}^{100}$ into $\mathbb{R}^{128}$ produces output confined to a set of measure zero, so the model and data distributions are almost surely disjoint at initialisation and remain so until they nearly coincide. Adding noise to both restores overlap and is a real remedy, at the cost of blurring what is being matched.

Why It Matters Here — link to the thesis

This is the theoretical case for WGAN, and it is why QuantGAN and FinGAN in this project use a WGAN-GP objective rather than the original cross-entropy formulation. Chapter 13 sets out the practical consequences — and the ways the Wasserstein formulation fails to deliver everything the theory promises.

12.8 Optimal Transport and the Wasserstein Distance

Situation — the everyday picture

Two piles of sand on a table, one the distribution of real returns and the other of synthetic returns. How different are they? Ask for the minimum work — mass times distance — needed to reshape the first pile into the second.

Problem — where it breaks or why it matters

KL and JS compare distributions *pointwise*, asking how the densities differ at each location, which is why they break when the supports do not meet. Transporting mass instead uses the geometry of the underlying space, so distributions that are close in position are close in value even when they never overlap.

Worked Arithmetic — concrete numbers

One dimension makes it easy. Sort both samples and pair by rank; W_1 is the average absolute gap.
Real $\{1, 3, 5\}$, synthetic $\{2, 3, 4\}$:

$$W_1 = \tfrac{1}{3}(|1-2| + |3-3| + |5-4|) = \tfrac{1}{3}(1+0+1) = 0.667.$$

Why sorted pairing is optimal. Any crossing pair can be uncrossed without increasing the cost. Pairing $1 \to 4$ and $5 \to 2$ costs $3+3=6$; uncrossing to $1 \to 2$ and $5 \to 4$ costs $1+1=2$. Uncrossing never hurts, so the sorted matching is optimal. ✓

The two point masses again. p at 0, q at θ: all the mass moves a distance θ, so $W_1 = |\theta|$ — matching the table in §12.7 and confirming the gradient is ± 1, never zero.

The shuffled control. $W_1 = 0.000\text{e+}00$ exactly. A permutation of the real data sorts to the identical sequence, so every paired gap is zero.

A floor that matters. Suppose a generator collapses to emitting the constant μ. Then W_1 becomes the mean absolute deviation of the real data — for BOVESPA, 0.00770. That is the *worst* score a constant-output model can achieve on this metric, and a genuine but imperfect generator can easily do worse than 0.00770. On W_1 alone, a collapsed model can beat a working one.

Formal Name — the mathematics

The **Wasserstein-p distance** is

$$W_p(P,Q) = \left(\inf_{\gamma \in \Gamma(P,Q)} \mathbb{E}_{(X,Y)\sim\gamma} \|X - Y\|^p \right)^{1/p},$$

the infimum over **couplings** γ — joint distributions whose marginals are P and Q. Each coupling is a transport plan; $\gamma(x,y)$ is the mass moved from x to y.
In one dimension with $p = 1$ this collapses to

$$W_1(P,Q) = \int_{-\infty}^{\infty} |F_P(x) - F_Q(x)|\, dx = \int_0^1 |F_P^{-1}(u) - F_Q^{-1}(u)|\, du.$$

W_p is a genuine metric: symmetric, zero only when $P = Q$, and satisfying the triangle inequality.

Where the Analogy Breaks

W_1 is permutation-invariant, so $W_1(\text{real}, \text{shuffled}) = 0$. That is correct — the two have identical distributions — and a hard limitation, since it is blind to ordering by construction. No refinement of the computation will make it detect a shuffle, which is why it belongs in the fidelity family and why the evaluation needs a second family alongside it.
The optimisation is also genuinely hard above one dimension: the infimum over couplings has no closed form and exact solution costs roughly $O(n^3)$. One dimension is the lucky case, and it is the case this project's metric occupies.

Why It Matters Here — link to the thesis

`wasserstein` is one of the seven ranked fidelity metrics, computed on the sorted one-dimensional samples. The MAD floor above is the reason it cannot be read in isolation: Chapter 15 shows how the composite design and the degeneracy guards together prevent a collapsed model from winning on it.

12.9 Kantorovich–Rubinstein Duality

Situation — the everyday picture

Two ways to price a removals job. Either plan every item's journey and total the work — hard, with a vast number of possible plans. Or ask what the most aggressive honest valuer would charge, given that their pricing cannot be arbitrarily steep. Remarkably, the two answers agree.

Problem — where it breaks or why it matters

The definition of W_1 is an infimum over all couplings, which is not something a neural network can compute. To train against Wasserstein distance we need a form involving a maximisation over *functions* instead — because maximising over functions is exactly what a network does.

Worked Arithmetic — concrete numbers

The dual form.

$$W_1(P,Q) = \sup_{\|f\|_L \le 1} \Big(\mathbb{E}_{x\sim P}[f(x)] - \mathbb{E}_{y\sim Q}[f(y)] \Big),$$

where the supremum runs over all 1-Lipschitz functions: those with $|f(a) - f(b)| \le |a - b|$ for all a, b.

Check it on the two point masses. P at 0, Q at $\theta > 0$. Then

$$\mathbb{E}_P[f] - \mathbb{E}_Q[f] = f(0) - f(\theta).$$

The Lipschitz condition caps this: $|f(0) - f(\theta)| \le |\theta|$. The bound is attained by $f(x) = -x$, which is exactly 1-Lipschitz and gives $f(0) - f(\theta) = \theta$. So the supremum is $\theta = W_1$. ✓

Why the constraint is essential. Without it, take $f(x) = -Kx$ for large K: the difference becomes $K\theta$, unbounded. The supremum would be $+\infty$ for any two distinct distributions and would measure nothing. The Lipschitz bound is what makes the dual finite and meaningful.

A three-point check. P uniform on $\{0, 2\}$, Q uniform on $\{1, 3\}$. Primal: pair $0 \to 1$ and $2 \to 3$, each distance 1, so $W_1 = 1$. Dual with $f(x) = -x$:

$$\mathbb{E}_P[f] - \mathbb{E}_Q[f] = \tfrac{1}{2}(0 + (-2)) - \tfrac{1}{2}((-1) + (-3)) = -1 - (-2) = 1. \quad ✓$$

Formal Name — the mathematics

A function f is K**-Lipschitz** if

$$|f(a) - f(b)| \le K\ \|a - b\| \quad \text{for all } a, b.$$

For differentiable f this is equivalent to $\|\nabla f\| \le K$ everywhere.
Kantorovich–Rubinstein duality:

$$W_1(P,Q) = \sup_{\|f\|_L \le 1} \Big(\mathbb{E}_P[f] - \mathbb{E}_Q[f] \Big).$$

The maximising f is called the **Kantorovich potential**; in a GAN it is approximated by the **critic**.

Where the Analogy Breaks

Duality is exact; the neural approximation is not. A network with a Lipschitz constraint is not the full space of 1-Lipschitz functions, so the supremum is taken over a restricted family and the result is a lower bound on W_1. Nor is the supremum reached: the critic is trained for a finite number of steps and merely approaches it. What the generator descends is therefore an estimate of an estimate — a gap Chapter 13 takes seriously.

Why It Matters Here — link to the thesis

This is where $\|\nabla D\|_2 = 1$ comes from, and every piece is now in place. Enforcing 1-Lipschitz continuity means bounding the critic's gradient norm by 1 (§5.5). WGAN-GP enforces it softly, as the penalty of §10.7, adding

$$\lambda\, \mathbb{E}\Big[\big(\|\nabla D(\hat{x})\|_2 - 1\big)^2\Big]$$

with $\lambda = 10$ throughout this project. The target is 1 rather than "at most 1" because the optimal critic saturates the constraint — as $f(x) = -x$ did above, achieving exactly slope 1 where it matters.

12.10 Diffusion Models and Score Matching

Situation — the everyday picture

A drop of ink in still water spreads until the glass is uniformly grey. Running that backwards — grey water reassembling into a drop — is impossible in physics. But if you had recorded, at every instant, which way the ink was spreading, you could retrace it.

Problem — where it breaks or why it matters

GANs avoid likelihoods at the cost of adversarial instability (§10.8). An alternative keeps a tractable training objective and no second player at all, by learning to reverse a noising process step by step.

Worked Arithmetic — concrete numbers

Forward: destroy the data. Add Gaussian noise in small steps:

$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\, \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, I).$$

With $\beta_t = 0.01$ held fixed, the signal retained after t steps is $(1-0.01)^{t/2}$:

t	signal retained $0.99^{t/2}$	
10	0.951	mostly intact
100	0.605	degraded
1000	0.0067	essentially pure noise

After 1,000 steps the data is gone and the distribution is $\mathcal{N}(0, I)$ — which we can sample from trivially.

Backward: learn to undo one step. Train a network ε_θ to predict the noise that was added:

$$L = \mathbb{E}_{t,x_0,\varepsilon}\left[\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2\right].$$

This is an ordinary squared-error regression. No adversary, no minimax, no Lipschitz constraint — just a target the network can be scored against directly.

Generation. Start from pure noise and apply the learned reverse step 1,000 times. The cost is 1,000 network evaluations per sample against a GAN's one, which is the central practical trade.

Formal Name — the mathematics

A **denoising diffusion probabilistic model** (Ho et al. 2020) defines a forward chain adding Gaussian noise over T steps and learns the reverse. The training objective is a variational bound on the log-likelihood that reduces to the simple noise-prediction regression above. The **score** of a density is $\nabla_x \log p(x)$, and **score matching** learns it directly; predicting the added noise is equivalent to estimating the score up to a scale factor. Sampling then follows the score uphill toward regions of high density.

Where the Analogy Breaks

Diffusion models trade instability for cost. Training is stable — an ordinary regression — and sampling is one to three orders of magnitude slower than a GAN, since it requires hundreds or thousands of forward passes. They also assume the Gaussian noising process is appropriate to the data, which for heavy-tailed financial returns is a real question rather than a formality: the forward process attenuates precisely the extreme values that matter most here.

Why It Matters Here — link to the thesis

Diffusion-TS is the intended fourth baseline for this benchmark and is not yet implemented. The evaluation framework is deliberately model-agnostic — it scores samples, never likelihoods or internals — so adding a diffusion model requires no change to the metrics. The parity question of §10.6 will need care, though: "equal generator updates" is well defined for the three GANs, and a diffusion model has no adversarial generator step, so a defensible common budget would have to be argued rather than assumed.

Chapter 13

Generative Adversarial Networks

13.1 The GAN Idea

Situation — the everyday picture

A counterfeiter tries to make fake currency. A bank teller learns to detect fakes. Each tries to fool or catch the other. Over time, both get better — the counterfeiter produces more convincing fakes, the teller develops sharper detection skills. Eventually, the fakes may be indistinguishable from real currency.

Problem — where it breaks or why it matters

We want a machine that generates realistic market return sequences. A single neural network trained directly is hard to evaluate: how do you define a loss function that measures "realism" for a complex, high-dimensional sequence? The GAN answer: train a second network to play the role of the expert bank teller. It provides the loss signal automatically.

Formal Name — the mathematics

A **GAN** (Goodfellow et al., 2014) consists of two neural networks:

- **Generator** $G : \mathcal{Z} \to \mathcal{X}$: maps random noise $z \sim p_z$ to a synthetic sample $\tilde{x} = G(z)$
- **Discriminator** $D : \mathcal{X} \to [0, 1]$: classifies inputs as real ($D(x) = 1$) or synthetic ($D(\tilde{x}) = 0$)

Training is a minimax game:

$$\min_G \max_D \; \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p_z}[\log(1 - D(G(z)))]$$

At the Nash equilibrium: $p_G = p_{\text{data}}$ and $D(x) = 1/2$ everywhere.

Where the Analogy Breaks

The Nash equilibrium is a theoretical ideal. In practice, GAN training is unstable: mode collapse (generator produces only a few outputs), discriminator dominance (generator gradients vanish), and oscillation without convergence are common. WGAN-GP addresses the most severe of these issues.

Why It Matters Here — link to the thesis

All three generators in this thesis (TimeGAN, QuantGAN, FinGAN) are GANs. The GAN framework is chosen because it implicitly defines the target distribution through adversarial training rather than requiring an explicit density. This is essential for market returns, whose true density is unknown and non-Gaussian.

13.2 WGAN and the Wasserstein Distance

Situation — the everyday picture

The original GAN discriminator outputs a probability $\in [0, 1]$. When the generator's distribution is far from the real one, the discriminator saturates: it assigns near-zero probability to fake samples, and the gradient of $\log(1 - D(G(z)))$ vanishes. The generator cannot learn which direction to improve.

Problem — where it breaks or why it matters

Vanishing gradients in early training. Mode collapse (generator produces a single realistic example to fool the fixed discriminator). Training instability. These problems worsen when the real and generated distributions do not overlap — which is always the case early in training.

Formal Name — the mathematics

Wasserstein GAN (Arjovsky et al., 2017) replaces the discriminator with a **critic** $f : \mathcal{X} \to \mathbb{R}$ (no sigmoid) and minimises the Wasserstein-1 distance:

$$W_1(p_{\text{data}}, p_G) = \sup_{\|f\|_L \le 1} \mathbb{E}_{x \sim p_{\text{data}}}[f(x)] - \mathbb{E}_{z \sim p_z}[f(G(z))]$$

by the Kantorovich-Rubinstein duality, where the supremum is over 1-Lipschitz functions.
WGAN-GP (Gulrajani et al., 2017) enforces the Lipschitz constraint via a gradient penalty:

$$\mathcal{L}_{GP} = \lambda \cdot \mathbb{E}_{\hat{x} \sim p_{\hat{x}}} \left[(\|\nabla_{\hat{x}} f(\hat{x})\|_2 - 1)^2 \right]$$

where $\hat{x} = \varepsilon x_{\text{real}} + (1 - \varepsilon) x_{\text{fake}}$, $\varepsilon \sim U[0, 1]$. Standard values: $\lambda = 10$, $n_{\text{critic}} = 5$.
Why $n_{\text{critic}} = 5$: the critic must accurately estimate W_1 before providing a useful gradient to the generator. Fewer updates underfit the critic, giving a biased gradient signal. 5 is the empirically validated minimum (Gulrajani et al., 2017).
Why $\beta_1 = 0$ in Adam: in adversarial training, the correct gradient direction for the generator reverses each time the critic is updated. Momentum ($\beta_1 > 0$) accumulates the previous direction and corrupts the current gradient. $\beta_1 = 0$ uses only the current gradient.

Where the Analogy Breaks

The gradient penalty is computed at interpolated points, not directly on training data. This means the Lipschitz constraint is only approximately enforced (not everywhere). In practice, WGAN-GP is far more stable than weight clipping (Arjovsky 2017) or spectral normalisation, and remains the standard.

Why It Matters Here — link to the thesis

Both QuantGAN and FinGAN use WGAN-GP with $n_{\text{critic}} = 5$, $\lambda_{gp} = 10$, Adam $\beta_1 = 0$, $\beta_2 = 0.9$. These hyperparameters are floors, not arbitrary choices — using $n_{\text{critic}} = 1$ or $\lambda_{gp} = 1$ would invalidate the Wasserstein approximation. This is documented in ED Table 3 of the paper.

13.3 TimeGAN

Situation — the everyday picture

You want to teach a machine to write convincing diary entries. First, you teach it to encode each entry into a compressed representation and decode back (autoencoder). Then you teach it to predict the next entry from the previous ones (supervisor). Only after these skills are established do you introduce the adversarial game.

Formal Name — the mathematics

TimeGAN (Yoon et al., 2019) has four components:

- **Embedder** $e : \mathcal{X} \to \mathcal{H}$: encodes real sequences to latent space
- **Recovery** $\hat{r} : \mathcal{H} \to \mathcal{X}$: decodes latent back to sequence
- **Supervisor** $s : \mathcal{H} \to \hat{\mathcal{H}}$: predicts next latent step from current
- **Generator** $G : \mathcal{Z} \to \mathcal{H}$: maps noise to latent sequences

Four training phases:

1. **Phase 1 — Autoencoder:** $\min \|X - \hat{r}(e(X))\|^2$
2. **Phase 2 — Supervisor:** $\min \|H_{t+1} - s(H_t)\|^2$ (temporal causality in latent space)
3. **Phase 3 — Joint adversarial:**

$$\mathcal{L}_G = \mathcal{L}_U + 100\sqrt{\mathcal{L}_S} + 100\,\mathcal{L}_V$$

   where $\mathcal{L}_U$ = unsupervised GAN loss, $\mathcal{L}_S$ = supervised latent loss, $\mathcal{L}_V$ = moment-matching on mean/std
4. **Phase 4 — Fine-tuning:** refine recovery with generated sequences

Backbone: GRU (Gated Recurrent Unit). Hidden dim = 24, 3 layers, lr = 10^{-3}.

Where the Analogy Breaks

The latent-space supervisor is the key innovation: by training the supervisor to predict the next latent state, TimeGAN forces the latent space to encode temporal causality. However, this requires many gradient steps per phase to work. With `epochs=5` (as in the current smoke-test pipeline), each phase gets approximately 2 gradient steps — far too few for the GRU to form any useful representation. TimeGAN results in the current draft should be interpreted as lower bounds.

Why It Matters Here — link to the thesis

TimeGAN's poor temporal ranking in the current results reflects training budget, not architectural weakness. The recommended training is $\geq 5{,}000$ iterations per phase (or ≥ 50 epochs at the current batch size). The RunPod production run will use adequate epochs; the current paper's TimeGAN results are explicitly flagged as undertrained.

13.4 QuantGAN

Situation — the everyday picture

Instead of processing a sequence step-by-step (like a recurrent network), imagine looking at the whole sequence at once with multiple "zoom levels": a short zoom sees the last 2 days, a medium zoom sees the last week, a long zoom sees the last month. The network combines all zoom levels simultaneously to capture structure at every timescale.

Formal Name — the mathematics

QuantGAN (Wiese et al., 2020) uses a **Temporal Convolutional Network** (TCN) with *causal dilated convolutions*:

- **Causal:** the convolution at time t only sees inputs at times $\leq t$ (no future leakage)
- **Dilated:** dilation d means the filter sees steps $t, t-d, t-2d, \dots$ (stride-d subsampling in the receptive field)

With K residual blocks of dilations $1, 2, 4, \dots, 2^{K-1}$, the receptive field $= 2^K$ steps.
Three TCN blocks with dilations 1/2/4: receptive field = 8 steps. For a 128-step sequence, this covers the full window.
WGAN-GP with $n_{\text{critic}} = 5$, $\lambda_{gp} = 10$. Noise dim = 100, lr = 10^{-4}.

Where the Analogy Breaks

The parallel multi-scale processing is the key advantage over RNNs: a GRU must "remember" information across many steps using a hidden state of fixed size, leading to gradient issues for long horizons. A dilated TCN accesses all timescales in one forward pass without vanishing gradients. The tradeoff: TCN is less natural for autoregressive modelling (each step predicts the next), and its inductive bias may over-smooth long-range structure.

Why It Matters Here — link to the thesis

QuantGAN's TCN architecture is well-suited to the multi-scale structure of BRICS volatility: intraday effects (1–5 days), weekly patterns (5 days), and monthly cycles (20 days) are all covered by dilations 1/2/4 in a 128-step window. QuantGAN achieves the best composite ranking in current (smoke-test) results.

13.5 FinGAN

Situation — the everyday picture

Image-generating GANs start from a small noisy image (4×4 pixels) and progressively upsample it to full resolution (256×256), adding detail at each step. FinGAN adapts this idea to sequences: start from a short noise vector and upsample via transposed convolutions to a full 128-step return sequence.

Formal Name — the mathematics

FinGAN (this paper): CNN deconvolution WGAN-GP.

- Generator: $3\times$ `ConvTranspose1d` with stride 2, base channels = 64
- Critic: mirror architecture with `Conv1d` and LayerNorm (not BatchNorm)

- seq_len must be divisible by 8 ($= 2^3$ from 3 upsampling blocks)

Why LayerNorm in the critic, not BatchNorm: the WGAN-GP gradient penalty is computed at a single interpolated point $\hat{x}$. BatchNorm normalises across the batch, introducing a dependency on other batch members that corrupts the per-point gradient norm calculation. LayerNorm normalises across channels at each point independently.

Where the Analogy Breaks

The image-GAN inductive bias assumes hierarchical structure: coarse features first, fine details added later. Market return sequences may not have this structure — the fine-scale noise on any given day is not a refinement of the coarse-scale trend. This may explain FinGAN's weaker performance relative to QuantGAN, whose temporal convolutions are better suited to sequential dependencies.

Why It Matters Here — link to the thesis

FinGAN serves as a baseline for CNN-based approaches to financial time series. Its architecture is deliberately simple (3 upsampling blocks, no attention, no recurrence) to isolate the effect of the upsampling inductive bias. The sq_len divisibility-by-8 constraint must be checked before training on any new dataset.

Chapter 15

Evaluation: Metrics, Taxonomy, and the Shuffled-Control Audit

15.1 The Metric Taxonomy

Situation — the everyday picture

You built a machine that generates fake playing cards. You want to check whether the fakes are good. You could count the card values (does the fake deck have the right number of aces?) — that's a *fidelity* check. But you could also shuffle the deck and see if the pattern looks natural — that's a *temporal* check. Both are necessary; neither alone is sufficient.

Formal Name — the mathematics

We classify all evaluation metrics into three groups:

Fidelity metrics (7, permutation-invariant): measure the marginal distribution of r_t only. A metric m is permutation-invariant if $m(\pi(r)) = m(r)$ for any permutation π. These metrics score the shuffled control perfectly by construction.

- `mean_diff`: $|\bar{r}_{\text{real}} - \bar{r}_{\text{syn}}|$
- `std_diff`: $|\text{sd}(r_{\text{real}}) - \text{sd}(r_{\text{syn}})|$
- `wasserstein`: $W_1(p_{\text{real}}, p_{\text{syn}})$
- `quantile_mse`: $K^{-1}\sum_k (Q_{\text{real}}(\alpha_k) - Q_{\text{syn}}(\alpha_k))^2$
- `tail_index_diff`: $|\hat{\alpha}_{\text{real}} - \hat{\alpha}_{\text{syn}}|$ (Hill estimator, top 5%)
- `extreme_events_diff`: $|P_{\text{real}}(|r| > 2\sigma) - P_{\text{syn}}(|r| > 2\sigma)|$
- `energy_distance`: $2\mathbb{E}\|X - Y\| - \mathbb{E}\|X - X'\| - \mathbb{E}\|Y - Y'\|$

Temporal metrics (7, ordering-sensitive): measure properties that depend on the sequence of r_t. The shuffled control scores poorly on all of these.

- `acf_returns_mae`: ACF-MAE of r_t at lags 1–20 (tests fact 2)
- `acf_absolute_mae`: ACF-MAE of $|r_t|$ (tests volatility clustering, fact 3)
- `acf_squared_mae`: ACF-MAE of r_t^2 (tests ARCH effects, fact 6)
- `hurst_diff`: $|H_{\text{real}} - H_{\text{syn}}|$ applied to $|r_t|$ (tests fact 4)
- `resid_kurtosis_diff`: $|\kappa_{\text{real}} - \kappa_{\text{syn}}|$ for GARCH residuals (tests fact 7)
- `discriminative_auc_dist`: $|\text{AUC} - 0.5|$ from logistic classifier
- `discriminative_auc_absz`: $|(\text{AUC} - 0.506)/0.084|$ (z-score vs empirical null)

Descriptive metrics (7, excluded from composite): computed and displayed for diagnosis only.

- `kurtosis_diff`: $\mathbb{E}[X^4] = \infty$ for $\hat{\alpha} < 4$; sample values diverge with n

- `skewness_diff`: $\mathbb{E}[X^3] = \infty$ for $\hat{\alpha} < 3$ (4/5 BRICS markets)
- `arch_pvalue_diff`: saturates on real data ($|\Delta p| = 0.000000$)
- `arch_stat_diff`: null sd 46.4 on simulated GARCH; too noisy to rank
- `var_coverage_error`: unconditional VaR; cannot distinguish Gaussian noise from real
- `garch_persistence_diff`: sd 0.31 across seeds; range 0.000–0.985
- `discriminative_auc_raw`: raw AUC value (superseded by dist/absz variants)

Why It Matters Here — link to the thesis

The three-group taxonomy is the central methodological contribution of this paper. The composite score = (fidelity_rank + temporal_rank)/2 gives equal weight to both families regardless of how many individual metrics each contains. The shuffled control is always shown in the results table with NaN ranks — it serves as the adversarial baseline.

15.2 The Shuffled-Control Audit

Situation — the everyday picture

You buy a set of playing cards supposedly shuffled randomly. To test the shuffler, you check: (1) are the right cards present (4 of each value)? (2) is the ordering random? Test (1) passes trivially — you can count cards without looking at order. Test (2) is the harder test that actually verifies the shuffle.

Worked Arithmetic — concrete numbers

Shuffled control construction:

```
rng = numpy.random.default_rng(42)
shuffled = rng.permutation(pooled_real_array).copy()
```

Scores of shuffled control on Fidelity metrics (pooled test set):

Metric	Shuffled control
wasserstein	0.000e+00 (exact zero)
kurtosis_diff	1.78×10^{-15} (floating-point noise)
energy_distance	2.97×10^{-5}
mean_diff	$< 10^{-14}$
quantile_mse	$< 10^{-14}$
acf_absolute_mae	0.076 (correctly penalised)
hurst_diff	0.052 (correctly penalised)

Under the old unweighted 19-metric composite: avg_rank = 1.24 (shuffled) vs 2.47 (real-like generator). The shuffled deck won.

Formal Name — the mathematics

A metric m is **permutation-invariant** if for all $r \in \mathbb{R}^n$ and all permutations $\pi \in S_n$:

$$m(\pi(r_1, \ldots, r_n)) = m(r_1, \ldots, r_n)$$

Any metric that depends only on the *sorted* values (empirical CDF, moments, quantiles, order statistics) is permutation-invariant. This includes: Wasserstein, energy distance, KS statistic, kurtosis, skewness, mean, variance, tail index, extreme event rate.

A metric m is **ordering-sensitive** if $\exists\, r, \pi$ such that $m(\pi(r)) \neq m(r)$. ACF, Hurst exponent, GARCH-based metrics, and the discriminative classifier (which uses time-window features) are all ordering-sensitive.

Where the Analogy Breaks

The shuffled control is not a bad generator — it is the best possible fidelity generator (it *is* the real data). Its failure on temporal metrics is not a bug; it is the point. Any evaluation that does not include the shuffled control as a baseline cannot claim to have tested temporal structure.

Why It Matters Here — link to the thesis

The shuffled control is a permanent entry in the evaluation table (with NaN composite rank). It plays the same role as a positive control in a biology experiment: it is expected to score well on fidelity and poorly on temporal. A GAN that scores worse than the shuffled control on temporal metrics has learned nothing about market dynamics.

15.3 Discriminative AUC: Null Calibration and Direction Bug

Situation — the everyday picture

You train a classifier to tell real returns from synthetic returns. If it achieves 50% accuracy (random guessing), the synthetic data is indistinguishable from real. If it achieves 90%, the synthetic data is clearly different.

Worked Arithmetic — concrete numbers

Direction bug: under ascending rank (lower is better), AUC = 0.30 ranked above AUC = 0.50. But AUC = 0.50 is *better* (indistinguishable), and AUC = 0.30 is worse (the classifier is *anti-predictive* — it can tell real from fake by doing the opposite of its prediction). Fix: rank on $|\mathrm{AUC} - 0.5|$.
Null calibration: even with two samples from the same distribution, the classifier achieves AUC $\neq 0.5$ exactly. Empirical null (15 real-vs-real half-splits): 0.506 ± 0.084.
An observed AUC = 0.62: $z = (0.62 - 0.506)/0.084 = 1.36$, $p = 0.17$. Not significant.
CV shuffle bug: 20-day windows overlap by 19 observations. Adding `shuffle=True` to the K-fold cross-validation puts near-duplicate windows in both train and test, creating look-ahead leakage. Measured null shift: $0.506 \to 0.584$ with shuffled CV. Unshuffled CV is correct.

Formal Name — the mathematics

AUC (Area Under the ROC Curve) for a binary classifier: $\mathbb{P}(\hat{p}(x_{\mathrm{real}}) > \hat{p}(x_{\mathrm{syn}}))$.
The optimal value for a generator is AUC = 0.5 (chance level).
Our two ranking metrics:

$$\texttt{discriminative\_auc\_dist} = |\mathrm{AUC} - 0.5|$$

$$\texttt{discriminative\_auc\_absz} = |(\mathrm{AUC} - 0.506)/0.084|$$

Both are minimised at the GAN optimum and entered into the Temporal family.

Why It Matters Here — link to the thesis

The three bugs found in the discriminative AUC implementation (direction, null calibration, CV shuffling) are documented in ED Table 2 of the paper. They represent common pitfalls in synthetic data evaluation that affect published benchmarks. The corrected implementation is in cell `aaef7bd2` of `3_4_integrated_pipeline.ipynb`.

Chapter 16

Downstream Utility: VaR Backtesting and QLIKE

16.1 Value at Risk

Situation — the everyday picture

A bank risk manager wants to answer: "What is the most I should expect to lose on any single trading day, with 95% confidence?" The answer is the **Value at Risk** (VaR) at the 5% level. If the daily VaR is 2%, it means the bank should expect to lose more than 2% on at most 1 day in 20.

Worked Arithmetic — concrete numbers

Unconditional VaR (5th percentile of the empirical distribution): For BOVESPA 20-year data, the 5th percentile of $r_t \approx -2.7\%$. $\mathrm{VaR}_{0.05} = -2.7\%$ means: on 5% of days, expect worse than a -2.7% return.
Conditional VaR (GARCH-based):

$$\hat{\sigma}_t^2 = \hat{\omega} + \hat{\alpha} r_{t-1}^2 + \hat{\beta}\hat{\sigma}_{t-1}^2$$
$$\mathrm{VaR}_t = \hat{\mu} + \hat{\sigma}_t \cdot z_{0.05}, \quad z_{0.05} = -1.645$$

This VaR changes each day based on yesterday's volatility. On a quiet day, $\mathrm{VaR}_t \approx -1.5\%$. After a large move, $\mathrm{VaR}_t \approx -3.5\%$.
Why unconditional VaR is useless as an evaluation metric: Gaussian iid noise (white noise with σ = historical vol) achieves coverage error = 0.0276. Real GARCH parameters on real data also achieve coverage error = 0.0259. The two are nearly identical because unconditional VaR tests only the marginal distribution, which is already covered by Wasserstein and KS.

Formal Name — the mathematics

Value at Risk at level α:

$$\mathrm{VaR}_\alpha = \inf\{x : \mathbb{P}(r_t \le x) \ge \alpha\}$$

For a GARCH model with normal innovations:

$$\mathrm{VaR}_{\alpha,t} = \hat{\mu} + \hat{\sigma}_t \Phi^{-1}(\alpha)$$

Coverage error: $|\hat{v} - \alpha|$ where $\hat{v}$ = fraction of test days with $r_t < \mathrm{VaR}_{\alpha,t}$. Correct model: $\hat{v} \approx \alpha = 5\%$. Observed for real GARCH on real BRICS data: $\hat{v} = 7.59\%$, so coverage error $= |0.0759 - 0.05| = 0.0259$.

Where the Analogy Breaks

VaR captures the 5th percentile only. It says nothing about what happens *beyond* the 5th percentile (Expected Shortfall / CVaR covers this). We use VaR because Kupiec and Christoffersen backtests are specifically designed for it, and because the thesis focuses on evaluating the GARCH structure, not the full tail.

Why It Matters Here — link to the thesis

The real-GARCH coverage error of 0.0259 is the reference baseline in `var_coverage_error`. A synthetic GARCH that transfers well should produce VaR forecasts with similar coverage. This metric is in DESCRIPTIVE (not TEMPORAL) because it probes only the marginal coverage rate, not the temporal dynamics of whether violations cluster together (which is what Christoffersen tests).

16.2 Kupiec Backtest

Situation — the everyday picture

If your VaR model is correct, exceedances (days where the loss exceeds VaR) should happen exactly 5% of the time. If you observe 40 exceedances in 200 test days (20%), something is wrong. The Kupiec test asks: is the observed exceedance rate statistically distinguishable from the claimed 5%?

Worked Arithmetic — concrete numbers

Let n = test days, x = observed exceedances, α = claimed coverage (5%).
Under the null hypothesis ($p = \alpha$):

$$LR_{\text{Kupiec}} = -2\ln\left[\frac{\alpha^x(1-\alpha)^{n-x}}{\hat{p}^x(1-\hat{p})^{n-x}}\right], \quad \hat{p} = x/n$$

$LR_{\text{Kupiec}} \sim \chi^2(1)$ under H_0.
Power at different n: $n = 125$ (5-year per-market): power = 17% at true $p = 7\%$ — barely above random. $n = 496$ (20-year per-market): power $\approx 56\%$. $n = 2{,}480$ (20-year pooled): power $\approx 99\%$.
With 5-year data, Kupiec is essentially powerless. With 20-year pooled data, it is meaningful.

Formal Name — the mathematics

Kupiec (1995) unconditional coverage test:

$$H_0 : p = \alpha, \quad H_1 : p \neq \alpha$$

Likelihood ratio statistic:

$$LR_{uc} = 2\ln\bigl[\hat{p}^x(1-\hat{p})^{n-x}\bigr] - 2\ln\bigl[\alpha^x(1-\alpha)^{n-x}\bigr] \sim \chi^2(1)$$

Rejects at 5% significance if $LR_{uc} > 3.84$.

Where the Analogy Breaks

Kupiec tests only the *unconditional* coverage: is the total fraction of exceedances right? It does not test whether exceedances cluster in time (which would indicate the model fails in precisely the turbulent periods that matter most). That is the Christoffersen test.

Why It Matters Here — link to the thesis

Kupiec p-values are reported in the downstream utility section but are descriptive only at per-market n. At pooled $n \approx 2{,}480$, they are informative. The primary ranking metric is QLIKE because it is a proper scoring rule with no minimum-sample-size issue.

16.3 Christoffersen Independence Test

Situation — the everyday picture

Your VaR model has 5% coverage (right number of exceedances). But all 5% happen in a single turbulent week. After that week, perfect calm. The model is correct on average but wrong when it matters: it fails during exactly the periods banks need accurate risk estimates.

Formal Name — the mathematics

Christoffersen (1998) independence test: tests whether VaR exceedances are independent over time.
Let $I_t = \mathbf{1}[r_t < \mathrm{VaR}_{\alpha,t}]$ be the indicator of an exceedance. Define transition counts: n_{ij} = number of days where $I_{t-1} = i$ and $I_t = j$.
Under independence:

$$LR_{\text{ind}} = 2\ln\left[\frac{\hat{\pi}_{01}^{n_{01}}(1-\hat{\pi}_{01})^{n_{00}}\hat{\pi}_{11}^{n_{11}}(1-\hat{\pi}_{11})^{n_{10}}}{\hat{\pi}^{n_{01}+n_{11}}(1-\hat{\pi})^{n_{00}+n_{10}}}\right] \sim \chi^2(1)$$

where $\hat{\pi}_{ij} = n_{ij}/(n_{i0} + n_{i1})$ and $\hat{\pi} = (n_{01} + n_{11})/(n_{00} + n_{01} + n_{10} + n_{11})$.

Why It Matters Here — link to the thesis

The combined Christoffersen test ($LR_{\text{cc}} = LR_{\text{uc}} + LR_{\text{ind}} \sim \chi^2(2)$) tests both unconditional coverage and independence simultaneously. For synthetic GARCH parameters to be useful, the VaR forecasts they produce must have both correct coverage and independent exceedances. This is a substantially harder bar than Kupiec alone.

16.4 QLIKE Loss

Situation — the everyday picture

You have two weather forecasters. Forecaster A says "tomorrow will be a 2% return day" (one number). Forecaster B says "tomorrow's volatility will be 1.5%" (a distribution). You observe the actual return. How do you score Forecaster B, who gave you a distribution, not a point forecast?

Worked Arithmetic — concrete numbers

QLIKE (Patton 2011):

$$\mathrm{QLIKE}_t = \log \hat{\sigma}_t^2 + r_t^2/\hat{\sigma}_t^2$$

A volatility forecast $\hat{\sigma}_t$ that is too high penalises $\log \hat{\sigma}_t^2$ (unnecessarily large). A forecast too low penalises $r_t^2/\hat{\sigma}_t^2$ (missed a large move).
TRTR (train and test on real data) baseline: QLIKE ≈ -8.134, sd 0.000. A synthetic GARCH that produces parameters close to the true ones will have QLIKE close to -8.134.
Per-market inversion (5-year data, $n \approx 126$): Gaussian noise: QLIKE $= -6.888$. Real data: QLIKE $= -6.876$. Gaussian noise beats real data — QLIKE is meaningless at this n. Shuffled control sd across seeds: 3.354 — random variation dominates.
Pooled (20-year, $n \approx 2{,}480$): QLIKE is stable and interpretable. Per-market ($n \approx 496$) also stable.

Formal Name — the mathematics

QLIKE (Quasi-Likelihood): a proper scoring rule for volatility forecasts.

$$\mathrm{QLIKE}(r, \hat{\sigma}) = \frac{1}{T}\sum_{t=1}^{T}\left[\log \hat{\sigma}_t^2 + \frac{r_t^2}{\hat{\sigma}_t^2}\right]$$

A scoring rule is **proper** if it is minimised in expectation when $\hat{\sigma}_t^2 = \mathbb{E}[r_t^2|\mathcal{F}_{t-1}]$ (the true conditional variance). Patton (2011) shows QLIKE is robust to the use of imperfect variance proxies (here: r_t^2 instead of the true σ_t^2).

Where the Analogy Breaks

QLIKE is minimised (more negative = better) when the volatility forecast matches the conditional variance. It cannot test whether the GARCH model is the right *specification* (e.g., whether leverage effects matter), only whether the estimated parameters are close to the true conditional variance. A synthetic GARCH with wrong parameters but right unconditional volatility might score well on QLIKE by accident — Kupiec and Christoffersen provide additional checks.

Why It Matters Here — link to the thesis

QLIKE is the primary downstream utility ranking signal because: (1) it is a proper scoring rule, (2) it directly tests whether the GARCH structure learned from synthetic data transfers to real data, and (3) it is continuous with no minimum-power issues at pooled $n \approx 2{,}480$. The TSTR protocol (Train on Synthetic, Test on Real) uses synthetic GARCH parameters to forecast real-data volatility and computes QLIKE on the real test set.

16.5 Walk-Forward Validation

Situation — the everyday picture

You want to evaluate a forecasting model on historical data. The naive approach: train on all data up to 2020, test on 2020–2026. But this uses only one test period. If 2022–2026 happened to be unusual (it was, for MOEX), your evaluation might be misleading.
Walk-forward validation uses multiple test periods, each with a fresh training set.

Formal Name — the mathematics

Rolling-origin walk-forward (CTBench protocol):

- Divide data into K folds (here: $K = 5$)
- For fold k: train on data up to fold k's start, test on fold k
- Report metric mean $\pm$ std across 5 folds

Unlike K-fold cross-validation, the order of folds is never shuffled. Later folds always use more training data (expanding window) or a shifted training window (rolling window). We use rolling origin.

Why not K-fold with shuffling? Financial returns have serial dependence. Shuffling folds puts future data in the training set (look-ahead leakage). The discriminative AUC case showed this explicitly: shuffling 20-day windows shifts the null from 0.506 to 0.584.

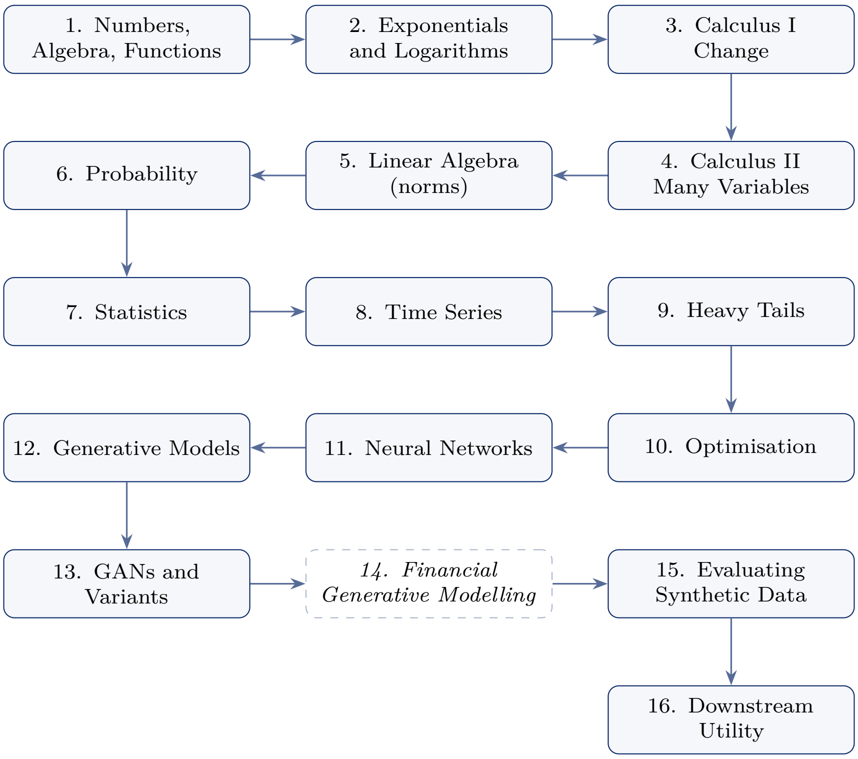
Why It Matters Here — link to the thesis

Five rolling folds per market give mean $\pm$ std across folds rather than a single metric estimate. This enables: (1) confidence intervals, (2) detection of fold-specific anomalies (MOEX fold 3 covers the 2022 invasion), (3) comparison of temporal stability across architectures. The CTBench protocol (Liao et al., 2025) requires this design for standardised reporting.

Symbol Glossary

Symbol	Meaning
r_t	Log return on day t: $\ln(P_t/P_{t-1})$
σ_t	Time-varying conditional standard deviation at t
σ	Unconditional (long-run) daily volatility
μ	Long-run mean return (close to zero)
α (GARCH)	ARCH coefficient: weight on the squared previous shock
β (GARCH)	GARCH coefficient: weight on previous variance
ω	GARCH intercept: ensures long-run variance is positive
$\hat{\alpha}$ (Hill)	Tail index estimate from Hill estimator
H	Hurst exponent ($H > 0.5$ implies long memory)
W_1	Wasserstein-1 (Earth Mover's) distance
$\mathbb{E}$	Expectation operator
Var	Variance operator
$\mathcal{N}(\mu, \sigma^2)$	Normal distribution with mean μ and variance σ^2
$\mathbb{P}(\cdot)$	Probability of an event
z_α	α-quantile of standard normal: $z_{0.05} = -1.645$
AUC	Area under the ROC curve
GAN	Generative Adversarial Network
WGAN-GP	Wasserstein GAN with Gradient Penalty
GRU	Gated Recurrent Unit
TCN	Temporal Convolutional Network
CNN	Convolutional Neural Network
BCE	Binary Cross-Entropy loss
GARCH	Generalised ARCH
VaR	Value at Risk
QLIKE	Quasi-likelihood loss for volatility forecast evaluation

Dependency graph and roadmap. Chapters are strictly ordered: nothing depends on a concept introduced later. The foundations (Part I) are prerequisites for everything that follows.



Solid boxes are written. Dashed italic boxes are planned and not yet drafted; their numbers are reserved so that cross-references remain stable as the book is completed. The critical path this book exists to support runs

$$\text{derivative} \to \text{partial derivative} \to \text{gradient} \to \text{norm} \to \text{Lipschitz continuity} \to \text{Wasserstein distance} \to \text{WGAN} \to \text{critic} \to \text{gradient penalty},$$

and Chapters 3–5 supply every link in it up to and including the norm.